

GRAS Determination for the Use of Mycoprotein in Selected Conventional Foods

Prepared for:
Division of Food Ingredients
Office of Pre-Market Additive Safety
Office of Food Chemical Safety, Dietary Supplements, and Innovation
Human Foods Program
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GRAS Submission Date: February 24, 2025

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FOREWORD

MoreMeat (Guangzhou) Biotech Co., Ltd. (“MoreMeat”) is dedicated to discovering and developing strains of fungi capable of efficiently producing target proteins. Utilizing advanced microbial and synthetic biology technology platforms, MoreMeat aims to meet the growing demand for alternative proteins and health foods by producing fungal proteins and health foods at lower costs and with greater sustainability. MoreMeat specializes in producing mycoprotein, which is the subject of this GRAS notification. Mycoprotein can be used as an ingredient or macronutrient.

At MoreMeat's request, Anchor Center for Certification (ACC) convened an Expert Panel to complete an independent safety evaluation of this mycoprotein preparation.

ACC primarily focused on composite safety information, i.e., scientific procedures with corroboration from the history of use. The review covered the production strains and identity information, safety/toxicity studies of mycoprotein, the history of use of mycoprotein, and compositional details, specifications, and method of manufacture of MoreMeat mycoprotein. A thorough search of the scientific and regulatory literature was conducted through February 8, 2025, with particular attention paid to adverse effect reports and those that supported safety conclusions. Pertinent references from this review are listed in Part 7. The integration of safety/toxicity studies, supported by the history of use and dietary exposure information, ultimately forms the specific scientific basis for the GRAS conclusion.

Part 1 SIGNED STATEMENTS AND CERTIFICATION

1.1 Submission of GRAS Notice

MoreMeat is hereby submitting a GRAS notice in accordance with subpart E of 21 CFR part 170.

1.2 Name and Address of The Sponsor

MoreMeat (Guangzhou) Biotech Co., Ltd
Room 101, Building C1,
No. 11, Kaiyuan Avenue, Huangpu District,
Guangzhou City, Guangdong Province, China

As the responsible party, MoreMeat accepts the responsibility for the GRAS conclusion that has been made for this mycoprotein ingredient as described in the subject safety evaluation; consequently, the mycoprotein ingredient having an acceptable composition that meets the conditions described herein, is not subject to the premarket approval requirements for food ingredients.

1.3 Common or Usual Name

MoreMeat plans to market this product as "mycoprotein". MoreMeat recognizes that "mycoprotein" is commonly used in the industry referring to FCC 14 Monograph of "Mycoprotein" as the "cellular mass at approximately 25% (w/w) total solids obtained from the filamentous fungus *Fusarium venenatum* that occurs as a paste substance". Due to the substantial similarity of products, MoreMeat will adopt this terminology to describe its product in this GRAS notice. MoreMeat also plans to market this product as "mycelium protein" or "fungal protein."

1.4 Trade Secret or Confidential Information

This notification does not contain any trade secret or confidential information.

1.5 Intended Use

Mycoprotein is intended to be used as a source of dietary protein in a variety of products, except in egg, meat, poultry, and fish products, as detailed in Part 3, at levels determined in this safety review and current Good Manufacturing Practices (cGMP). This ingredient is not intended for use in products regulated under the U.S. Department of Agriculture. Furthermore, the mycoprotein ingredient is not intended to be used in infant formula.

1.6 Basis of Exclusion from the Requirement for Premarket Approval Pursuant to Subpart E of 21 CFR 170.30¹

MoreMeat (Guangzhou) Biotech Co., Ltd has concluded that its mycoprotein preparation, which meets the specifications described herein, is GRAS in accordance with the Federal Food, Drug, and Cosmetic Act (FD&C Act). The GRAS determination is based on scientific procedures as described in the following sections. The evaluation accurately reflects the intended conditions of use for the designated ingredient.

1.7 Availability of Information

The data and information that serve as the basis for this GRAS determination will be available for review and copying at reasonable times at the office of MoreMeat (Guangzhou) Biotech Co., Ltd:

Contact: Luqin Yin
MoreMeat (Guangzhou) Biotech Co., Ltd
Room 101, Building C1,
No. 11, Kaiyuan Avenue, Huangpu District,
Guangzhou City, Guangdong Province, China
Email: info-reg@moremeat.com
Or will be sent to FDA upon request.

1.8 Freedom of Information Act (FOIA)

Parts 2 through 7 of this notification do not contain data or information that is exempt from disclosure under FOIA.

1.9 Information Included in the GRAS Notification

To the best of our knowledge, the information contained in this GRAS notification is complete, representative, and balanced. It contains both favorable and unfavorable information, known to the Sponsor and ACC, and pertinent to the evaluation of the safety and GRAS status of MoreMeat's mycoprotein.

Signed:



Name: —Luqin Yin————

Date: February 24, 2025

¹ See 81 FR 54960, 17 August 2016. Accessible at: <https://www.govinfo.gov/content/pkg/FR-2016-08-17/pdf/2016-19164.pdf>
(Accessed 2025/02/08)

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Signed:



Agent for MoreMeat (Guangzhou) Biotech Co., Ltd
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Date: February 24, 2025

Part 2 IDENTITY, METHOD OF MANUFACTURE, SPECIFICATIONS, AND PHYSICAL OR TECHNICAL EFFECT

This section reviews and cites publicly available information, alongside the information provided by the manufacturer, MoreMeat (Guangzhou) Biotech Co., Ltd, pertaining to the composition, manufacturing process, and specifications of this mycoprotein, which is the subject of this GRAS determination.

2.1 Identity of Ingredient

MoreMeat's mycoprotein ingredient is derived from submerged liquid aerobic fermentation of *Fusarium compactum*. This mycoprotein consists of the mycelial network of the fungal microorganism *F. compactum* MM-135. Under well-controlled cultivation conditions, *F. compactum* MM-135 grows into a filamentous mat of mycelial biomass, referred to as a “biomat”, with a texture profile similar to that of muscle fibers.

F. compactum MM-135 was isolated from soil of Shennongjia, Hubei Province, China and subsequent identification was performed in-house by MoreMeat (as described further, below). Morphological study shows that fungal cells are arranged in a filamentous pattern to form hyphae and that there are separations between the cells. The average length of hyphae is 800~1500 μm , the hyphae are usually branched, and the large conidia are sickle-shaped or long cylindrical, with more transverse septums. The hyphae may cross and entangle to form a reticular structure (Figure 1).

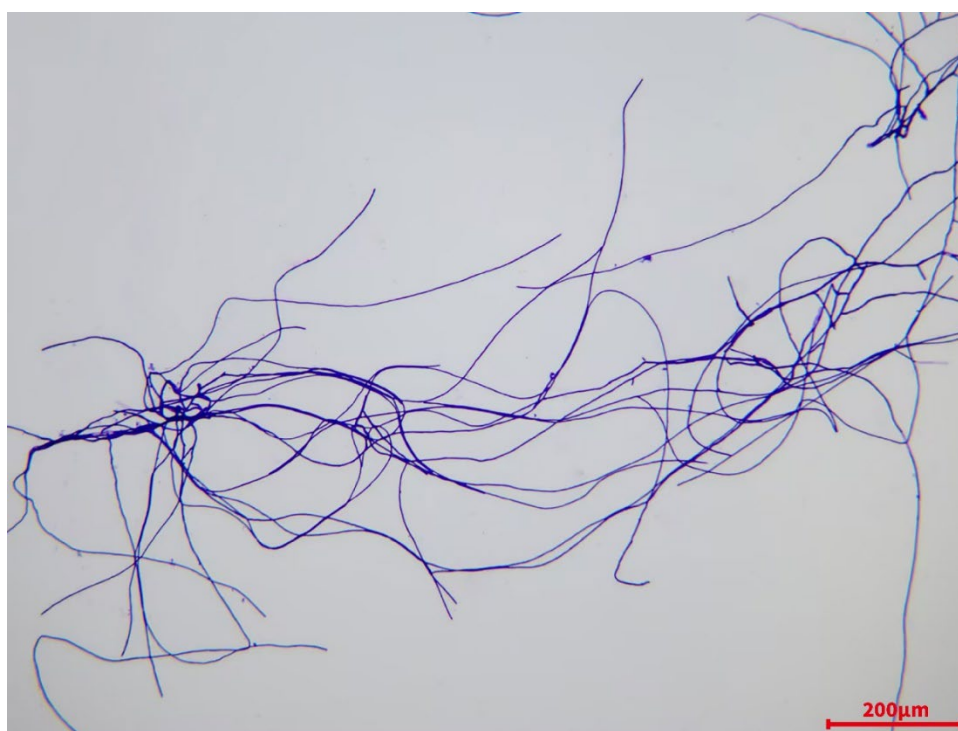


Figure 1 Mycelial Network derived from *Fusarium compactum* MM-135 under Light Microscope

Numerous molecular phylogenetic studies over the past two decades have indicated that morphological species recognition frequently fails to distinguish many fusaria that have been discovered by employing genealogical concordance phylogenetic species recognition (GCPSR sensu Taylor et al. 2000); therefore, MoreMeat has taken a polyphasic approach to further verify the strain through sequence analysis of four *Fusarium*-ID genes. These four genes include internal transcribed spacer region (ITS), translation elongation factor I (TEF1), RNA polymerase I (RPB1), and RNA polymerase II (RPB2) genes (Geiser et al. 2004; Park et al. 2010; O'Donnell et al. 2010; O'Donnell et al., 2013). MoreMeat employed the Polyphasic identification from the FUSARIOID-ID database to identify *F. compactum* MM-135 using these four genes. Result of DNA sequence comparison from these four genes shows that *F. compactum* MM-135 shares the greatest sequence homology with *F. compactum* LLC 2178, with a similarity of 99.82%.

Furthermore, MoreMeat collected TEF1, RPB1, RPB2, and ITS sequences from all strains of multiple subspecies in the FUSARIOID-ID database (www.fusarium.org) and analyzed the evolutionary position of the *F. compactum* MM-135 by constructing a multigene phylogenetic tree. The subspecies includes *Fusarium incarnatum-equiseti* species complex (FIESC, to which *F. compactum* belongs to), *Fusarium sambucinum* species complex (FSAMSC, to which *F. venenatum* belongs to), and *Fusarium solani* species complex (FSSC, to which *F. strain. flavolapis* belongs to). The steps for multigene phylogenetic tree analysis are as follows: First, use MAFFT (version v7.310) to perform multiple sequence alignment for each gene; then, use GBLOCKS (version 0.91b) to prune the multi-sequence alignment result file; and finally use AliView to manually prune. After pruning, the sequences of the four genes were concatenated. IQ-TREE (version 2.2.5) was used to construct a multigene phylogenetic tree based on the concatenated sequences. The tree below (Figure 2) is drawn by ggtree. As the closest species of MM-135 is *Fusarium compactum*, MoreMeat verifies and concludes that the production organism indeed is *Fusarium compactum*.

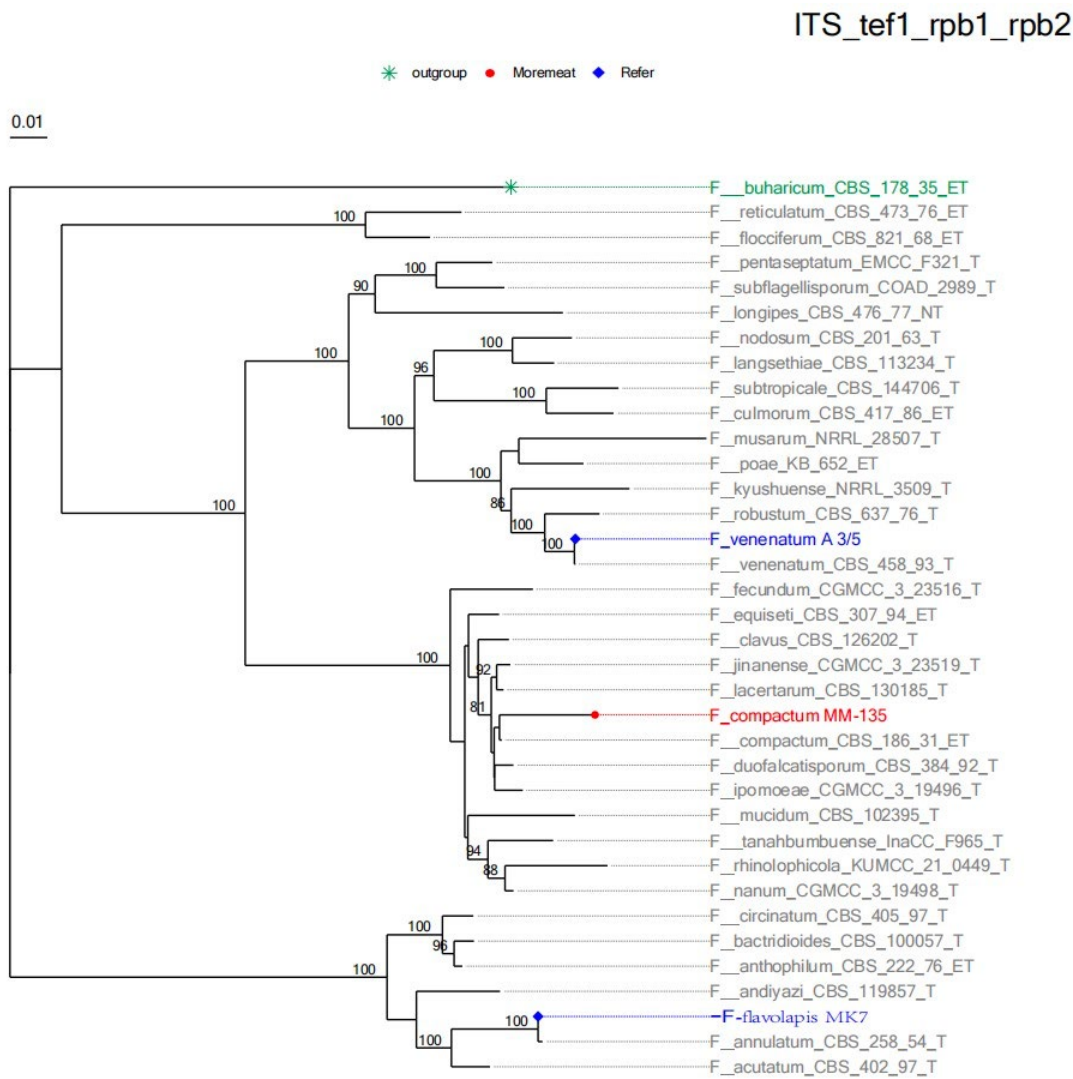


Figure 2 Phylogenetic tree mapping of *F. compactum* MM-135.

The phylogenetic tree in Figure 2 shows that *F. compactum* MM-135 (production organism of the current notification), *F. venenatum* (production organism of GRN 000945 and GRN 000091), and *F. strain flavolapis* (production organism of GRN 000904) belong to neighboring groups of *Fusarium* species. Therefore, they have close evolutionary relationship, sharing large genetic similarities. Figure 2 also shows that *F. compactum* is located between *F. venenatum* and *F. strain flavolapis*, a position closer to *F. venenatum*, which implies that *F. compactum* is much closer to *F. venenatum* than *F. strain flavolapis*, in terms of genetic similarity.

Table 1 shows that by phylogenetic taxonomy classification, *Fusarium compactum* belongs to the family Nectriaceae and is a member of the Fungi kingdom. The production organism *F. compactum* MM-135 is deposited in the China General Microbiological Culture Collection Center (CGMCC) with the designation number of 41312.

Table 1 Taxonomic Classification of *Fusarium Compactum*

Classification	Scientific Classification
Kingdom	Fungi
Phylum	Ascomycota
Class	Sordariomycetes
Order	Hypocreales
Family	Nectriaceae
Genus	<i>Fusarium</i>
Species	<i>compactum</i>
Strain	MM-135
Obtained from: https://www.catalogueoflife.org/data/taxon/6JSRT (Accessed on 2024/9/14)	

The production organism utilized by MoreMeat bears resemblance to those described in prior GRAS Notifications where FDA subsequently issued "no questions" letters. Table 2 lists the GRAS Notices with other species of *Fusarium* that have received such acknowledgment from FDA.

Table 2 GRAS Notices with Similar Production Organisms as Compared with MoreMeat's

GRN Number	Production Organism	Manufacturer
Current GRAS Notification	<i>Fusarium compactum</i> MM-135	MoreMeat (Guangzhou) Biotech Co., Ltd
GRN 000945	<i>Fusarium venenatum</i> (originally referred to as <i>Fusarium graminearum</i> A 3/5)	3F Bio Limited
GRN 000904	<i>Fusarium strain flavolapis</i> (formerly known as <i>Fusarium novum. yellowstonensis</i>)	Nature's Fynd (The Fynder Group, Inc.)
GRN 000091	<i>Fusarium venenatum</i> PTA-2684	Marlow Foods Ltd.

In summary, the phylogenetic tree and DNA sequence comparison shows a similar phylogenesis and high genetic relationship between *F. compactum* MM-135, *F. venenatum*, and *Fusarium strain flavolapis*. *F. compactum* MM-135 shows greater genetic similarity to *F. venenatum* A 3/5 than to *Fusarium strain flavolapis* MK7 when comparing genetic sequences. Furthermore, there is no genetic modification process done on the production organism. MoreMeat confirms that *F. compactum* is non-pathogenic and non-toxicogenic.

2.2 Manufacturing Process

A production batch of mycoprotein is defined as a product manufactured using the same batch of raw materials, the same batch of production organism, and the same fermentation batch. The production organism is not genetically modified and there are no antibiotics added in the fermentation process. The production process for MoreMeat's mycoprotein is shown in Figure 3, followed by a detailed description. The steps employed for fermentation are essentially the same as those described in GRN

000904 (page 10) and GRN 000945 (page 11).

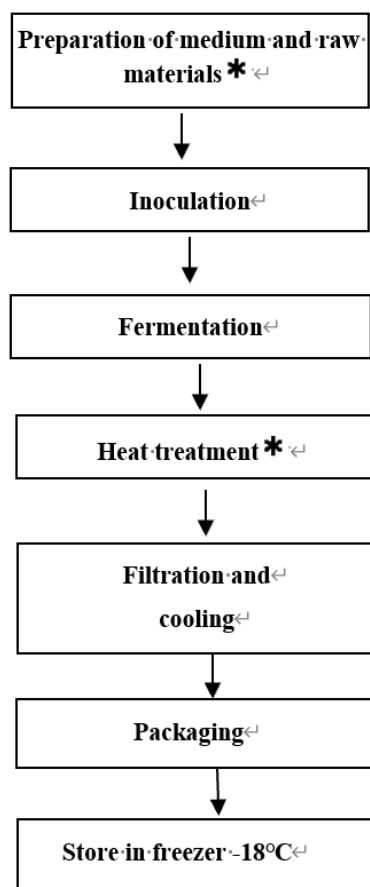


Figure 3 Manufacturing process for mycoprotein derived from *F. compactum* MM-135.

Note: * Represents critical control points

Preparation of medium and raw materials

The raw materials used in mycoprotein fermentation are commonly used in the food, fermentation, and enzyme production industries. All raw materials are of food-grade standards and the filters used are food-contact safe and/or listed in 21 CFR as approved food-contact materials. None of the materials used in the production of MoreMeat's mycoprotein are derived from major allergens.

This process is the first critical control point of the whole production process. All raw materials are inspected batch by batch according to internal procedures and review of the corresponding certificate of analysis provided by the supplier before release to production. Subsequently, raw materials are filtered through a 60-mesh stainless steel sieve to remove impurities; this step prevents physical hazards from raw materials entering the production process. After, the fermentation medium is either sterilized at 115-121°C for 20-30 minutes or filtered through a 0.22 µm filter membrane, which prevents microbial contamination from raw materials entering the production process. Only the production strain is allowed to enter the fermentation system so that food safety and product quality is controlled and managed in this critical control point. All raw materials and processing aids are listed

in Table 3.

Table 3 List of Ingredients, Processing Aids and Food Contact Surfaces Used in the Manufacturing Process

Materials Name	Function	Regulatory Status or Grade
Glucose	Carbon source of seed medium	21 CFR 168.121
Yeast extract	Nitrogen source of seed medium	21 CFR 184.1983
Corn starch	Fermentation carbon source	FCC
Citric Acid	pH Regulator	21 CFR 184.1033
Ammonia	Fermentation nitrogen source, pH Regulator	FCC
Ammonium sulfate	Fermentation nitrogen source	21 CFR 184.1143
Potassium dihydrogen phosphate	Nutrients for seed medium and fermentation	GRAS (SCOGS 32)
Phosphoric Acid	Nutrients for fermentation	21 CFR 182.1073
Magnesium sulfate	Inorganic salt for seed medium and fermentation	21 CFR 184.1443
Calcium Chloride	Inorganic salt for seed medium and fermentation	21 CFR 184.1193
Sodium hydroxide	pH Regulator	21 CFR 184.1763
α -amylase	Catalyze the hydrolysis of corn starch	21 CFR 172.892
Glucoamylase	Catalytic macromolecule	21 CFR 172.892
Zinc Sulfate	Microelement	21 CFR 182.8997
Manganese sulfate	Microelement	21 CFR 184.1461
Ferrous sulfate	Microelement	21 CFR 184.1315
Thiamine Hydrochloride	Vitamin	21 CFR 184.1875
Riboflavin	Vitamin	21 CFR 184.1695
Niacin	Vitamin	21 CFR 184.1530
Pyridoxine hydrochloride	Vitamin	21 CFR 184.1676
D-Biotin	Vitamin	21 CFR 182.8159
Calcium pantothenate	Vitamin	21 CFR 184.1212
Folic acid	Vitamin	21 CFR 172.345
Ascorbic Acid	Vitamin	21 CFR 182.3013
Soybean oil	Defoaming agent	FCC
60-mesh Stainless Steel Sieve	Filter	cGMP Compliance Food Contact Surface Material

Inoculation

The production strain master and working banks are prepared under sterile conditions (in the Biosafety

cabinet) and tested for microbial purity by plating and observing the morphology of the resulting colonies. The production strain is inoculated into seed culture medium and enlarged by two runs of seed culturing. The first run is conducted in a 50-mL bioreactor, and the second run is then conducted in a 1000-mL bioreactor, respectively. The seed culture medium includes glucose as major carbon source, yeast extract as major nitrogen source, as well as potassium dihydrogen phosphate, magnesium sulfate, and calcium chloride. Strain purity is monitored by morphological tests on individual colonies that have been picked from the seed cultures and grown on plates. The entire operation is carried out using aseptic technology under sterile conditions.

Fermentation

Before fermentation begins, the pre-sterilized fermentation tank is configured with a temperature range of 20–35°C, an agitation speed of 50–300 rpm, a ventilation rate of 0.5–2 VVM, and a pH range of 4.0–7.0. The whole fermentation process takes place in a closed system to prevent any cross-contamination. Fermentation medium includes corn starch (10-100 g/L), citric acid (1-10 g/L), ammonia (1-20 mL/L), ammonium sulfate (1-10 g/L), phosphoric acid (0.1-10 g/L), sodium hydroxide (0.1-10g/L), α -amylase (1 g per 1 kg corn starch), glucoamylase (1 g per 1 kg corn starch), and vitamins and minerals listed in Table 3.

Heat treatment

After fermentation, the fermentation broth is transferred to a heat treatment tank and is treated at 60-90°C for 10-30 minutes. The heat treatment inactivates the *F. compactum* and enzymes, if any, while reducing the RNA content. The final product is tested for RNA content and microbial limits to ensure no live *F. compactum* is present in the final product.

Heat treatment processing is the second critical control point. The production organism *F. compactum* is inactivated in this process and, as a result, MoreMeat's final product mycoprotein does not contain live *F. compactum* or active enzymes. The control limits are the heat treatment's temperature and time parameter. If the production process deviates from the set parameters, all involved product will be quarantined and evaluated by the quality management team.

Filtration and cooling

After heat treatment, the fermentation broth is filtered with a vacuum drum filter to remove excess water and to cool the product. The vacuum drum filter has a pore size of 400 mesh and operates at 0.04-0.08 mPa pressure. The residues of the raw material are removed via the filtration process.

Packaging

The product is packaged in an aluminum foil bag and vacuum sealed, or other food-grade packaging material provided the stability results meet the required standards.

Store in freezer at -18°C

The product is stored at -18°C before shipping to customers.

Disposal of production wastes

Fusarium species are well researched with many published studies. It is of interest because numerous species are important plant pathogens. (Balajee et al. 2009, Geiser et al. 2013, Nelson, 1981). For this reason, MoreMeat implements stringent HACCP plan to control and screen mycotoxins in its production process, as well as strict waste management to prevent environmental contamination. All solid and liquid wastes contaminated with *F. compactum* are sterilized prior to disposal. All solid waste from production is collected by qualified third-party contractors for disposal. Wastewater is collected and sent to the local wastewater treatment facility for proper treatment as per the local regulations. The release of treated wastewater complies with National Standard GB 18918 Discharge Standard of Pollutants for Municipal Wastewater Treatment Plant.

MoreMeat manufactures mycoprotein using a non-genetically modified *F. compactum*. Furthermore, the absence of production organism and control of RNA residue is verified in three non-consecutive batches (See Table 5). These precautions taken are to ensure that the mycoprotein product is of the highest quality and safe for human consumption.

2.3 Quality Control

2.3.1 In-process Quality Control

MoreMeat's mycoprotein is manufactured in compliance with Current Good Manufacturing Practices (cGMP) and Hazard Analysis Critical Control Point (HACCP) principles. MoreMeat's prerequisite programs are written according to the cGMP components listed in US FDA 21 CFR Part 117 Subpart B. These GMP programs establish conditions and practices that MoreMeat follows when producing mycoprotein under sanitary conditions.

In addition, the food safety plan for MoreMeat's mycoprotein is developed in compliance with the requirements defined in the US FDA Food Safety Modernization Act (FSMA). Potential hazards arising from raw materials/processing aids and each manufacturing step are reviewed and evaluated.

Based on MoreMeat's HACCP plan, there are two identified critical control points for the production process. The first critical control point is the preparation of medium and raw materials. Materials are either heat sterilized or undergo sterile filtration before going into production. The purpose of this process is to make sure no contaminating microorganism enters the production system. The sterile conditions during fermentation are tightly monitored by production operators and online quality assurance personnel. The second critical control point is the heat treatment. During the heat treatment process, the production organism *F. compactum* is inactivated and, as a result, MoreMeat's final product mycoprotein does not contain live *F. compactum* or active enzymes.

Allergen management and control procedures are implemented and executed at the warehouse, production, and packaging areas. The quality system established at MoreMeat is certified by the globally recognized food safety verification program Food Safety System Certification FSSC 22000.

(Appendix 1)

During the production process, operating parameters such as temperature, fermentation time, pH, and pressure are controlled to ensure product quality. MoreMeat conducts hazard analysis for manufacturing processes, raw materials, products, packaging materials, equipment, and food contact surfaces.

The HACCP plan for the whole manufacturing process includes in-process controls to reduce potential impurities to the lowest level technically possible. All raw materials, food additives, processing aids, and food contact surfaces used meet applicable US food-grade standards. Table 3 summarize the fermentation ingredients and processing acids used in the production process.

Furthermore, MoreMeat uses city water as their production water source. Production water is dispatched to a third-party laboratory for testing annually. The test parameters, methods, and acceptance criterion meet China's National requirement for drinking water. MoreMeat also conducts internal monitoring of production water monthly, with test parameters including turbidity, odor and taste, pH, conductivity, total plate counts, and total coliform.

Standard operating procedures and work instructions are followed, batch records are retained, numerous in-process controls are performed, and the final mycoprotein product is controlled by certificates of analysis and batch release routines. For every production lot, protein content is assayed and recorded in the certificate of analysis by MoreMeat. This practice complies with the requirements of current GMPs for food products (21 Code of Federal Regulations §117).

2.3.2 Product Specification

Table 4 summarizes the product specifications for MoreMeat's mycoprotein derived from *F. compactum*. Food Chemical Codex (FCC) 14 has released an official monograph for mycoprotein derived from *F. venenatum*. The specification for MoreMeat's mycoprotein is adopted from the FCC monograph with a few minor changes tailored to MoreMeat's mycoprotein. Thus, the specifications for MoreMeat's mycoprotein are similar to the FCC monograph except for protein content, mycotoxins, and microbial testing. Given that MoreMeat's mycoprotein is derived from *F. compactum*, which is a different fusarium species than that specified in the FCC monograph (i.e., *F. venenatum*), mycotoxins specific to *F. compactum* (See Section 2.3.3.1 for mycotoxins testing) are analyzed. Additionally, MoreMeat's specifications include additional microbial limits for total plate count, yeasts and molds, *Listeria monocytogenes*, *Escherichia coli*, *Staphylococcus aureus*, *Salmonella*, and viable fusarium. In short, MoreMeat believes the minor differences in specifications as compared to the FCC monograph do not impact product safety, and the additional test parameters help to ensure that potential contaminants are characterized and controlled.

MoreMeat's mycoprotein product specifications include protein content, ash, moisture, heavy metals, mycotoxins, and microbial limits. Two mycotoxins (enniatins and beauvericin) are included in the

product specification based on the safety evaluation of *Fusarium compactum* (a detailed discussion is presented in section 6.7 of this notification). The product specification has <2% RNA in dry matter; measurements were made with a sensitive spectrophotometry-based assay widely used in academic research and industry (Nucleic acid assay method: spectrophotometry, 2023). MoreMeat's mycoprotein RNA content test and acceptance criteria is aligned with that of the mycoprotein product derived from *F. venenatum*, referring to FCC 14 Monograph for Mycoprotein from *Fusarium venenatum*.

To ensure product quality, MoreMeat conducts analytical tests on every batch of mycoprotein product. Each batch must meet the acceptance criteria before release. Three non-consecutive lots of mycoprotein were manufactured and tested. Test results are summarized in Table 5 and demonstrate that all three batches of the ingredient meet the product specifications. The collection of these reports demonstrates that the ingredient is well-characterized and meets the established limits.

Table 4 MoreMeat's Product Specification for Mycoprotein

Test Items	Specification for MoreMeat Mycoprotein	Test Methods
Appearance	Off white to light yellow, dough-like matter.	Organoleptic
Flavor	Odorless	
Taste	Tasteless	
Moisture	≤78%	AOAC 930.04/ 934.01
Protein	≥43%, on dried basis	AOAC 920.87/ 979.09
Total fiber	≥25.0%, on dried basis	AOAC 2011.25
Ash	<5.0%	AOAC 942.05
Cadmium	<0.1 mg/kg, on dried basis	AOAC 2015.01
Lead	<0.1 mg/kg, on dried basis	AOAC 2015.01
Arsenic	<0.1 mg/kg, on dried basis	AOAC 2015.01
Mercury	<0.1 mg/kg, on dried basis	AOAC 2015.01
Enniatins	<40 µg/kg	Published HPLC-MS/MS method ^a
Beauvericin	<10 µg/kg	Published HPLC-MS/MS method ^a
Total Aerobic Count	<10,000 CFU/g	USP General Chapter 61 <2022,2023>
Yeasts and Molds	≤100 CFU/g	USP General Chapter 61 <2022,2023>
<i>Listeria monocytogenes</i>	Not detected /25 g	FDA BAM Chapter 10

<i>Escherichia coli</i>	Not detected/g	USP General Chapter 62 <2022,2023>
<i>Staphylococcus aureus</i>	Not detected/g	USP General Chapter 62 <2022,2023>
<i>Salmonella</i>	Not detected/25 g	FDA BAM Chapter 5
Viable Fusarium	Not detected/g	Internal method: enrichment & subculture + ITS ^b
RNA in Dry Matter	≤2%	Nucleic acid assay method: spectrophotometry ^c

^a HPLC-MS/MS method is well established in publication <Development of high performance liquid chromatography tandem-mass spectrometry method for determination of beauvericin and enniatins in corn and wheat and their products> (Han, 2017)

^b Description of test method: Take 25 g mycoprotein sample into a sterilized homogenizing bag containing 225 mL NaCl-peptone buffer (pH = 7.0). Use homogenizer to mix the sample for 1-2 minutes. Pipette 10 mL sample solution into 100 mL Sabouraud Dextrose Broth and culture at 28 ± 2 °C for 24 to 48 hours. Subculture 10 µL of the enrichment broth to a plate of Sabouraud Dextrose Agar or Malt Extract Agar to culture for 2 to 5 days at 28 ± 2 °C. If *Fusarium* typical colony is observed, use ITS rRNA sequence for identification.

^c Description of test method: this test method is based on measuring the absorbance of ultraviolet light at 260 nm. The purine and pyrimidine bases in the molecular structure of RNA have a conjugated double bond system, which can absorb the ultraviolet light of 250-280 nm with the maximum absorption at 260 nm.

Table 5 Analytical Results of Three Batches of MoreMeat Mycoprotein

Test Items	Specification for current notification	Test Results		
		Batch #MM13524004	Batch # MM13524007	Batch # MM13524009
Appearance	Off-white to light yellow, dough-like matter	Light yellow, dough-like matter	Light yellow, dough-like matter	Light yellow, dough-like matter
Flavor	Odorless	Odorless	Odorless	Odorless
Taste	Tasteless	Tasteless	Tasteless	Tasteless

Moisture	≤78%	76.3%	74.1%	75.7%
Protein	≥43%, on dried basis	55.7%	57.5%	54.3%
Total fiber	≥25.0%, on dried basis	34.0%	39.1%	38.4%
Ash	<5.0%	0.44%	0.54%	0.50%
Cadmium ^a	<0.1 mg/kg, on dried basis	ND	ND	ND
Lead ^a	<0.1 mg/kg, on dried basis	ND	ND	ND
Arsenic ^a	<0.1 mg/kg, on dried basis	ND	ND	ND
Mercury ^a	<0.1 mg/kg, on dried basis	ND	ND	ND
Enniatins ^c	<40 µg/kg	0.04878 µg/kg	2.3087 µg/kg	ND
Beauvericin ^c	<10 µg/kg	5.955 µg/kg	8.895 µg/kg	0.7116 µg/kg
Total Aerobic Count	<10,000 CFU/g	740 CFU /g	< 10 CFU /g	< 10 CFU /g
Yeasts and Molds	≤100 CFU/g	< 10 CFU /g	< 10 CFU /g	< 10 CFU /g
<i>Listeria monocytogenes</i>	Not detected /25g	ND	ND	ND
<i>Escherichia coli</i>	Not detected/g	ND	ND	ND
<i>Staphylococcus aureus</i>	Not detected /g	ND	ND	ND
<i>Salmonella</i>	Not detected /25g	ND	ND	ND
Viable Fusarium	Not detected/g	ND	ND	ND
Ribonucleic Acid in Dry Matter ^b	≤2%	ND	ND	ND

^a Limit of detection for lead is 3.5 µg/kg, cadmium is 0.5 µg/kg, arsenic is 1.6 µg/kg , and mercury is 3.0 µg/kg.

^b Limit of detection for RNA is 0.016%

^c Limit of detection for mycotoxins are enniatins: 0.22 µg/kg; beauvericin: 0.04 µg/kg.

ND: not detected

2.3.3 Additional Testing

In addition to the parameters listed in MoreMeat’s product specifications, additional analysis was conducted on three non-consecutive batches of mycoprotein, and included expanded mycotoxin testing, nitrates, amino acid composition, nutritional profile, and nutrients. Considering that the manufacturing of mycoprotein complies with cGMP, MoreMeat believes that the risk of exposure to environmental hazards is negligible.

2.3.3.1 Mycotoxins

To ensure product safety, MoreMeat conducted a mycotoxins screening to cover all mycotoxins identified from previous mycoprotein GRAS notifications that received “no question” letters from FDA, related mycotoxins identified from the literature search, and those identified via bioinformatics analysis. The mycotoxins analysis was conducted by an ISO/IEX 17025: 2005 accredited third-party laboratory. Table 6 summarizes the mycotoxin analysis results of three non-consecutive batches of MoreMeat’s mycoprotein. The test results show that mycotoxin content is consistent across lots.

Table 6 Mycotoxins Screening Results of Three Non-consecutive Batches of Mycoprotein

Test Items	Test method	LOD (µg/kg)	Test Results (µg/kg)		
			Batch # MM1352400 4	Batch # MM13524007	Batch # MM13524009
3-Acetyldeoxynivalenol (3-AcDon)	GB 5009.111	10	ND	ND	ND
15-Acetyldeoxynivalenol (15-AcDon)	GB 5009.111	10	ND	ND	ND
Deoxynivalenol (DON)	GB 5009.111	10	ND	ND	ND
Fumonisin (sum of B1, B2 and B3)	GB 5009.240	13	ND	ND	ND
Fusarenon X (FUSX)	RWX_OP_016	10	ND	ND	ND
Nivalenol	RWX_OP_016	10	ND	ND	ND
Diacetoxyscirpenol (DAS)	RWX_OP_016	10	ND	ND	ND
Neosolaniol (NEO)	RWX_OP_016	10	ND	ND	ND
Zearalenone (ZEN)	GB 5009.209	5	ND	ND	ND
HT2 Toxin	NY/T 4124	2	ND	ND	ND

T2 Toxin	NY/T 4124	2	ND	ND	ND
Ochratoxin A	GB 5009.96	0.3	ND	ND	ND
Aflatoxin B1	GB 5009.22	0.03	ND	ND	ND
Aflatoxin (sum of B1, B2, G1 and G2)	GB 5009.22	0.08	ND	ND	ND
Aflatoxin M1	GB 5009.24	0.005	ND	ND	ND
Patulin	GB 5009.185	12	ND	ND	ND
Enniatin (sum of A, A1, B and B1)	HPLC-MS/MS method	/	0.04878	2.3087	ND
Enniatin A	HPLC-MS/MS method	0.02	0.04878	0.1138	ND
Enniatin A1	HPLC-MS/MS method	0.09	ND	0.1518	ND
Enniatin B	HPLC-MS/MS method	0.01	ND	1.0122	ND
Enniatin B1	HPLC-MS/MS method	0.10	ND	1.0309	ND
Beauvericin	HPLC-MS/MS method	0.04	5.955	8.895	0.7116
Fusaric acid	LS/T 6133-2018 HPLC MS/MS method	/	ND	ND	ND
Aurofusarin	LS/T 6133-2018 HPLC MS/MS method	/	ND	ND	ND
8-acetylneosoianiol	LS/T 6133-2018 HPLC MS/MS method	/	ND	ND	ND
T2 Triol	LS/T 6133-2018 HPLC MS/MS method	/	ND	ND	ND
Verrucarol	LS/T 6133-2018 HPLC MS/MS method	/	ND	ND	ND
Monoacetoxyscirpenol	LS/T 6133-2018 HPLC MS/MS method	/	ND	ND	ND
8,15-Isoneosoianiol	LS/T 6133-2018 HPLC MS/MS method	/	ND	ND	ND

Moniliformin	LS/T 6133- 2018 HPLC MS/MS method	/	ND	ND	ND
Culmorin	LS/T 6133- 2018 HPLC MS/MS method	/	ND	ND	ND
2(5H)-Furanone (BUTENOLIDE)	LS/T 6133- 2018 HPLC MS/MS method	/	ND	ND	ND
Fusarin C	LS/T 6133- 2018 HPLC MS/MS method	/	ND	ND	ND
Fusaproliferin	LS/T 6133- 2018 HPLC MS/MS method	/	ND	ND	ND
Cyclonerodiol	LS/T 6133- 2018 HPLC MS/MS method	/	ND	ND	ND

ND: not detected; LOD: limit of detection; “/” means not specified.

Of the thirty-five (35) mycotoxins tested, only two mycotoxins were detected (Enniatin and Beauvericin). Diacetoxyscirpenol (“DAS”), which is one of the most toxic of the trichothecene mycotoxins (Whitacre, 2014), was not detected. Enniatin and Beauvericin were tested with less than 2.5 µg/kg and 9.0 µg/kg respectively, which are within safe levels of human consumption. (See Section 3.5 for a detailed discussion). MoreMeat will implement a biannual screening of DAS in its final product as part of the quality assurance practice. If DAS is detected, quality assurance team will pull all retain samples from the most recent batch that passed the test and the current batch that failed. All retain samples will undergo DAS testing. Based on the results of the DAS tests, MoreMeat’s quality management team will evaluate product safety and conduct a root cause analysis to investigate the source of the DAS contamination. As a result of this analysis, more frequent sampling and testing may be implemented to prevent future issues.

2.3.3.2 Amino Acids

MoreMeat’s mycoprotein is a complete protein, consisting of all naturally occurring amino acids. It includes the nine essential amino acids that cannot be made by the human body and must be supplemented by food. The nine essential amino acids are isoleucine, leucine, histidine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. Based on analyses of three non-consecutive production batches of mycoprotein, the average amino acid content is tabulated Table 7, wherein the essential amino acids make up 20-24% of the total dry weight, the branched-chain amino acids (leucine, isoleucine, and valine) make up around 9% of the dry weight, and the total amino acids

make up 44%-49% of the total dry weight. The essential and non-essential amino acids combined comprise all the organic nitrogen content in the protein.

Table 7 Amino Acids in MoreMeat's Mycoprotein

Amino Acids	Batch # MM13524 004 (% of total, dry weight basis)	Batch # MM13524 004 (g/ 100g protein)	Batch # MM13524 007 (% of total, dry weight basis)	Batch # MM13524 007 (g/ 100g protein)	Batch # MM13524 009 (% of total, dry weight basis)	Batch # MM13524 009 (g/ 100g protein)	Average (g/ 100g protein)
Alanine	3.05	5.76	3.31	6.06	3.00	5.82	5.88
Arginine	3.04	5.75	3.46	6.33	3.04	5.89	5.99
Aspartic acid	4.72	8.92	5.07	9.28	4.63	8.98	9.06
Glutamic acid	5.39	10.19	5.82	10.65	5.30	10.27	10.37
Glycine	2.21	4.18	2.40	4.39	2.17	4.21	4.26
Histidine	1.11	2.10	1.18	2.16	1.09	2.11	2.12
Isoleucine	2.91	5.50	2.72	4.98	2.84	5.51	5.33
Leucine	3.69	6.97	4.30	7.87	3.65	7.08	7.30
Lysine	3.70	6.99	4.12	7.54	3.61	7.00	7.18
Phenylalanine	2.28	4.31	2.57	4.70	2.23	4.32	4.44
Proline	1.99	3.76	1.94	3.55	1.95	3.78	3.70
Serine	2.34	4.42	2.46	4.50	2.25	4.36	4.43
Threonine	2.39	4.52	2.56	4.68	2.33	4.52	4.57
Tyrosine	1.46	2.76	1.63	2.98	1.48	2.87	2.87
Valine	2.74	5.18	3.05	5.58	2.76	5.35	5.37
Methionine	0.82	1.55	1.64	3.00	0.82	1.59	2.05
Cystine and Cysteine	0.42	0.79	0.51	0.93	0.42	0.81	0.85
Tryptophan	0.73	1.38	0.71	1.30	0.73	1.42	1.36
Total Amino Acids	44.99		49.45		44.30		

Note: amino acid g/100 g protein is calculated as amino acid % of total / protein content / (1 – moisture

in dry weight). For example: Alanine (g/100 g protein) = $3.05 / 55.7 / (1 - 5\%) * 100$. Protein content for Batch MM13524004 is 55.7%, and moisture in dry weight is <5%.

2.3.3.3 Essential Nutrients

Essential nutrient analyses were conducted on three non-consecutive batches of MoreMeat mycoprotein. Table 8 summarize the vitamin and mineral content (values reported on a wet weight basis).

Table 8 Vitamins and Minerals in Mycoprotein

Test Items	Test method	Test Results (wet weight basis)		
		Batch # MM13524004	Batch # MM13524007	Batch # MM13524009
Vitamin B1 (mg/100g) ^a	GB 5009.84-2016 First method	ND	ND	ND
Vitamin B2 Riboflavin (mg/100g) ^b	GB 5009.85-2016 First method	<0.05	ND	<0.05
Niacin (µg/100g) ^c	GB 5009.89-2023 First method	ND	ND	ND
Ascorbic Acid (mg/100g) ^d	GB 5009.86-2016 1st method	ND	ND	ND
Pantothenic Acid (mg/100g) ^e	GB 5009.210-2023 First method	ND	ND	ND
Biotin (µg/100g)	R-Biopharm VitaFast® Biotin P1003 2021	0.80	0.82	0.46
Folic acid µg/100g	R-Biopharm VitaFast® Folic Acid P1001 2016	0.80	2.66	1.69
Phosphorus (mg/kg)	BS EN 16943:2017	1150	1380	1050
Zinc (mg/kg)	BS EN 16943:2017	41.5	47.4	40.8
Chloride (%) ^f	GB 5009.44-2016 third method	0.014	ND	0.015
Iron (mg/kg)	BS EN 16943:2017	18.1	15.8	17.6
Copper (mg/kg)	BS EN 16943:2017	<1	<1	<1
Magnesium (mg/kg)	BS EN 16943:2017	152	155	124
Calcium (mg/kg)	BS EN 16943:2017	222	184	194
Sodium (mg/kg)	BS EN 16943:2017	25.1	35.4	23.5
Potassium (mg/kg)	BS EN 16943:2017	412	454	506
Manganese (mg/kg)	BS EN 16943:2017	4.81	5.10	4.99

- ^a limit of detection for Vitamin B1 is 0.03 mg/100 g
^b limit of detection for Vitamin B2 is 0.02 mg/100 g
^c limit of detection for niacin is 70 µg/100 g
^d limit of detection for ascorbic acid is 0.5 mg/100 g
^e limit of detection for Pantothenic acid is 0.025 mg/100 g
^f limit of detection for Chloride is 0.008%

2.3.3.4 Nitrate and Nitrite

The nitrate and nitrite content of MoreMeat mycoprotein were evaluated for three non-consecutive batches. Table 9 below summarizes the nitrate and nitrite analyses (values reported on a wet weight basis). The estimated daily intake of nitrate and nitrite are discussed further in Section 3.6, which demonstrates both nitrate and nitrite levels in mycoprotein are within an acceptable level of safety.

Table 9 Nitrate and Nitrite in Mycoprotein

Nitrate & Nitrite	Test method	Limit of Quantitation	Test Results		
			Batch # MM13524004	Batch # MM13524007	Batch # MM13524009
Nitrate (mg/kg)	GB 5009.33-2016 First method	1.1	3.4	12	3.5
Nitrite (mg/kg)	GB 5009.33-2016 First method	0.6	ND	ND	ND

2.4 Nutritional Profile

MoreMeat has conducted full nutritional profile tests on mycoprotein. The result shows that on an as-is basis, the finished product contains 9.17% dietary fiber, 13.8% protein, and 9.0% carbohydrate, and is low in sugar (<0.1%), fat (~1.36%) and cholesterol (<1%). The caloric value is approximately 100 kcal/100 g. Table 10 summarizes the test results of nutritional profiles for three nonconsecutive lots of MoreMeat mycoprotein.

Table 10 Nutritional Profile of Mycoprotein

Nutrients	Test Methods	Test Results (on wet basis)			Average Test Results
		Batch # MM13524004	Batch # MM13524007	Batch # MM13524009	
Total Sugars (g/100 g)	AOAC 982.14	<0.1	<0.1	<0.1	<0.1
Dietary fiber (%)	AOAC 2011.25	8.06	10.12	9.34	9.17
Total fat (g/100 g)	AOAC 996.06	0.996	1.92	1.17	1.36
Saturated fat (g/100 g)	AOAC 996.06	0.185	0.352	0.233	0.26
Trans fat (g/100 g)	AOAC 996.06	<0.024	0.0315	<0.024	0.027
Protein (%)	AOAC 920.87 / 979.09	13.2	14.9	13.2	13.8
Total carbohydrates (g/100 g)	21 CFR 101.9	9.0	8.5	9.4	9.0
Cholesterol (mg/100 g)	AOAC 994.10	ND	ND	ND	ND
Calories (kcal/100 g)	21 CFR 101.9	98	111	101	103.3

ND: not detected; Limit of detection for Cholesterol is 0.3 mg/100g

Protein Quality

MoreMeat's mycoprotein contains a high-quality protein as indicated by an average Protein Digestibility Adjusted Amino Acid Score (PDCAAS) of 1.00 out of 1.00. (See Appendix 4 for MoreMeat's PDCAAS report). The nutritive value of a protein is determined by its capacity to provide nitrogen and amino acids in amounts sufficient to meet human needs. PDCAAS is a method of evaluating the quality of a protein based on the amino acid requirements of humans and their ability to digest it. The PDCAAS for MoreMeat's mycoprotein was measured by *in vivo* methods recommended by the Food and Agriculture Organization of the United Nations (FAO) and World Health Organisation (WHO) for evaluating protein quality.

The quality of protein present in MoreMeat's mycoprotein was assessed, and the PDCAAS was determined to be 1.0 for children (ages 3 to 10 years old) and adults (ages 18 years and older). The highest possible PDCAAS score is 1.0, which indicates that MoreMeat's mycoprotein is a source of quality protein.

2.5 Stability

Product stability testing was conducted on three nonconsecutive lots of MoreMeat mycoprotein. According to MoreMeat's product storage conditions, final products should be stored in a freezer (-18°C). Hence, the ongoing long-term stability study is being conducted at -20°C ± 5°C and the accelerated stability study was conducted at 5°C ± 3°C. Three nonconsecutive lots of mycoprotein were tested for protein, moisture, and microbial limits throughout the accelerated stability period of 3 months. The test results of the completed 3-month accelerated stability study and the 3-month long-term stability study (on-going), as summarized in Table 11 and Table 12, respectively, indicate that the product quality does not change significantly and that mycoprotein is stable under the described conditions.

MoreMeat's own stability studies support a shelf life for mycoprotein derived from *F. compactum* of up to 12 months at frozen conditions. This concurs with the shelf-life claims of mycoprotein derived from *F. venenatum* (GRN 000945 page 48). The currently available stability data supports a shelf-life of 12 months at -18°C when stored in a vacuum-sealed aluminum bag. The stability data shall be updated upon the availability of further data.

Table 11 Stability Test Result for Mycoprotein at Accelerated Conditions (5°C±3°C)

Lot #	Test Month	Appearance	Protein(g /100g as wet basis)	Protein (g/100g on dried basis)	Moisture (g/100g)	Total Plate Count (CFU /g)	Yeast & Mold (CFU /g)	<i>L. monocytogenes</i>	<i>E. coli</i>	<i>S. aureus</i>	<i>Salmonella</i>
MM135 24002	0	Off-white, dough-like matter	13.8	50.2	72.5	<10	<10	ND	ND	ND	ND
	1	Off-white, dough-like matter	12.9	49.0	73.7	<10	<10	/	/	/	/
	2	Off-white, dough-like matter	14.2	50.0	71.6	<10	<10	/	/	/	/
	3	Off-white, dough-like matter	14.2	52.6	73.0	<10	<10	ND	ND	ND	ND
MM135 24004	0	light yellow, dough-like matter	13.2	55.7	76.3	<10	<10	ND	ND	ND	ND
	1	light yellow, dough-like matter	13.5	51.1	73.6	<10	<10	/	/	/	/
	2	light yellow, dough-like matter	13.2	50.2	73.7	<10	<10	/	/	/	/
	3	light yellow, dough-like matter	13.9	59.7	76.7	<10	<10	ND	ND	ND	ND
MM135 24009	0	light yellow, dough-like matter	13.2	54.3	75.7	<10	<10	ND	ND	ND	ND
	1	light yellow, dough-like matter	13.0	61.3	77.5	<10	<10	/	/	/	/

	2	light yellow, dough-like matter	13.0	54.2	76.0	<10	<10	/	/	/	/
	3	light yellow, dough-like matter	12.8	57.4	77.7	<10	<10	ND	ND	ND	ND

ND: not detected.

Table 12 Stability Test Result for Mycoprotein at Long-Term Conditions (-20°C±5°C)

Lot #	Test Month	Appearance	Protein (g/100 g) as wet basis	Protein (g/100 g) on dried basis	Moisture (g/100 g)	Total Plate Count (CFU /g)	Yeast & Mold (CFU /g)
MM13524002	0	Off-white, dough-like matter	13.8	50.2	72.5	<10	<10
	3	Off-white, dough-like matter	14.0	52.2	73.2	<10	<10
MM13524004	0	light yellow, dough-like matter	13.2	55.7	76.3	<10	<10
	3	light yellow, dough-like matter	13.5	61.6	78.1	<10	<10
MM13524009	0	light yellow, dough-like matter	13.2	54.3	75.7	<10	<10
	3	light yellow, dough-like matter	13.5	53.4	74.7	<10	<10

Part 3 DIETARY EXPOSURE**3.1 Intended Uses of Mycoprotein**

MoreMeat's mycoprotein is intended for use in processed food products (excluding infant formulas and meat and poultry products) to increase the level of protein in a product, serving as a substitute for other animal or vegetable protein. MoreMeat's mycoprotein is primarily composed of water (~75%), protein (~15%), and fiber (~10%). MoreMeat's mycoprotein is intended for use as a food ingredient and macronutrient in a variety of food categories and the range of intended incorporation rates of MoreMeat mycoprotein in various foods is presented in Table 13. Because MoreMeat's mycoprotein is primarily composed of water (~75%), the intended use levels are presented on a wet weight basis, with the anticipated minimum and maximum use levels provided.

Table 13 Intended Food Use Categories

Food Category	Proposed Food Use	Minimum Use Level MoreMeat's Mycoprotein (% weight)	Maximum Use Level MoreMeat's Mycoprotein (% weight)
Baked Goods and Baking Mixes	Breads, muffins, cakes, brownies, cookies, doughnuts, biscuits, pie, pancakes, waffles, tortillas, breakfast/granola bars, crackers, and doughs	6	50
Dairy Product Analogs	Non-dairy yogurt	8	70
	Non-dairy milk, creamer, cheese, and milk-based meal replacements	8	50
Fats and Oils	Salad dressings, margarine, non-dairy cheese sauce, condiments, mayonnaise, and gravies	8	30
Grain Products and Pastas	Dried pasta noodles	6	50
Plant Protein Products	Mycoprotein-based SPAM, meat ball, and sausage; Mycoprotein-based chicken flavor nugget, fish flavor fillet, and pork flavor chop; Meatless chicken nuggets, veggie	10	90

Food Category	Proposed Food Use	Minimum Use Level MoreMeat's Mycoprotein (% weight)	Maximum Use Level MoreMeat's Mycoprotein (% weight)
	burgers, lunch meat, meal replacement protein bars and instant breakfasts, fishless sticks, meatless fillets, meatless roast, meatless sharp cheese cutlets, meatless breakfast patties, meatless cutlets, meatless dippers, and meatless mince.		
Processed Fruits and Fruit Juices	High protein juices	8	60
Processed Vegetable and Vegetable Juices	High protein juices	8	60
Soups and Soup Mixes	Frozen or shelf-stable prepared soups	8	60
Other	Frozen or shelf stable prepared meals; Quick-frozen prepared cuisine products	8	60

3.2 Estimated Daily Intake of MoreMeat's Mycoprotein

The estimated daily intake (EDI) of MoreMeat's mycoprotein was conservatively calculated based on its the maximum intended incorporation rate in representative foods drawn from the intended food categories, in combination with food consumption data provided in the National Health and Nutrition Examination Survey (NHANES) for 2021 – 2023². NHANES 2021-2023 dietary survey data were collected from a sampling of the U.S. civilian noninstitutionalized population selected from within households in dwelling units. Additional information regarding the sample design and data collection

² The NHANES is a program of studies conducted by the National Center for Health Statistics (NCHS) within the Centers for Disease Control and Prevention (CDC) and is designed to assess the health and nutritional status of adults and children in the United States. One component of the NHANES is a 2-day dietary survey of the study participants. The data from this survey is made publicly available and can be used to conduct intake estimates, see <https://www.cdc.gov/nchs/nhanes/index.htm>.

is available from the Centers for Disease Control and Prevention.³

Data management, statistical analysis, and calculations for the mean and 90th percentile *per capita* and consumer-only intakes, reported per person and per kilogram body weight, were performed using Dazult Ltd.'s DaDiet Software, Version 24.09.

The individual food codes that best reflect the proposed uses were selected for this intake assessment and are listed in Appendix 2.

Per capita and consumer-only estimates of the total daily intake of MoreMeat's mycoprotein from the proposed food uses at the maximum use levels listed in Table 13 were determined on an absolute basis (g/person/day; Table 14) and on a per kilogram body weight basis (g/kg bw/day, Table 15).

Table 14. Estimated Daily Intake of MoreMeat's Mycoprotein (g/day)

Population Group	Age Group (Years)	Per Capita Intake (g/day)		Consumer-Only Intake (g/day)			
		Mean	90 th Percentile	%	N	Mean	90 th Percentile
Children	2-5	88	184	97.2	308	90	184
Children	6-11	102	207	99.2	473	103	207
Female Teenagers	12-19	88	204	96.2	327	91	207
Male Teenagers	12-19	101	239	96.1	288	105	252
Female Adults	20+	86	194	98.2	2368	88	194
Male Adults	20+	95	217	97.3	1830	98	220
Total Population	2+	92	205	97.7	5594	94	207

n= sample size; g = grams

Table 15 Estimated Daily Intake of MoreMeat's Mycoprotein (g/kg bw/day)

Population Group	Age Group (Years)	Per Capita Intake (g/kg bw/day)		Consumer-Only Intake (g/kg bw/day)			
		Mean	90 th Percentile	%	N	Mean	90 th Percentile
Children	2-5	5.1	10.6	97.2	306	5.3	11.9
Children	6-11	3.2	6.9	99.2	473	3.2	6.9
Female Teenagers	12-19	1.5	3.4	96.2	325	1.6	3.5
Male Teenagers	12-19	1.6	3.8	96.1	288	1.7	3.9

³ Available online at: <https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/OverviewBrief.aspx?Cycle=2021-2023>

Female Adults	20+	1.2	2.6	98.2	2345	1.2	2.6
Male Adults	20+	1.1	2.7	97.3	1816	1.1	2.7
Total Population	2+	1.5	3.4	97.7	5553	1.6	3.5

n= sample size; g = grams; kg = kilogram; bw = body weight

The percentage of users was high among all evaluated age groups in the intake assessment and was >96% of the population groups for products for which MoreMeat's mycoprotein would be incorporated under the proposed uses listed in Table 13. It should be noted that the estimated daily intakes are considered conservative, as MoreMeat's mycoprotein is unlikely to be incorporated into every proposed food product in the marketplace and/or at the maximum proposed use levels.

Among the total population (ages 2 years and older), the mean and 90th percentile consumer-only intakes of MoreMeat's mycoprotein were determined to be 94 g/day and 207 g/day, respectively. Male teenagers (ages 12 to 19 years) were determined to have the greatest consumer-only mean and 90th percentile intakes of MoreMeat's mycoprotein, at 105 g/day and 252 g/day, respectively. The lowest consumer-only mean and 90th percentile intakes were for female adults (ages 20 years and older), at 88 g/day and 194 g/day, respectively.

Among the total population (ages 2 years and older) and on a body weight basis, the mean and 90th percentile intakes of MoreMeat's mycoprotein were determined to be 1.5 g/kg bw/day and 3.4 g/kg bw/day, respectively. Children (ages 2-5 years) were determined to have the greatest mean and 90th percentile consumer-only intakes at 5.3 g/kg bw/day and 11.9 g/kg bw/day, respectively. The lowest mean and 90th percentile consumer-only intakes were for male adults (ages 20 years and older), at 1.1 g/kg bw/day and 2.7 g/kg bw/day, and female adults (ages 20 years and older) at 1.2 g/kg bw/day and 2.6 g/kg bw/day, respectively.

3.3 Estimated Daily Intake of Protein from MoreMeat's Mycoprotein

The protein digestibility-corrected amino acid score (PDCAAS) is used to assess the dietary protein value for human nutrition. The quality of protein present in MoreMeat's mycoprotein was assessed, and the PDCAAS was determined to be 1.0 for children (ages 3 to 10 years old) and adults (ages 18 years and older). The highest possible PDCAAS score is 1.0, which indicates that MoreMeat's mycoprotein is a source of quality protein.

For comparison to dietary reference intakes for protein, the estimated total intake of protein (as wet basis) from consumption of MoreMeat's mycoprotein from the proposed uses was calculated (see Table 16), using a maximum protein content of 15% for MoreMeat's mycoprotein, based on MoreMeat's specification for mycoprotein, as provided in Table 4.

Table 16 Estimated Daily Intake of MoreMeat’s Mycoprotein as Protein from the Proposed Uses (g/day)

Population Group	Age group (Years)	Per Capita Intake ^a (g/day)		Consumer-Only Intake ^a (g/day)			
		Mean	90 th Percentile	%	N	Mean	90 th Percentile
Children	2-5	13	28	97.2	308	14	28
Children	6-11	15	31	99.2	473	15	31
Female Teenagers	12-19	13	31	96.2	327	14	31
Male Teenagers	12-19	15	36	96.1	288	16	38
Female Adults	20+	13	29	98.2	2368	13	29
Male Adults	20+	14	33	97.3	1830	15	33
Total Population	2+	14	31	97.7	5594	14	31

n= sample size; g = grams

^a Calculated as: (Estimated daily intake of MoreMeat’s mycoprotein) * (maximum protein content, 15%)

The conservative consumer-only mean and 90th percentile estimate of the daily consumption of protein from MoreMeat’s mycoprotein are therefore ~14 g/day and 31 g/day, respectively, for the total population (2 years and older). Of the individual consumer-only population groups, male teenagers (ages 12 to 19 years and older) were determined to have the greatest mean and 90th percentile intakes of protein, at 16 g/day and 38 g/day, respectively. Female adults (ages 20 years and older) were determined to have the lowest mean consumer-only intakes of protein, at 13 g/day, while children (ages 2 to 5 years) were determined to have the lowest 90th percentile consumer-only intakes of protein, at 28 g/day.

Based on the information presented herein, the conservative 90th percentile consumer-only estimate of the daily consumption of MoreMeat’s mycoprotein from the proposed uses is 207 g/day for the total population (2 years and older). Assuming a maximum protein content of 15% for MoreMeat’s mycoprotein, the conservative 90th percentile consumer-only estimate of the daily consumption of protein from the proposed uses of MoreMeat’s mycoprotein is 31 g/day for the total population (ages 2 years and older).

Background Intake Levels of Protein

Dietary reference intakes (DRIs) for protein have been established by Food and Nutrition Board of the Institute of Medicine (IOM) of the National Academy of Science (IOM, 2005), as summarized in Table 17.

Table 17 DRIs for Protein Established by IOM

Population Group	Age Group (Years)	Estimated Average Requirement (EAR) (g/kg bw/day)	Recommended Dietary Allowance (RDA) (g/day)
Children	1-3	0.87	13
Children	4-8	0.76	19
Children	9-13	0.76	34
Female Teenagers	14-18	0.71	46
Male Teenagers	14-18	0.73	52
Female Adults	19+	0.66	46
Male Adults	19+	0.66	56

IOM (2005) noted that there are insufficient data to establish a Tolerable Upper Intake Level (UL) for total protein. Further, while the “risk of adverse effects resulting from excess intake of protein from foods appears to be very low”, IOM notes that data on adverse effects associated with high protein intakes are limited, therefore “caution may be warranted.”

FDA has also established a daily reference value (DRV) for protein of 50 g/day based on a 2,000-calorie diet.⁴

Based on consumption data obtained from the 2007-2008 NHANES survey, the average daily intake of protein by adult men and women is 100 g and 69 g, respectively (Wright and Wang, 2010, as reported by NIH⁵). More recently, Berryman et al. (2018) evaluated protein intake trends from 2001-2014 using NHANES survey data. Protein intakes ranged from an average of ~55 g/day in children ages 2-3 years to ~88 g/day in adults ages 19-30 years. Relative protein intakes were reported to be ~ 3.6 g/kg bw/day for children ages 2-3 years and ~1.1 g/kg bw/day for adults ages 71 years and older. It was noted that relative protein intakes were above the estimated average requirement (EAR) in all demographic groups. Berryman et al. (2018) concluded that the majority of U.S. consumers exceed minimum daily recommendations for protein intake.

Based on the EDI for MoreMeat’s mycoprotein presented herein, the estimated 90th percentile consumer-only intake of protein from all proposed uses for the total population (i.e., 31 g/day protein from MoreMeat’s mycoprotein for consumers 2 years and older) is lower than or within the range of the recommended and/or background intake levels of protein in the diet. Furthermore, MoreMeat’s mycoprotein is intended to substitute for other protein sources in the diet, therefore the proposed uses

⁴ See:

https://www.accessdata.fda.gov/scripts/InteractiveNutritionFactsLabel/assets/InteractiveNFL_Protein_October2021.pdf
(Accessed February 8, 2025)

⁵ Available online at: <https://ods.od.nih.gov/factsheets/ExerciseAndAthleticPerformance-HealthProfessional/#protein>
(Accessed February 8, 2025)

of MoreMeat’s mycoprotein are not expected to result in an overall increase in protein consumption in the diet. MoreMeat concurs with the conclusion that: “...it is reasonable to expect that most of the U.S. population’s intake of protein is expected to remain in the form of unprocessed foods including meat, poultry, fish, and legumes,” as presented on p. 24 in Part 3.2.2 of GRN 001125.

3.4 Estimated Daily Intake of Fiber from MoreMeat’s Mycoprotein

In addition to being a dietary source of protein, MoreMeat mycoprotein also contains ~10% dietary fiber, as noted in Table 10. For comparison to dietary reference intakes for dietary fiber, the estimated total intake of dietary fiber from consumption of MoreMeat mycoprotein from the proposed uses was calculated (Table 18), assuming a dietary fiber content of approximately 10% for MoreMeat mycoprotein, based on representative batch results provided in Table 10.

Table 18 Estimated Daily Intake of Dietary Fiber from the Proposed Uses of MoreMeat Mycoprotein (g/day)

Population Group	Age Group (Years)	Per Capita Intake ^a (g/day)		Consumer-Only Intake ^a (g/day)			
		Mean	90 th Percentile	%	N	Mean	90 th Percentile
Children	2-5	9	18	97.2	308	9	18
Children	6-11	10	21	99.2	473	10	21
Female Teenagers	12-19	9	20	96.2	327	9	21
Male Teenagers	12-19	10	24	96.1	288	11	25
Female Adults	20+	9	19	98.2	2368	9	19
Male Adults	20+	10	22	97.3	1830	10	22
Total Population	2+	9	21	97.7	5594	9	21

n= sample size; g = grams

^a Calculated as: (Estimated daily intake of MoreMeat’s mycoprotein) * (approximate dietary fiber content, 10%)

The conservative consumer-only mean and 90th percentile estimate of the daily consumption of dietary fiber from MoreMeat’s mycoprotein are therefore ~ 9 g/day and 21 g/day, respectively, for the total population (2 years and older).

Background Intake Levels of Fiber

Dietary reference intakes (DRIs) for dietary fiber have been established by the Food and Nutrition Board of the Institute of Medicine (IOM) of the National Academy of Science (IOM, 2005), as summarized in Table 19.

Table 19 DRIs for Protein Established by IOM

Population Group	Age group (Years)	Total Fiber Adequate Intake (AI) (g/day)
Children	1-3	19
Children	4-8	25
Female Children	9-13	26
Male Children	9-13	31
Female Teenagers	14-18	26
Male Teenagers	14-18	38
Female Adults	19+	21-25
Male Adults	19+	30-38

IOM (2005) noted that there are insufficient data to establish a UL for dietary fiber. IOM (2005) also noted that the median intake of dietary fiber ranges from 16.5 to 17.9 g/day for men and 12.1 to 13.8 g/day for women.

FDA has also established a DRV for dietary fiber of 28 g/day.⁶

Based on consumption data obtained from NHANES surveys from 1999-2008, the mean daily intake of dietary fiber for adults (ages 18 years and older) from 1999-2000 was 15.6 g/day, from 2001-2002 was 16.1 g/day, from 2003-2004 was 15.5 g/day, from 2005-2006 was 15.8 g/day, and from 2007-2008 was 15.9 g/day (King et al., 2012). These values are well below the recommended AI for dietary fiber for adults, as listed in Table 19. Furthermore, in 2021, the American Society for Nutrition reported that “[o]nly 5% of men and 9% of women are getting the recommended daily amount of dietary fiber.”⁷

Based on the EDI for MoreMeat’s mycoprotein presented herein, the estimated 90th percentile consumer-only intake of dietary fiber from all proposed uses for the total population (i.e., 21 g/day dietary fiber from MoreMeat’s mycoprotein for consumers 2 years and older) is lower than or within the range of the recommended and/or background intake levels of dietary fiber in the diet.

3.5 Estimate of Dietary Exposure to Mycotoxins

As aforementioned, various *Fusarium* species have the potential to produce mycotoxins as secondary metabolites. To ascertain the presence of such toxins in *F. compactum*, a comprehensive investigation, integrating genetic sequencing and rigorous analytical examinations of the final product were conducted. The genomic sequencing and analysis indicate the potential for mycotoxin production, including beauvericin and enniatins. (See Section 6.7 for a detailed discussion). Diacetoxyscirpenol

⁶ See: <https://www.fda.gov/food/nutrition-facts-label/daily-value-nutrition-and-supplement-facts-labels> (Accessed February 8, 2025)

⁷ <https://nutrition.org/most-americans-are-not-getting-enough-fiber-in-our-diets/> (Accessed February 8, 2025)

(DAS) was identified as one of the potential mycotoxins in *F. compactum* in the literature search. (See Section 6.7 for a detailed discussion). Applicable analytical methods were employed to verify if *F. compactum* produces these mycotoxins under the specified manufacturing conditions for mycoprotein. The results of analytical testing of three representative batches of MoreMeat’s mycoprotein for the potential mycotoxins are summarized in Table 20. The tests conclusively show that these mycotoxins either are not present or are below levels that would present a safety concern when consuming mycoprotein under the proposed uses and use levels. (See Section 6.7 for safety discussion of mycotoxins)

Table 20 Mycotoxins Analysis of Three Batches of Mycoprotein (On Wet Weight Basis)

Test Items	Test method	LOD (µg/kg)	Test Results (µg/kg)		
			Batch # MM13524004	Batch # MM13524007	Batch # MM13524009
Diacetoxyscirpenol (DAS)	RWX_OP_016	10	ND	ND	ND
Enniatin (sum of A, A1, B and B1)	HPLC-MS/MS method	/	0.04878	2.3087	ND
Enniatin A	HPLC-MS/MS method	0.02	0.04878	0.1138	ND
Enniatin A1	HPLC-MS/MS method	0.09	ND	0.1518	ND
Enniatin B	HPLC-MS/MS method	0.01	ND	1.0122	ND
Enniatin B1	HPLC-MS/MS method	0.10	ND	1.0309	ND
Beauvericin	HPLC-MS/MS method	0.04	5.955	8.895	0.7116

“/” means not specified. ND: not detected

3.5.1 Diacetoxyscirpenol (DAS)

DAS was not detected with a limit of detection of 10 µg/kg for three representative batches of mycoprotein derived from *F. compactum*. Using the limit of detection to calculate the maximum possible EDI of DAS from consumption of MoreMeat’s mycoprotein, the EDI can therefore be approximated on a “worst case scenario” basis for a 90th percentile consumer as follows:

$$\frac{10 \mu\text{g}}{\text{kg}} * \frac{207 \text{ g}}{\text{day}} * \frac{1 \text{ kg}}{1,000 \text{ g}} = 2.07 \mu\text{g/day}$$

If DAS is present in MoreMeat’s mycoprotein, it is at levels below the detection limit, and would result in a calculated mean and 90th percentile EDI of less than 0.013 µg/kg bw/day and 0.030 µg/kg bw/day,

respectively (assuming 70 kg bw/person).

As requested by the European Commission, the EFSA Panel on CONTAM assessed the risk of DAS to human and animal health related to its presence in food and feed. Very limited information was available on toxicity and on toxicokinetics in experimental and farm animals. Due to the limitations in the available data set, human acute and chronic health - based guidance values (HBGV) were established based on data obtained in clinical trials of DAS as an anticancer agent (anguidine) after intravenous administration to cancer patients. The CONTAM Panel considered these data as informative for the hazard characterization of DAS after oral exposure. The main adverse effects after acute and repeated exposure were emesis, with a NOAEL of 32 µg DAS/kg bw and haematotoxicity, with a NOAEL of 65 µg DAS/kg bw, respectively. An acute reference dose of 3.2 µg DAS/kg bw and a tolerable daily intake of 0.65 µg DAS/kg bw were established. Based on over data from 15,000 occurrences, the highest acute and chronic dietary exposures were estimated to be 0.8 and 0.49 µg DAS/kg bw/ day, respectively, and were not of health concern for humans.

The maximum possible dietary exposure of DAS from MoreMeat's mycoprotein is negligible compared to the CONTAM Panel exposure estimates.

3.5.2 Enniatins

Enniatins were detected at low levels during analytical testing. The three-batch analysis of MoreMeat's mycoprotein product reported in Table 20 indicate an average enniatins content of 0.86 µg/kg.⁸ The EDI of enniatins from consumption of MoreMeat's mycoprotein can therefore be calculated for a 90th percentile consumer as follows:

$$\frac{0.86 \mu\text{g}}{\text{kg}} * \frac{207 \text{ g}}{\text{day}} * \frac{1 \text{ kg}}{1,000 \text{ g}} = 0.178 \mu\text{g/day}$$

Enniatins have been evaluated by the European Food Safety Authority's (EFSA) Panel on Contaminants in the Food Chain (CONTAM). Part of CONTAM's review included the determination of the existing intake of enniatin from food across 17 European countries based on the submission of occurrence data. Although the CONTAM panel calculated existing upper- and lower-bound values for both chronic and acute dietary exposures, we have focused on the chronic data as being the most relevant. The following Table 21 summarizes the calculated mean and 95th percentile chronic exposures.

Table 21 Existing Dietary Intake of Enniatin (EU)

	EU Intake Mean Consumer, µg/kg bw/day	EU Intake 95th percentile, µg/kg bw/day
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⁸ The average enniatins content is calculated as the average of (0.04878 + 2.3087 + 0.22). Since enniatins were not detected in Batch MM13524009, the value of 0.22 is used, based on the LOD for the combination of enniatins A, A1, B, and B1.

Enniatins	0.42-1.82	0.91-3.28
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We have identified no intake estimate of enniatins for U.S. consumers and, given the relative similarity of the European and U.S. diets, we have considered it reasonable to utilize the EU data regarding current intakes of enniatins to be representative of the U.S. population. This approach is utilized in GRN 000904 (page 26) for the dietary intake discussion of mycotoxins in fungal protein derived from *Fusarium strain flavolapis*; FDA had “no questions” regarding about this approach. Therefore, the same approach is applied here to evaluate the dietary intake of enniatins from consumption of MoreMeat’s mycoprotein. Indeed, as the EU data calculated a 95th percentile intake (rather than 90th percentile), the use of this data is conservative. We have, therefore, calculated the Cumulative Estimated Daily Intake (CEDI) of enniatins based on the EU intake data in combination with the calculated intakes from MoreMeat’s mycoprotein. We first calculated the enniatins EDIs (mean and 90th percentile consumer) on a body weight basis assuming an individual body weight of 70 kg (Table 22).

Table 22 Enniatins EDIs

	Mean EDI µg/day	90th percentile EDI, µg/day	Mean EDI, µg/kg bw/day	90th percentile EDI, µg/kg bw/day
Enniatins	0.081	0.178	0.001	0.003

These intakes were added to the upper-bound intake values for the mean and 95% percentile consumers from the CONTAM evaluation to calculate the CEDIs. Thus, the CEDIs are summarized in Table 23 below.

Table 23 Summary of Enniatins EDI and CEDI

	Mean EDI, µg/kg bw/day	90th Percentile EDI, µg/kg bw/day	CONTAM Mean EDI, µg/kg bw/day	CONTAM 95th Percentile EDI, µg/kg bw/day	CEDI Mean, µg/kg bw/day	CEDI 95th Percentile, µg/kg bw/day
Enniatins	0.001	0.003	0.91	3.28	0.911	3.283

3.5.3 Beauvericin

Beauvericin was detected at low levels in MoreMeat’s mycoprotein during analytical testing. The three-batch analysis reported in Table 20 indicates an average beauvericin content of 5.187 µg/kg. The EDI of beauvericin from consumption of MoreMeat’s mycoprotein for a 90th percentile consumer is calculated as follows:

$$\frac{5.187 \mu\text{g}}{\text{kg}} * \frac{207 \text{ g}}{\text{day}} * \frac{1 \text{ kg}}{1,000\text{g}} = 1.074 \mu\text{g/kg}$$

Beauvericin was evaluated by the EFSA CONTAM. One component of the CONTAM’s review

determined the existing intake of beauvericin from food across 17 European countries based on the submission of occurrence data. Although the CONTAM panel calculated existing upper- and lower-bound values for both chronic and acute dietary exposures, we have focused on the chronic data as being the most relevant. The following Table 24 summarizes the calculated mean and 95th percentile chronic exposures.

Table 24 Existing Dietary Intake of Beauvericin (EU)

	EU Intake Mean Consumer, µg/kg bw/day	EU Intake 95th Percentile, µg/kg bw/day
Beauvericin	0.003-0.05	0.006-0.93

We have identified no intake estimate of beauvericin for U.S. consumers and given the relative similarity of the European and U.S. diets, we have considered it reasonable to utilize the EU data regarding current intakes of beauvericin to be representative of the U.S. population. This approach is utilized in GRN 000904 (page 26) for the dietary intake discussion of beauvericin; FDA had “no questions” regarding about this approach. Therefore, the same approach is applied here to evaluate the dietary intake of beauvericin from consumption of MoreMeat’s mycoprotein. Indeed, as the EU data calculated a 95th percentile intake (rather than 90th percentile), the use of this data is conservative. We have, therefore, calculated the CEDI to beauvericin based on the EU intake data in combination with the calculated intakes from MoreMeat’s mycoprotein. We first calculated the beauvericin EDIs (mean and 90th percentile consumer) on a body weight basis assuming an individual body weight of 70 kg (Table 25).

Table 25 Beauvericin EDI

	Mean EDI, mg/day	90th Percentile EDI, mg/day	Mean EDI, µg/kg bw/day	90th Percentile EDI, µg/kg bw/day
Beauvericin	0.488	1.074	0.007	0.015

These intakes were added to the upper-bound intake values for the mean and 95% percentile consumers from the CONTAM evaluation to calculate the CEDIs. Thus, the CEDIs are summarized in Table 26 below.

Table 26 Summary of Beauvericin EDI and CEDI

	Mean EDI, µg/kg bw/day	90th Percentile EDI, µg/kg bw/day	CONTAM Mean EDI, µg/kg bw/day	CONTAM 95th percentile EDI, µg/kg bw/day	CEDI Mean, µg/kg bw/day	CEDI 95th Percentile, µg/kg bw/day
Beauvericin	0.007	0.015	0.05	0.93	0.057	0.945

Summary of Intake of Mycotoxin From MoreMeat's Mycoprotein

As enniatins and beauvericin have been identified and quantified in the three-batch analyses, as summarized in Table 27, a rigorous analytical testing program using LC-MS/MS methodology has been established by MoreMeat and maximum specifications have been established for these mycotoxins, thereby ensuring that consumer exposure remains consistently within safe levels.

Table 27 Summary of Mycotoxin EDIs and CEDIs

	Average test results for 3 batches (µg/kg)	Mean EDI, µg/kg bw/day	90th Percentile EDI, µg/kg bw/day	CEDI Mean, µg/kg bw/day	CEDI 95th Percentile, µg/kg bw/day
Enniatins	0.86	0.001	0.003	0.911	3.283
Beauvericin	5.19	0.007	0.015	0.057	0.945

3.6 Estimate of Dietary Exposure to Nitrates

Ammonia is used as fermentation ingredient, which means nitrates may be generated during the fermentation process and are the only possible source of input identified in mycoprotein production. The three-batch analysis shown in section 2.3.3.4 indicates a mean nitrates residue of 6.3 mg/kg (on a wet weight basis). Based on mycoprotein daily mean and 90th percentile intake values of 94 g/day and 207 g/day, respectively, this equates to mean and 90th percentile EDIs for nitrates of 0.592 mg/person/day and 1.304 mg/person/day, respectively, or 0.008 mg/kg bw/day and 0.019 mg/kg bw/day, respectfully, assuming a 70 kg individual.

Calculation of mean intake: $(6.3 \text{ mg NO}_3/\text{kg mycoprotein}) \times (1 \text{ kg}/1000 \text{ g}) \times (94 \text{ g mycoprotein}/\text{day}) = 0.592 \text{ mg}/\text{day}$

Calculation of 90th percentile intake: $(6.3 \text{ mg NO}_3/\text{kg mycoprotein}) \times (1 \text{ kg}/1000 \text{ g}) \times (207 \text{ g mycoprotein}/\text{day}) = 1.304 \text{ mg}/\text{day}$

Given the various potential sources of nitrates intake from the U.S. diet and the fact that the EDIs above are a tiny fraction of the typical intake of nitrates in the diet, we have not calculated a CEDI. In 2002, JECFA determined the acceptable daily intake (ADI) for nitrate to be 0-3.7 mg nitrate ion/kg bw/day (JECFA, 2002); therefore, the mean intake of 0.008 mg/kg bw/day nitrates from consumption of MoreMeat's mycoprotein under the proposed uses and use levels is negligible.

Part 4 SELF-LIMITING LEVELS OF USE

The use of the MoreMeat's mycoprotein as a food is limited by the level that can technically be added to a given food. For example, when mycoprotein is used in liquid products up to a certain amount, the liquid base might become viscous, and as a result there is a self-limiting use level.

Part 5 EXPERIENCE BASED ON COMMON FOOD USE IN FOOD BEFORE 1958

The statutory basis for MoreMeat’s conclusion of the GRAS status of mycoprotein in this document is not based on common use in food before 1958. The GRAS conclusion is based on scientific procedures.

Part 6 NARRATIVE

Extensive information relating to product identity, nutritional composition, digestibility, potential contaminants and by-products, allergenicity, and toxicology have been assessed to determine the safety of MoreMeat's mycoprotein and the basis for its GRAS status. Safety studies, including toxicology studies commissioned by MoreMeat, as well as published literature on similar *Fusarium*-derived products, have indicated that there are no acute or chronic exposure concerns from the consumption of mycoprotein. Additionally, the available literature provides evidence that compositionally similar *Fusarium*-derived products support normal growth and development in animals and humans. An allergenicity evaluation was conducted through a literature search in concert with testing conducted by MoreMeat, the results of which have been reviewed by an independent allergenicity expert. It has been concluded that *F. compactum* does not present additional significant risk for allergenicity to foods containing mycoproteins. Furthermore, there are no impurities or secondary metabolites that present a safety concern from the consumption of mycoprotein, even at relatively high intake levels. The culmination of these data supports the conclusion that MoreMeat's mycoprotein is GRAS for use in food, as detailed in the following sections.

6.1 Substantial Similarity of Mycoprotein to The Subject of GRN 000904 and GRN 000945

In exploring the potential use of fungal proteins in food ingredients, it is important to recognize the significant impact of different production strains on product properties, nutritional content, and by-products. The production microorganism *F. compactum* belongs to the same genus as the *Fusarium strain flavolapis* (formerly known as *F. novum. yellowstonensis*) described in GRN 000904 and which received a "no questions" letter from FDA in 2021, and the *F. venenatum* described in GRN 000945 and which received a "no questions" letter from FDA in 2022. For this reason, we compared the similarity between the *Fusarium*-derived ingredients described in the two previous GRAS notifications to MoreMeat's *F. compactum* production organism in terms of strain, manufacturing process, and product specifications. The safety studies of GRN 000904 and GRN 000945 can be used as pivotal safety information for MoreMeat's mycoprotein.

6.1.1 DNA Sequencing Comparison

The phylogenetic tree described in Part 2 demonstrates that there are genetic similarities between *F. compactum* MM-135, *F. venenatum*, and *Fusarium strain flavolapis*. *F. compactum* MM-135 is closer in genetic similarity to *F. venenatum* (the production organism described in GRN 000091 and GRN 000945) than *F. strain flavolapis* (the production organism described in GRN 000904) is to *F. venenatum*.

6.1.2 Manufacturing Process Comparison

When comparing the production processes of these three fungal proteins, it was found that their fermentation procedures, heat deactivation treatments, and storage environments are essentially the

same; thus, it can be reasonably deduced that the three products have a high degree of similarity in some of their basic properties, as follows:

1. The types and proportions of metabolites accumulated by the three fungi during fermentation are very similar. This means that the three proteins have a high degree of similarity in the composition of their nutritional and functional components.
2. The same heating method was used to inactivate the fungus and denature the protein to prevent it from deterioration during storage and use. The physicochemical changes to the mycoprotein during heating are expected to be similar.
3. Storage temperature below -18°C minimizes quality changes due to differences in storage conditions. All three mycoprotein products are stable under their storage conditions, as demonstrated by the stability studies detailed in Section 2.5 of the current notification.

There are some minor differences in the manufacture of the ingredients, such as the volume and model of the reactors in the fermentation and harvesting processes. However, the impact of these differences on the properties of mycoproteins is minimal. Therefore, it can be concluded that these three mycoproteins have a high degree of similarity in terms of product properties under similar fermentation procedures, heat inactivation treatments, and storage environments.

6.1.3 Product Specification Comparison

MoreMeat's specifications for mycoprotein are provided in Table 4. The specifications for MoreMeat's mycoprotein are consistent with industry standard for similar preparations (i.e., FCC 14, GRN 000945, etc.) and include protein, moisture, appearance, relevant mycotoxins (i.e., enniatins and beauvericin), and microbial testing.

MoreMeat's mycoprotein ingredient specification is also compared to GRN 000904 and GRN 000945 in the following Table 28. MoreMeat's mycoprotein ingredient specification is essentially the same as those specifications established in GRN 000904 and GRN 000945. MoreMeat performs additional tests for appearance compared with GRN 000904 and GRN 000945. Protein and total fiber levels are slightly different for the three products, which is the result of differences in their manufacturing processes and culture mediums. MoreMeat's mycoprotein is derived from *F. compactum*, which is a different *Fusarium* species compared with those described in GRN 000904 and GRN 000945, as stated above, which requires specifications for different mycotoxins as well. In conclusion, MoreMeat believes that the minor differences in product specifications do not affect the safety of its mycoprotein ingredient compared with those described in GRN 000904 and GRN 000945.

Table 28 Comparison of Product Specification of the Current Notification, GRN 000904, and GRN 000945.

Test Items	MoreMeat	GRN000904	GRN000945
Appearance	Off-white to light yellow, dough-like matter.	NS	NS
Flavor	Odorless	NS	NS
Taste	Tasteless	NS	NS
Moisture	≤78%	70-80%	NS
Protein	≥43%, on dried basis	>45% dry weight	>42%
Total fiber	≥25.0%, on dried basis	25-35%	22-26%
Ash	<5.0 %	<5%	<5%
Cadmium	<0.1 mg/kg, on dried basis	NS	<0.1 mg/kg
Lead	<0.1 mg/kg, on dried basis	NS	<0.1 mg/kg
Arsenic	<0.1 mg/kg, on dried basis	NS	<0.1 mg/kg
Mercury	<0.1 mg/kg, on dried basis	NS	<0.1 mg/kg
Enniatins	<40 µg/kg	NS	NS
Beauvericin	<10 µg/kg	NS	NS
Diacetoxyscirpenol	NS	NS	<5 µg/kg
Deoxynivalenol	NS	NS	<10 µg/kg
3-Acetyldeoxynivalenol	NS	NS	<10 µg/kg
Nivalenol	NS	NS	<5 µg/kg
Fusarenone X	NS	NS	<5 µg/kg
Neosolaniol	NS	NS	<10 µg/kg
Total Aerobic Count	<10,000 CFU/g	<10,000 CFU/1g	<100 CFU/g
Yeasts and Molds	≤100 CFU/g	<100 CFU/g	<100 CFU/g
<i>Listeria monocytogenes</i>	Not detected /25 g	NS	ND in 25g
<i>Escherichia coli</i>	Not detected /g	ND	<10 CFU/g
<i>Staphylococcus aureus</i>	Not detected /g	NS	<20 CFU/g

<i>Salmonella</i>	Not detected /25 g	ND	ND in 25g
Viable <i>Fusarium</i>	Not detected/g	NS	ND/g
Ribonucleic Acid in Dry Matter	≤2%	<2.0%	<2.0%

NS: not specific; ND: not detected

Considering the substantial similarities in the production process and product specifications, it can be concluded that MoreMeat's mycoprotein is quantitatively and qualitatively equivalent to the subject of GRN 000904 and GRN 000945. Therefore, the safety of MoreMeat's mycoprotein is also supported by the toxicology studies summarized in GRN 000904 (pages 31-49) and GRN 000945 (pages 26-31).

6.2 Toxicology Studies Commissioned by MoreMeat

6.2.1 Genetic Toxicity Studies

As part of the safety evaluation for a new food ingredient, genotoxicity studies are performed to determine if the ingredient has the potential to cause genetic damage. MoreMeat has assessed mycoprotein for potential genotoxic effects using an AMES test, an *in vivo* Mammalian Erythrocyte Micronucleus test, and an *in vitro* Mammalian Chromosome Aberration test. The three tests unanimously concluded that MoreMeat's mycoprotein is nongenotoxic under the tested conditions. In addition to these tests, a review of the genotoxicity evaluations for other *Fusarium*-derived products also provides no evidence of genotoxic effects of mycoproteins.

6.2.1.1 Bacterial Reverse Mutation Test (Ames) of MoreMeat's Mycoprotein

To assess the mutagenic potential of mycoprotein, an Ames assay was conducted at the GLP compliant laboratory - Chemical Toxicity Testing Institute of the Shanghai Municipal Center for Disease Control and Prevention. The study was conducted in accordance with the OECD Guideline for the Testing of Chemicals No. 471. The assay utilized plate incorporation and pre-incubation methods to evaluate the mutagenic potential of mycoprotein on *Salmonella typhimurium* strains TA97a, TA98, TA100, TA102, and TA1535, all of which are histidine-dependent. The evaluation was conducted with and without metabolic activation using a chemical-induced rat liver S9 mix. The test substance was formulated as a solution in sterile water (0.0158, 0.05, 0.158, 0.5, 1.58, 5, 15.8 and 50 mg/mL) to provide corresponding dose levels of up to 5000 µg/plate.

The pre-experiment with plate incorporation method was conducted with mycoprotein at dose levels of 15.8, 50, 158, 500, 1580 and 5000 µg/plate, but serious precipitate occurred in the 5000 µg/plate. Therefore, the initial test with the plate incorporation method was conducted with mycoprotein at dose levels of 15.8, 50, 158, 500, and 1,580 µg/plate. For the confirmatory test, a preincubation modification of the plate incorporation method was used. In this approach, the test or control substances were first

incubated with the bacterial suspension and S9/substitution buffer for approximately 30 min at approximately 37 °C with gentle shaking to ensure thorough mixing. After the pre-incubation period, the mixture was combined with overlay agar and poured on minimal agar plates. The plates were handled in the same manner as in the initial test, with incubation at approximately 37 °C until adequate growth was observed. The confirmatory test followed the same study design as the initial test, including the use of the same bacterial strains and number of replicates. The highest concentration of the test substance in the confirmatory test was also 1,580 µg/plate, followed by lower concentrations of 316, 63.2, 12.6, and 2.5 µg/plate.

Five different concentrations were tested for all strains, thus adequately evaluating bacterial mutagenicity. The results show no concentration-related increases or substantial increases in revertant colonies with strains TA1535, TA97a, TA98, TA100, or TA102 when S9 was used or not, using either the plate incorporation or pre-incubation method. Therefore, no genotoxicity was observed in any strain at all dose concentrations. In conclusion, mycoprotein derived from *F. compactum* MM-135 did not induce any gene mutations and is therefore concluded to be non-mutagenic under the experimental conditions employed for this study.

6.2.1.2 *In Vivo* Mammalian Erythrocyte Micronucleus Test

The *in vivo* mammalian erythrocyte micronucleus test for mycoprotein derived from *F. compactum* MM-135 was conducted according to the OECD Guideline for the Testing of Chemicals No. 474 Mammalian Erythrocyte Micronucleus Test (2016) by the GLP compliant laboratory - Shanghai Municipal Center for Disease Control and Prevention. The *in vivo* micronucleus assay in mammals was performed to assess the genotoxic potential of the mycoprotein from *F. compactum* MM-135 by detecting any chromosome damage or effects on the mitotic apparatus. Mice were weighed and the prepared test solutions of different concentrations and the positive control solution were administered to each group of animals at a dose of 10 mL/kg body weight. Comparisons between the micronucleated cell frequencies in the treated groups and the negative control revealed no significant differences. Consequently, mycoprotein derived from *F. compactum* MM-135 yielded a negative outcome in the mammalian erythrocyte micronucleus test, indicating no *in vivo* genotoxicity under the conditions of the study. It is concluded that mycoprotein from *F. compactum* MM-135 does not exhibit mutagenicity related to clastogenicity and/or aneugenicity based on the *in vivo* mammalian cell micronucleus test.

6.2.1.3 *In Vitro* Mammalian Chromosome Aberration Test

The *in vitro* mammalian chromosome aberration test of mycoprotein was conducted at the GLP compliant laboratory - Chemical Toxicity Testing Institute, Shanghai Municipal Center for Disease Control and Prevention following the OECD Guideline No. 473. Chinese hamster lung cells (CHL, purchased from Wuhan Prosel Life Science and Technology Co., Ltd) were digested with 0.25% Trypsin-EDTA and prepared into a 1×10^6 /mL suspension with MEM culture medium containing 10% fetal bovine serum. During the test, each concentration of the test solution was added to the test system

(5 mL) at 10% (V/V), and the final concentrations of the test solution were 5000 µg/mL, 2500 µg/mL, 1250 µg/mL, 625 µg/mL, and 312.5 µg/mL, respectively.

The study was conducted as a formal test and then an additional test, but were separate experiments, not as an extension of the formal test. The formal test was done with and without S9 mixture and a 4-hour exposure period and then an additional test was done without S9 and a 24-hour exposure period. In the absence of S9 mixture, the dose resulting in 45% ±5% Relative Increase in Cell Count (RICC) was 1,250 µg/mL. Therefore, without S9 mixture, the concentrations of the test article were 1,250 µg/mL in the high dosage group, 625 µg/mL in the middle dosage group, and 312.5 µg/mL in the low dosage group.

When compared to the negative control group, the P value of the mammalian chromosome aberration rate for the high, medium, and low doses of the test sample was greater than 0.05, indicating no significant difference. Therefore, the result of the *in vitro* mammalian chromosome aberration test for mycoprotein from *Fusarium compactum* MM-135 was negative under the conditions of the study.

6.2.2 14-Day *In Vivo* Toxicity Studies

This preliminary study aimed to assess the general toxicity of mycoprotein derived from *F. compactum* MM-135 in rats after a minimum of 14 days of dietary intake. This study was completed in a GLP compliant laboratory and the study followed OECD Guideline No. 407. Forty SPF Sprague-Dawley rats (20 males and 20 females, purchased from Zhejiang Vital River Laboratory Animal Technology Co., Ltd.) were randomly divided into four groups (5/sex/group) and fed diets containing 0, 50,000, 100,000, and 150,000 ppm of mycoprotein, i.e., 3866, 8343, and 12,150 mg/kg/day for males, and 3834, 8434, and 12,499 mg/kg/day for females, respectively. Body weights were measured on Days -5 and -1. During the experiment, twice-daily cage-side observations (morning and afternoon) were conducted, with detailed clinical observations once daily. Rats were weighed on Days 3, 7, 10, 14, and 15 (pre-necropsy). Food consumption was measured daily, and food efficiency was calculated as Food efficiency = (Mean Daily Body Weight Gain) / (Mean Daily Food consumption). Clinical pathology testing was performed on Day 15 for all groups, with blood samples collected by abdominal aortic puncture for hematology, coagulation, and serum biochemistry analysis. Necropsy and gross observation were performed on Day 15, with tissues and organs collected, weighed, and fixed for pathological examination. Treatment groups were compared with the control group using pair-wise comparisons for all indicators except leukocyte count, which was analyzed using log transformation of group pair-wise comparisons.

There were no mortalities during the course of the study. Blood biochemical indicators of total cholesterol were slightly decreased in all dose groups of the test substances. Because the changes were dose-related and showed slight differences, they were considered to be test substance-related. This could be due to the fiber content in mycoprotein. Comparable effects on cholesterol reduction have been noted in both subchronic and chronic animal studies of mycoprotein from *F. venenatum* (Miller, 2001) and *F. strain flavolapis* (Furey, 2022), and in human clinical trials (Coelho, 2021), all of which

explored the inclusion of mycoprotein in the diet. Other test parameters showed no test substance-related changes, for example, in clinical observations, body weight, food consumption, food efficiency, clinical pathology, and necropsy observations. There were no toxicological concerns related to mycoprotein administration.

6.3 Published Toxicity Studies

6.3.1 Genotoxicity Studies on Mycoprotein

MoreMeat has reviewed the publicly available literature on genotoxicity studies for similar food ingredients. To obtain a current, thorough, and comprehensive understanding of publicly available studies that support the safe use of mycoprotein, literature searches were conducted using both PubMed and GoogleScholar to identify published genotoxicity studies. The search terms “mycoprotein AND genotoxicity”, and “fungal protein AND genotoxicity” were used to identify recently published toxicology studies conducted with mycoprotein up to February 8, 2025.

Two relevant studies were identified regarding the genotoxicity of mycoprotein derived from *Fusarium* species. These two studies are summarized in Table 29 below. The results of both studies demonstrated that mycoproteins derived from *F. venenatum* and *F. strain flavolapis* are non-mutagenic. These results are corroborated by those obtained for mycoprotein derived from MoreMeat’s production strain *F. compactum* MM-135, which exhibits similar genotoxic characteristics.

Table 29 Summary of Genotoxicity Studies on Mycoprotein Derived from Other *Fusarium* Species.

Study Type	Study design	Results	Summarized in	Reference
Reverse mutation test	Bacterial reverse mutation test (Ames) in <i>S. typhimurium</i> (strains not specified).	The mycoprotein product was determined to be non-mutagenic in both the presence and absence of S9 mix metabolic activation.	GRN 000091 (page 17)	Miller et al. (2001)
Bacterial reverse mutation test (Ames)	Bacterial strains used: <i>S. typhimurium</i> TA 98, TA 100, TA 1535, and TA 1537 and <i>E. coli</i> WP2 uvrA in both the absence and presence of S9 mix, at levels up to 5000 µg/plate. Study conducted in accordance with OECD Guidelines No. 471. The study was conducted with mycoprotein derived from <i>F. strain flavolapis</i> at levels of 0.0158, 0.05, 0.158, 0.5, 1.58, 5, 15.8, and 50 mg/mL which correspond to 1.58, 5.0, 15.8, 50, 158, 500, 1580, and 5000 µg/plate of active ingredient, with the high level being the standard limit for this test.	<i>F. strain flavolapis</i> did not elicit evidence of bacterial mutagenicity in the Ames assay.	GRN 000904 (page 32)	Furey et al. (2022)
<i>In vitro</i> micronucleus test in human lymphocytes	Used human peripheral blood lymphocytes both with and without S9 mix, at levels up to 50 µg/mL. Study conducted in accordance with OECD Guidelines No. 487. The study was conducted with mycoprotein derived from <i>F. strain flavolapis</i> at levels of 12.5, 25, and 50 µg/mL with a 4 h treatment and 44 h fixation interval both in the presence of and absence of metabolic activation.	The evaluation of micronuclei frequencies did not show a biologically relevant increase compared to the controls. <i>F. strain flavolapis</i> produced no cytotoxic effects up to the highest concentrations tested of 50 µg/mL, in both the presence and absence of S9 mix.		

6.3.2 Acute Toxicity of Mycoprotein

MoreMeat has reviewed the publicly available literature on acute oral toxicity studies for similar food products. To obtain a current, thorough, and comprehensive understanding of publicly available studies that support the safe use of mycoprotein, literature searches were conducted using both PubMed and GoogleScholar to identify acute oral toxicological studies that have been published. The search terms “mycoprotein AND acute study” were used to identify recently published toxicology studies conducted with mycoprotein as of February 8, 2025. A summary of those studies is tabulated in Table 30 below.

There were no adverse effects observed in both studies for mycoprotein derived from *F. venenatum* and *F. strain flavolapis*, which corroborates similar observations from the 14-day *in vivo* study conducted on MoreMeat’s mycoprotein derived from *F. compactum*.

Table 30 Summary of Acute Toxicological Studies of Mycoprotein Derived from Other *Fusarium* Species.

Study Type	Test System	Dose	Results	Summarized in	Reference
The estrogenic potential of mycoprotein on mouse and non-RNA reduced mycoprotein on Landrace pig	Mouse and Landrace pig	Not specified, mycoprotein derived from <i>Fusarium venenatum</i> PTA-2684	No estrogenic potential was observed in either species.	GRN 000091 (page 17)	Miller et al. (2001)
Short-term oral range-finding study over 14 days This study follows the US FDA Toxicological Principles for the Safety Assessment of Food Ingredients, Redbook 2000, IV.C. 4. a. (2007).	Sprague-Dawley rats	Doses of 0, 5%, 10%, and 15% of the diet, mycoprotein derived from <i>Fusarium strain flavolapis</i>	There were no abnormal physical observations resulting from the product. All animals were subject to gross necropsy and no macroscopic observations were identified. No significant differences were noted in body weight gain, feed	GRN 000904 (page 31-32)	Furey et al. (2022)

			consumption, or feed efficiency over the course of the study as compared to the control groups, and no clinical observations were noted in the groups fed mycoprotein, regardless of dose rate.		
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6.3.3 Subchronic and Chronic Oral Toxicity for Mycoprotein

To obtain a current, thorough, and comprehensive understanding of publicly available studies that support the safe use of mycoprotein, literature searches were conducted using both PubMed and GoogleScholar to identify subchronic and chronic oral toxicological studies that have been published. The search terms “mycoprotein AND subchronic study” and “mycoprotein AND chronic study” were used to identify published toxicology studies conducted with mycoprotein as of February 8, 2025. A summary of those studies is listed in Table 31 below.

Table 31 Summary of Subchronic and Chronic Toxicological Studies of Mycoprotein

Study Type	Species	Dose	Results	Reference
22-week oral feeding study	Rats	26-52%, mycoprotein derived from <i>Fusarium venenatum</i> PTA-2684	With the exception of caecal enlargement attributed to the high fiber content of the food, no significant findings were observed.	Miller et al. (2001)
13 - week oral feeding study with dried, cooked mycoprotein	Rats	13%, 35%, mycoprotein derived from <i>Fusarium venenatum</i> PTA-2684	With the exception of caecal enlargement, no significant differences were found in growth, blood parameters, organ weights, and histopathology.	Miller et al. (2001)
13-week oral feeding study with undried mycoprotein	Rats	13%, 35%, mycoprotein derived from <i>Fusarium venenatum</i> PTA-2684	No significant differences were found in all parameters studied, including the availability and balance of calcium, phosphorus, magnesium, iron, copper, and zinc.	Miller et al. (2001)
90-day oral feeding study, with <i>in utero</i> exposure, and 90-day	Male and female rats	20%, 40%, mycoprotein derived from <i>Fusarium</i>	All test groups demonstrated normal growth and development. At weaning, however, both male and female pups fed the high mycoprotein diet had	Miller et al. (2001)

studies in offspring		<i>venenatum</i> PTA-2684	lower body weights than the casein controls. While the proportional differences diminished during the 90-day feeding period, statistical differences remained. In addition, plasma cholesterol and triglycerides were lower in the rats fed with mycoprotein. Liver weights were also lower. All of these effects could be attributed to the reduction in food intake observed in these animals. To test this hypothesis, the physical form of the diet was changed to improve its palatability. The differences in food intake and weight gain were abolished in the subsequent study, confirming the hypothesis.	
13-week oral feeding study	Baboons	26%, 51.5%, mycoprotein derived from <i>Fusarium venenatum</i> PTA-2684	All baboons survived the study in good condition and there were no effects on growth, food and water intake, or hematology. Serum alkaline phosphatase activity was increased in the groups fed mycoprotein and serum alanine transaminase and aspartate transaminase activities were increased from normal levels in the casein control group. There were no changes in electrocardiograms. The test groups showed marginally higher liver weights than the casein control and colon and rectum weights were higher, but this reflected the increased contents. No histopathological changes were seen. It was concluded that mycoprotein was without adverse effects, but the casein diet produced a	Unpublished, summarized in GRN 000091 (page 25)

			minimal degree of liver damage.	
2-year carcinogenicity study (with <i>in utero</i> phase)	Rats	21%, 41% (at 41%, all dietary protein was derived from mycoprotein; mycoprotein derived from <i>Fusarium venenatum</i> PTA-2684	Although minor intergroup variations were observed in some parameters, no significant adverse effects were noted in either phase of the study on growth, survival, incidence, or onset of tumors or in hematological, urinary, or histopathological examinations, indicating that mycoprotein is not toxic or carcinogenic.	Miller et al. (2001)
1-year oral feeding study	Beagle dogs	20%, 40%, mycoprotein derived from <i>Fusarium venenatum</i> PTA-2684	The study examined growth, clinical conditions, urine, hematology, and histopathology. All diets supported good growth and no diet-related clinical signs were reported. The dogs fed with diets containing mycoprotein showed lower plasma cholesterol and triglycerides and female dogs showed marginally greater thyroid weight. No gross or microscopic pathological findings were made that were diet-related.	Miller et al. (2001)
2-generation reproductive and developmental toxicity study	Rats	12.5%, 25%, 50% (50% casein as control) mycoprotein derived from <i>Fusarium venenatum</i> PTA-2684	The mycoprotein diets supported good growth and maturation in both parental generations and offspring, and fertility was within normal limits. There were no changes in reproductive function that were attributable to mycoprotein, and there were no microscopic or pathological findings.	Miller et al. (2001)
Teratogenicity studies	Rats and rabbits	Not specified, mycoprotein derived	No evidence of teratogenic or embryo-toxic effects associated with mycoprotein-containing diets was	Miller et al. (2001)

		from <i>Fusarium venenatum</i> PTA-2684	observed.	
90-day dietary intake The study was performed in accordance with OECD Test No. 408.	Sprague Dawley rats	50,000 ppm, 100,000 ppm, 150,000 ppm of mycoprotein (15% in the diet), mycoprotein derived from <i>Fusarium strain flavolapis</i>	No test substance-related mortalities, no toxicologically relevant clinical signs, or changes in body weight were observed in the 90-day study.	Furey et al. (2022)

Additionally, the mycoprotein derived from *F. venenatum* has an over 35-year history of consumption as a food ingredient, having been consumed in 17 countries with an estimated 9 billion servings consumed worldwide, with low levels of reported adverse effects according to Marlow Foods official website.⁹

MoreMeat agrees with the conclusion presented on page 36 of GRN 904, to which FDA issued a “no questions” letter, that the “robust dataset strongly supports a conclusion that there will be no acute or chronic adverse effects” from the consumption of mycoprotein. Cumulatively, the safety studies conducted with mycoprotein derived from *F. venenatum* and *F. strain flavolapis* demonstrate and support the safe use of mycoprotein as a food ingredient, and in turn, also supports the safe use of mycoprotein derived from *F. compactum* as a food ingredient.

⁹ Official website of Quorn: <https://www.quorn.co.uk/intolerance>
Anchor Center for Certification (ACC)

6.4 Overarching Basis for Comparing Toxicological Profiles

Based on generally recognized methods for evaluating toxicity in *Fusarium*, MoreMeat has determined that *F. compactum*, *F. strain flavolapis*, and *F. venenatum* grown under controlled conditions possess the same toxicological potential or lack thereof. This conclusion is based on an assessment of similarities from a phylogenetic and compositional standpoint, as well as similarities and differences in their potential and actual production of toxic secondary metabolites. This determination is consistent with generally recognized methods for read-across of biological substances and is additionally based on well-established literature and methods for evaluating the production of toxic secondary metabolites in *Fusarium* as detailed below. MoreMeat's determination can readily be verified by the scientific community using generally recognized methods of toxicity assessment for *Fusarium* and mycotoxins, as well as publicly available genetic data on *F. compactum*, *F. strain flavolapis*, and *F. venenatum*. This conclusion is further corroborated by the results of unpublished studies on MoreMeat's mycoprotein derived from *F. compactum*, which include *in vitro* and *in vivo* genotoxicity studies and a 14-day *in vivo* study.

6.4.1 Read-across Precedent

Within FDA GRAS precedent, there is a well-accepted practice of leveraging toxicological data from a prior GRAS Notice whereas here -- there are clear compositional and phylogenetic similarities between the notified substance and the subject of prior GRAS Notices. In fact, numerous successful GRAS Notices have incorporated this approach to supporting their GRAS conclusions for novel proteins and other complex biological substances, see e.g., GRN 000737, GRN 000811 and GRN 000904. Thus, it is MoreMeat's view that our approach of referencing pivotal studies and data from GRN 000091 and GRN 000904 to support a GRAS conclusion for *F. compactum* in our present GRAS Notice aligns with FDA precedent.

Further, in recent successful instances of “read-across” in the GRAS context, FDA has accepted reasoning based, in part, on the generally recognized methodology outlined by Pariza and Johnson (2001). Although originally intended for assessing the safety of modified organisms for enzyme production, the decision-tree approach advanced by Pariza and Johnson can effectively apply more broadly to novel protein production, including the safety of secondary metabolites, as evidenced by GRN 000737 for soy leghemoglobin protein derived from *Pichia pastoris* (Jin et al., 2018).

While the decision-tree methodology from 2001 has largely focused on modified strains of the same species lineage, advancements in bioinformatics and “omics” that have occurred over the twenty years since Pariza and Johnson's publication now allow scientists to similarly assess distinct species for their common lineages and shared evolutionary relationships to inform toxicology assessments (Pielaat et al., 2013). Toxicology experts and regulatory agencies in the U.S. and other developed nations are increasingly incorporating these modern approaches, particularly metabolomics (the systematic identification and quantification of small molecule metabolic products in their safety assessments of

distinct species; Rivetti et al., 2019; Viant et al., 2019; Sauer et al., 2010).

As such, and consistent with methods outlined in the decision-tree approach and modern bioinformatics, our assessment of the safety of *F. compactum* has focused on understanding its phylogenetic and compositional relationship to *F. strain flavolapis* and *F. venenatum*, as well as a comparison of the toxicological potential of *F. compactum*, *F. strain flavolapis*, and *F. venenatum*. The evaluation described in further detail below is grounded in published data on: a) the genetic pathways and regulation of toxic secondary metabolites in *Fusarium*, b) analytical methods for quantifying mycotoxins, and c) established safe exposure limits to mycotoxins.

6.4.2 Overarching Similarity of *F. compactum*, *F. strain flavolapis*, and *F. venenatum*

Our multi-faceted comparison of *F. compactum*, *F. strain flavolapis*, and *F. venenatum* clearly indicate strong similarities between the species pertinent to understanding their toxicological potentials and supportive of “read-across” to pivotal toxicology studies on *F. strain flavolapis* and *F. venenatum*. These similarities are summarized as follows:

- Phylogenetics: *F. compactum*, *F. strain flavolapis*, and *F. venenatum* are all *Fusarium* spp. and members of closely related clades, as described in Section 2.1. and Section 6.1.1.
- Composition: Mycoprotein derived from *F. compactum*, *F. strain flavolapis*, and *F. venenatum* have similar nutritional and protein profiles, as discussed in Table 33 Nutritional Profile Comparison.
- Intended Use: The mycoprotein ingredients derived from *Fusarium* spp. are intended for use as a replacement for animal-based and plant-based proteins in the human diet, as discussed in Section Part 3 dietary exposure of this notification, though at different levels and different food categories.
- Toxicological Potential: Similarities and differences in the potential and actual production of toxic secondary metabolites by *F. compactum*, *F. strain flavolapis*, and *F. venenatum* are well understood through generally recognized methods of genetic analysis and analytical testing, as discussed in further detail below.
- Toxic Effect: The assessment of toxicological potential is corroborated by a battery of *in vitro* and *in vivo* toxicology studies, additionally discussed below.

6.4.3 Genetic and Analytical Basis of Similar Toxicological Profiles

As stated in GRN 000904’s response to FDA’s inquiry (Toxicology Question 1), GRN 000904 outlined a well-established phylogenetic, genomic, and analytical approach to support the general recognition of similar toxicological profiles between *Fusarium* species. MoreMeat has followed the same approaches for determining that *F. compactum* has the same toxicological potential or lack thereof as

F. strain flavolapis and *F. venenatum* by demonstrating that:

- Toxicity of *Fusarium spp.* is well understood by the scientific community,
- Toxicological inquiry in *Fusarium* is focused on the toxic secondary metabolites: mycotoxins,
- Mycotoxins in *Fusarium spp.* are well characterized with a limited domain of compounds,
- The genes that control the type/quantity of mycotoxins in *Fusarium* are well understood,
- Established genetic methods elucidate the toxicological potential of *F. spp.*, specifically *F. compactum*, *F. strain flavolapis*, and *F. venenatum*,
- The toxicological potential based on genetics is verifiable with established analytical methods, and
- *F. compactum*, *F. strain flavolapis*, and *F. venenatum* produce limited mycotoxins when grown under controlled manufacturing conditions.

The validity of this methodology can be verified by the scientific community leveraging the extensive set of published literature referenced in this discussion and the published genomes of *F. compactum*, *F. strain flavolapis*, and *F. venenatum*.

6.4.4 In vitro & in vivo Basis of Similar Toxicological Profiles

While there is no genetic or analytical evidence that *F. compactum* produces mycotoxins at levels of concern for food safety when MoreMeat's mycoprotein is consumed under the intended uses, there also is strong evidence from *in vitro* and *in vivo* studies that further corroborate our conclusion that *F. compactum* has a toxicological potential similar to or less than that of *F. strain flavolapis* and *F. venenatum*.

In vitro Basis of Similar Toxicological Profiles

As described in the above section, MoreMeat's mycoprotein has been tested using a battery of *in vitro* toxicity assays that have been shown to be sufficient to detect carcinogens and *in vitro* toxicity in rodents (Kirkland et al., 2011) and are used to support the safety of human consumption (FDA 2007). These tests would be expected to detect harmful effects from potentially unknown secondary metabolites commonly associated with filamentous fungi (Pariza and Johnson, 2001).

Testing found no significant increase in DNA mutation rate (Ames tests; OECD 471), similar to the results from the Ames test on mycoprotein derived from *F. strain flavolapis* and *F. venenatum* which showed non-mutagenicity (Furey, 2022; Miller, 2001), when both metabolically active and inactive bacterial cells were treated with the highest recommended concentration of mycoprotein derived from *F. compactum* (5 mg / plate). Additionally, no significant increase in cell death or increase in the number of micronuclei were observed in a mammalian erythrocyte micronucleus study (OECD 474) in which mammalian erythrocyte cells were treated with the highest concentration of mycoprotein

derived from *F. compactum* (10 mL/kg body weight).

Furthermore, MoreMeat's mycoprotein derived from *F. compactum* MM-135 has been tested using *in vitro* mammalian chromosome aberration test and it was concluded that MoreMeat's mycoprotein does not exhibit mutagenicity related to clastogenicity and/or aneugenicity.

Together, results of the Ames test, chromosome aberration test, and micronucleus tests show that mycoprotein derived from *F. compactum* has no mutagenic or cytotoxic effects under the conditions of the studies. Additionally, corroborative evidence has been published that mycoprotein derived from *F. strain flavolapis* and *F. venenatum* are also non-mutagenic, further supporting the genetic and analytical results detailed above.

In vivo Basis of Similar Toxicological Profiles

Finally, the published *in vivo* results for substantially similar mycoprotein ingredients, which indicate a lack of acute and chronic adverse effects, are further corroborated by the 14-day *in vivo* oral toxicity tests in rats fed with high amounts of MoreMeat's mycoprotein, as described in Section 6.2.2. The test identified no physical or gross morphological differences between controls and rats fed MoreMeat's mycoprotein up to 12,150 mg/kg/day for males, and 12,499 mg/kg/day for females.

6.4.5 Conclusion of Similar Toxicological Profiles

Based on the foregoing similarities and assessment of the extensive published and peer-reviewed data on *Fusarium* toxicity and *F. compactum* toxicity, as well as the data on *F. strain flavolapis* toxicity and *F. venenatum* toxicity, we have determined that *F. compactum*, *F. strain flavolapis*, and *F. venenatum* have the same toxicological potential or lack thereof, when grown under controlled conditions, and that the read-across to pivotal data cited in GRN 000091 and GRN 000904 for *F. venenatum* and *F. strain flavolapis* offers sound scientific support for our conclusion that the consumption of MoreMeat's mycoprotein under the proposed conditions of use is safe to a reasonable certainty.

6.5 Allergenicity

MoreMeat's mycoprotein is produced using *F. compactum* MM-135, a species of the *Fusarium incarnatum-equiseti* species complex. To comprehensively evaluate its potential allergenic risk, MoreMeat conducted a risk assessment, including:

- a) A literature search to identify published evidence of allergy to food and environmental protein sources of various *Fusarium* species, of IgE binding and allergy to specific proteins from these sources, or of cross-reactivity to other sources with high sequence identity to *Fusarium* proteins.
- b) The stability of the proteins in the strains was tested in simulated gastric fluid (SGF) and simulated intestinal fluid (SIF) assays.

- c) MoreMeat conducted a genome sequence of their production organism, and the genome sequence of this strain was evaluated to predict possible proteins and compared to AllergenOnline.org version 22 to identify matches to allergen identities.

The purpose of this allergenicity assessment is to identify any potential additional food allergy risk that may arise from consuming mycoprotein-containing foods that may be relevant to consumers who are allergic to other protein sources. Obviously, most food sources present some food allergy risk, so the results of this assessment are not expected to show that there is no risk of allergic reaction, but rather to predict risks arising from the consumption of MoreMeat's mycoprotein.

The genome of the *F. compactum* MM-135 production strain was sequenced by MoreMeat using an Illumina NextSeq 2000 Sequencer. The analysis indicated that the genome could express 12,548 proteins based on gene prediction and annotation of the DNA. In March 2024, we evaluated whether the predicted amino acid coding regions matched allergen sequences listed in the www.AllergenOnline.org database version 22 (AOLv22, sequences were collected from NCBI). We compared the predicted protein sequences using FASTA-35.4.12 to provide matches of >35% identity over 80 or more amino acids. We also compared the sequences to the NCBI non-redundant protein database (NR) using BLAST+ version v2.13.0.

The Codex Alimentarius Commission Guidance (2003) is utilized to evaluate the potential risk of food allergens, as well as to assess the toxicity and nutritional properties of genetically modified (GM) food crops with the addition of one or more genes. It is important to note that while the Guidance was followed in this risk assessment, it was not meant to assess a novel organism intended to be used as food ingredients and the mycoprotein reviewed is not a GM food crop. MoreMeat has adhered to the Codex Alimentarius process for assessment, consistent with the approach taken for other novel food ingredients, such as GRN 000904 and GRN 000945.

Through this assessment, MoreMeat has verified that *F. compactum* MM-135-derived mycoprotein demonstrates no high identity matches to proteins with very high identity matches to clinically severe allergens from common allergenic sources (e.g. Ara h 1, Ara h 2, Ara h 3, or Ara h 6 from peanut; tropomyosin from shrimp; or beta-lactoglobulin from milk). The Codex guidance also recommends reviewing published scientific studies on the allergenic potential of the novel protein with peer-reviewed published studies related to allergies and allergens. The highest identity matches identified were for a few proteins from the genus *Fusarium*. Reports of allergies to *Fusarium* and related species have been addressed in the literature review study. Importantly, these allergy reports are of rare cases of IgE-mediated allergy and there are no obvious severe reactions to foods.

6.5.1 Proteins Predicted from the Genome Analysis of *Fusarium* Strains.

The 12,548 proteins predicted from the genome sequence of *F. compactum* MM-135 were evaluated by MoreMeat for amino acid sequence identity matches of >35% to allergens listed in www.AllergenOnline.org version 22. These *Fusarium* protein sequences have not been confirmed to

be expressed in cultured fungi, but DNA sequences predict that they may be expressed. The predicted sequences were used to assess potential identity matches to known allergens.

Furthermore, because the genome is complex and it is assumed that food produced from cultures of this organism would produce a very complex set of proteins, we decided to evaluate sequences of related organisms for evidence of allergy. The *Fusarium* genus is complex because it is divided into multiple species complexes based on species ecology and evolutionary predictions based on genome sequences. Most *Fusarium* species live primarily on tissues of certain plant types or in soil and are important for decomposing plants. A few groups, such as the *Fusarium fujikura* species complex, are important plant pathogens restricted to certain host species (Niehaus et al., 2016). Some species or strains are able to infect human or other mammalian tissues and some are known to grow in human lungs, causing lung disease similar to that caused by certain *Aspergillus* species. A literature search using *Fusarium* AND allergy OR allergen found more than 188 publications. However, only a few publications showed a clear association with *Fusarium* proteins. The publication by Levetin et al. in 2016 provides an overview of fungal taxa that have been shown to cause allergies, including allergies triggered by inhalation, ingestion, or infection. A major change in the characterization of fungal species has been the use of DNA sequence analysis. By searching for *Fusarium*-related proteins in the Allergen Nomenclature database from WHO/IUIS, MoreMeat found that four *Fusarium* proteins were included in this database. These four proteins are AAL79930 (Fus c 1.0101), AJA79001 (Fus p 9.0101), AHY02994 (Fus p 4.0101), and AAL79931 (Fus c 2.0101), which are present in MoreMeat's production organism.

Five proteins have been identified as possible allergens in this genus in www.AllergenOnline.org. In addition, since a similar species, *Fusarium venenatum*, has been used as Quorn™ (a protein source in food products in the UK) since the mid-1980s and in the US since 2002, MoreMeat also collected the sequences of all proteins from this species for comparison with those in AllergenOnline.org as well as *Fusarium compactum* MM-135, which is under evaluation.

6.5.2 Proteins from *F. compactum* MM-135

MoreMeat used the Illumina NextSeq 2000 sequencer to perform paired-end 150 bp short-read sequencing on *F. compactum* MM-135, obtaining a total of 6.7G of raw data with an average Q30 of 85.57%. The raw data was cleaned using fastp (version 0.20) to remove low-quality reads, and then the genome was assembled using SPAdes (version 3.13.1). The size of the assembled genome is 38 Mb. Finally, funannotate (version 1.8.15) was used to predict genes.

The important thing to consider for allergy risk is how similar or different the protein in question is to proteins known to cause food allergens. Essentially all foods present some allergy risk that is specific to the proteins in the food source. Typically, only the most abundant proteins, such as Ara h 2, Ara h 6, Ara h 1, and Ara h 3 in peanuts, are likely to trigger clinical reactions. Yet a number of other proteins in peanuts may bind IgE in some peanut-allergic people. However, tests such as basophil activation or

skin tests are needed to prove allergic risk. It is also useful to consider the dose of exposure upon consumption.

Cross-reactions or cross-reactive IgE binding may be identified via laboratory testing using sera from subjects allergic to some fungi, including *Fusarium spp.* However, clinical reactions are not likely to occur during consumption of foods with mycoprotein derived from *F. venenatum* as reviewed in GRN 000091 and GRN 000945. Hoff et al. (2003) describes this protein as a common minor allergen of a number of mold species including *Fusarium culmorum* protein, though Quorn is from *F. venenatum*. Symptoms from ingestion in the UK were reported as 1 in 370,000 and in Switzerland as 1 in 80,000 subjects with symptoms of gastric cramping beginning one hour to four hours after ingestion. Based on skin prick tests of weak reactions and low level RAST IgE binding, they found apparent reactivity across a wide range of fungi. The one subject had a history of 30 years of perennial asthma. They performed double blind food challenge with mycoprotein from *F. venenatum* and also clear testing with cloned Fus c 1 and with mycoprotein extract. The IgE binding was cross-reactive but not equal and there is no connection with understanding clinical allergic risks. The low reactivity of mycoprotein from *F. venenatum*, which is a high protein content mycoprotein food ingredient, indicates little risk for the population. It is important to consider other fungal species that are closely related to *Fusarium* because they have high identity matches with similar fungi. Low identity matches of protein sequences are less likely to share IgE binding, and more diverse organisms are less likely to share cross-reactions. With the redefinition of fungal species as clusters of species that share common DNA and protein sequences, the *Fusarium* genus has been divided into groups of species called complexes. MoreMeat analyzed multiple fungal conserved genes (TEF1, RPB1, RPB2, ITS) and showed that *F. compactum* MM-135 is most closely related to the species *F. compactum* LLC 2178, which is a species in the *F. incarnatum-equiseti* species complex, a class of pathogens that only spread in plants

Protein sequences were translated from DNA sequences, and a total of 12,548 proteins were obtained. These sequences were compared with the www.AllergenOnline.org database version 22 in batches using FASTA 35.4.12. They were also tested using the sliding 80 amino acid match routines. The sequences were also searched using BLASTP in BLAST+ to compare the predicted proteins with protein sequences in the NCBI NR database.

6.5.3 Proteins from Quorn Strain *F. venenatum* A3/5

The genome sequence and protein sequences of mycoprotein derived from *F. venenatum* A3/5 were downloaded from the NCBI GenBank database (accession id: GCA_019425555.1). The downloaded genome is 38.6Mb in length and contains 13,946 annotated protein-coding sequences.

6.5.4 *Fusarium* sp. Proteins in Allergen Databases.

The WHO/IUIS Allergen Nomenclature Database (accessible at www.allergen.org) includes four *Fusarium*-related proteins that have been listed as known allergens by the committee, yet it is important to note that the WHO/IUIS Allergen Nomenclature Committee does not validate evidence of an allergy.

Instead, it relies on submitters' claims that their protein sequence is accurate and that there is IgE binding from individuals who report being allergic to the source. The four proteins identified are:

- a) **Fus c 1:** A ribosomal protein P2, approximately 11 kDa in size, derived from *Fusarium culmorum*.
- b) **Fus c 2:** A thioredoxin-like protein, approximately 13 kDa, also sourced from *Fusarium culmorum*. (Hoff M, 2003).
- c) **Fus p 4:** A transaldolase protein, approximately 37.5 kDa, obtained from *Fusarium proliferate*.
- d) **Fus p 9:** A vacuolar serine protease, approximately 36.5 kDa, derived from *Fusarium proliferatum*.

Additionally, the www.AllergenOnline.org database identified a fifth protein, which lacks a specific biological name and is referred to as a helix-loop-helix protein with 450 amino acids from *F. culmorum*. This protein exhibited weak IgE binding in immunoblots conducted on 3 of 52 subjects suspected of having an allergy to this mold. However, it did not undergo basophil activation testing, unlike Fus c 1 and Fus c 2, as noted in the publication by Hoff and Ballmer-Weber et al. (2003a).

6.5.5 Allergenic Protein Identification by Genetic Sequence

After analysis using the www.AllergenOnline.org website, 418 proteins that screened through specific filter criteria were identified. Since each protein may be aligned to multiple reference sequences, the reference sequence with the lowest *E* value was used as the best match for each protein, and then sorted by full-length identity. By comparing the genome and protein sequences of Quorn™ strain *F. venenatum* A3/5, we can see that 385 of the 418 proteins in *F. compactum* MM-135 are highly similar to those of the Quorn™ strain (identity larger than 60%), and the remaining 33 proteins are not present in Quorn™ strain *F. venenatum* A3/5. However, looking at the results of Sliding 80mer Window 35 Analysis, the highest identity of these 33 proteins is less than 60% and the sequence coverage (length) is low.

Through BLAST analysis, these 33 proteins have many homologous sequences in the NR database and are widely present in *Fusarium* and the evidence level of known allergic proteins is low. Therefore, even if *F. compactum* MM-135 does cause some IgE binding, whether *de novo* or cross-reactions, it is unlikely to be caused by these 33 proteins.

As for the 385 proteins that are similar to the Quorn™ strain *F. venenatum* A3/5, we are unable to verify if these proteins would lead to IgE binding or cross-reactions. Considering the possibility of mutations or other reasons, including cross-reactive carbohydrate determinants, we believe that proteins of *F. compactum* MM-135 are unlikely to elicit severe allergic reactions and are certainly no more likely to do so than the few self-reported allergens noted from consumption of Quorn™ (Jacobson and DePorter, 2018).

According to statistics models, MoreMeat's production strain may produce 17 proteins with full-length amino acid identities $\geq 70\%$ with the matching reference sequence, accounting for 4.07% of the total proteins identified (418 proteins); 46 proteins with identity $\geq 50\%$; 11% of the identified proteins with $< 70\%$ identity; and 355 proteins with identity less than 50%, accounting for 84.93% of the total protein identified (418 proteins).

Common allergenic proteins, such as those in sesame, wheat, and peanut, have protein sequences with less than 50% full-length identity. However, it should be noted that the CODEX guideline ($> 35\%$ identity over 80 amino acids) is designed to protect highly allergic people from cross-reactions to extremely strong allergens (Aalberse, 2000). Proteins with identity matches of less than 50% are generally not likely to show cross-reactions with IgE to allergens. In addition, when we consider the coverage with the matching sequence, we can see that these low-identity matches usually only cover half of the reference sequence length, and some are even only 10%, and therefore we believe that they have a lower risk of causing cross-reactions (Aalberse, 2000).

Among the proteins that are $\geq 50\%$ identical, most of them match fungal allergens, and based on the BLASTP to QuornTM *F. venenatum* A3/5 strain genome alignment, these protein sequences present in the QuornTM strain genome are not clear allergens.

The five genomes of the species with identities exceeding 80% are listed in Table 32 below, together with their potential allergenicity characteristics. These proteins are similar to allergic reactions to dust mites and other fungi in the air, not to food allergens, and they generally do not cause serious clinical reactions. One of the reasons why severe reactions are unlikely is that the sequence identities relative to human proteins as determined by BLASTP are restricted to *Homo sapiens*. Fus c 2 (41% ID to a human protein); Fus p 9: (35% ID to human protein); Fus p 4 (61% ID to human), Fus c 1 (65% ID to a human protein), Fus c 3 (is not similar to a human protein, but may have asparagine linked CCD as predicted by N-Glyc, though signal peptide is a question).

Table 32 Five Genomes Identified (exceeding 80% similarity) with Potential Allergenicity

ID	NCBI Accession	IUIS Allergen	Species	ID(%)	Allergenicity	Route of allergen exposure
MM135_000907-T1	AAL79931.1	Fus c 2.0101	<i>Fusarium culmorum</i>	95.90	IgE but no biological test	Airway
MM135_001316-T1	AJA79001.1	Fus p 9	<i>Fusarium proliferatum</i>	95.30	IgE but no biological test	Airway
MM135_002537-T1	AHY02994.1	Fus p 4.0101	<i>Fusarium proliferatum</i>	96.60	IgE but no biological test	Airway

MM135_008631-T1	AAL79930.1	Fus c 1.0101	<i>Fusarium culmorum</i>	94.50	IgE plus basophil+ or SPT+	Airway
MM135_000240-T1	AAN73248.1	Fus c 3	<i>Fusarium culmorum</i>	82%	IgE but no biological test	-

6.5.6 Summary of Allergenicity of *Fusarium compactum* MM-135

Bioinformatics analysis of the *F. compactum* MM-135 genome revealed that the potential allergenic proteins identified are commonly found in other *Fusarium* species, while their sequence similarity to proteins in non-fungal species is low. *F. compactum* MM-135 does not contain sequence-matching proteins in common allergens such as peanuts, sesame, soybeans, and seafood. The potential allergens are likely limited to fungi or *Fusarium spp.* Consumers who are allergic to these fungal sources will likely have allergy risks if they consume MoreMeat's mycoprotein. Therefore, products made with MoreMeat's mycoprotein should include appropriate food labels clearly indicating that they contain proteins derived from *Fusarium* fungi.

A search of the literature for evidence of allergy to *Fusarium* proteins showed that some people do develop allergic reactions to several proteins from this species when the fungus invades human tissues, most commonly the lungs or sinuses. In addition, based on the evidence from QuornTM, a small number of people may develop IgE-mediated allergic reactions to the ingestion of QuornTM proteins (derived from *F. venenatum*), which is used in large amounts in foods as a source of protein. These reactions are not surprising, as most foods containing large amounts of protein do cause allergies or adverse reactions similar to allergies in some consumers. Some cross-reactions are possible, as consumers become allergic to related proteins after inhaling homologous proteins from *Aspergillus* or *Penicillium*. Four proteins have been identified as allergens from both *Fusarium* species and are recognized by the WHO/IUIS Subcommittee on Allergen Nomenclature and are included in the AllergenOnline.org database. However, three of them are marked as putative allergens based on IgE binding only. There is also an additional protein identified from AllergensOnline, but not in WHO/IUIS databases, which is also recognized as a putative allergen based on IgE binding.

An important feature of reactions to QuornTM is the rare case of overt IgE-mediated allergy. There is evidence that two people may have died from consuming QuornTM: individuals who were previously sensitized to the fungus and who ate a patty-sized meal of QuornTM (Jacobson and DePorter, 2018). Based on the fragmentary information provided by Jacobson and DePorter and references therein, it is uncertain whether their deaths were due to an IgE-mediated reaction to the QuornTM proteins. The reaction could have also been due to other possible proteins in the food (e.g., from eggs, milk whey, etc.) Additionally, some individuals have reported occasional symptoms of vomiting or diarrhea after consuming QuornTM products, which may be attributed to the ingestion of large quantities of food that are difficult to digest due to its high chitin content (Chitin is an N-acetylglucosamine polymer, an

insoluble polysaccharide). IgE-mediated reactions would occur every time a large amount of the offending food is ingested, rather than occasionally (Sicherer, 2018).

In 2015, the York Health Economics Consortium conducted a systematic review of allergic reactions to mycoprotein. Among 30 experimental studies, only two reactions were confirmed. The consortium concluded that reported intolerance reactions to mycoprotein are very low relative to other common allergenic foods, however rare cases of true allergy to mycoprotein do occur, as for many common foods, e.g. wheat, eggs and potatoes.

Based on the results of bioinformatics analysis, literature reports on the risk of allergic reactions to *Fusarium*, and reports of allergic reactions to the existing *Fusarium* product derived from *F. venenatum* as reviewed in GRN 000091 and GRN 000945, there is no evidence that foods containing *Fusarium* pose a significant allergic risk to the general consumer, unless the consumer is allergic to *Fusarium* or closely related fungi.

Furthermore, MoreMeat contracted world-renowned immunologist Dr. Rick Goodman to evaluate the allergenicity of MoreMeat's mycoprotein derived from *F. compactum* MM-135. Dr. Goodman was a faculty member at the University of Nebraska-Lincoln in the Food Allergy Research and Resource Program in the Dept. of Food Science and Technology and managed the www.AllergenOnline.org database from 2004-2023. After a thorough evaluation of the product information, which included the information present in this GRAS dossier, as well as the *in vitro* digestibility report, and the literature research associated with *Fusarium sp.* including reports and papers on mycoprotein derived from *F. venenatum*, Dr. Goodman concluded that MoreMeat's mycoprotein does not represent a likely risk for food allergy based on history from literature, and lack of high identity matches to abundant and potent allergens. (See Appendix 3 for Dr. Goodman's evaluation report.)

6.6 Nutrition and Digestibility

The exploration of alternative proteins such as mycoprotein or fungal proteins is becoming increasingly important as the demand for sustainable and nutritious protein sources continues to grow. To provide a clearer understanding of the nutrient composition and amino acid content of MoreMeat's mycoprotein compared to mycoprotein from other *Fusarium* species, summaries of the comparisons are presented in Table 33 and Table 34 below. The results indicate that mycoprotein and fermented microbial protein have comparable nutrient compositions, consisting of a comprehensive array of macronutrients, vitamins, minerals, and other bioactive compounds. In addition, the *in vitro* PDCAAS test of MoreMeat's mycoprotein yielded an average score of 1.00, which demonstrates that mycoprotein is a quality protein similar to other protein sources, such as beef (PDCAAS score 0.92) and pea protein concentrates (PDCAAS score 0.67) (van Vliet, 2015).

In conclusion, MoreMeat's mycoprotein has a comparable nutritional profile and amino acid content when compared to the subjects of GRN 000904 and GRN 000945, and is a good quality protein source comparable to beef and pea protein concentrates.

Table 33 Nutritional Profile Comparison

Nutrient	Per 100 g (wet weight)		
	MoreMeat	GRN 000904 (page 19)	GRN 000091 (Miller, 2001)
Water %	75	75	75
Energy (kcal)	103.3	97	85
Protein (g)	13.8	12	12
Total fat (g)	1.4	1.1	3.0
Total Carbohydrates (g)	9.0	9.8 ^a	9.0 ^a
Sugars (g)	<0.1	<0.5	0.5
Dietary fiber (g)	9.17	7.4	6.0
Cholesterol (g)	<0.3	0 ^b	0 ^b

^a Total carbohydrates for GRN 000904 and GRN 000091 are calculated as: carbohydrates + dietary fiber. For GRN 000904: 2.4+7.4=9.8; GRN 000091: 3.0+6.0=9.0

^b Limit of detection for both results were not specified in the GRAS notification.

Table 34 Amino Acid Comparison

Amino acid	(g/100 g protein, dry weight basis)		
	MoreMeat	GRN 000904 ^a	GRN 000945 (page 16-17)
Isoleucine	5.33	5.16	4.9
Leucine	7.30	8.51	7.7
Histidine	2.12	2.60	2.5
Lysine	7.18	8.98	7.7
Methionine	2.05	1.96	2.0
Phenylalanine	4.44	4.73	4.7
Threonine	4.57	5.71	5.2
Tryptophan	1.36	1.89	1.4
Valine	5.37	11.38	5.9
Tyrosine	2.87	3.64	3.1
Serine	4.43	5.24	4.9
Proline	3.70	4.93	5.0
Glycine	4.26	5.11	4.7
Alanine	5.88	6.64	6.5
Glutamic Acid	10.37	12.24	11.9
Aspartic Acid	9.06	10.38	10.4
Arginine	5.99	6.58	6.9
Cystine and Cysteine	0.85	0.89	0.8

^a Amino acid (g/ 100g protein, dry weight basis) for GRN 000904 is derived from page 16-17 of GRN 000904. It is calculated as amino acid (% of total, dry weight basis) / protein level in dry

weight basis * 100. For example, Isoleucine is calculate as $2.32 / 45 * 100 = 5.16$ g/100 g protein.

As for digestibility, several studies have shown that specifically treated mycoprotein exhibits good digestion and absorption both in animal and *in vitro* experiments. For example, a protein band of approximately 30 kDa was detected up to 30 minutes of digestion in pepsin, and it was reduced to fragments of less than 5 kDa after 60 minutes. Afterword, pancreatin digested all proteins to below 15 kDa and most were gone, or absent by 60 minutes. The results demonstrated that the *Fusarium* mycoproteins are readily digestible (Goodman, 2019).

According to the digestibility test conducted by MoreMeat, tests of stability of the mycoproteins in the simulated gastric fluid showed that most of the proteins were digested in pepsin in relatively short times. Thus, MoreMeat's mycoprotein is expected to have good digestion in the human body following consumption.

Reports of adverse gastrointestinal effects from consumption of mycoprotein (Jacobson and DePorter, 2018), such as bloating, gas, constipation, and/or diarrhea, may be linked to the fiber content of the product. Based on a 2,000-calorie diet, FDA's adult daily value for fiber is 28 g per day (21 CFR 101.9 (c)(9)). MoreMeat's mycoprotein contains dietary fiber with a mean level of 10%. As previously discussed in Part 3., mean consumption of dietary fiber from mycoprotein under the proposed uses and use levels is 9 g/day, with a 90th percentile intake of 21 g/day. These EDIs are lower than or within the range of the recommended and/or background intake levels of dietary fiber in the diet. Therefore, it is very unlikely that usual intakes of mycoprotein from the proposed uses would result in either adverse gastrointestinal or physiological effects.

6.7 Mycotoxins

MoreMeat has reviewed the publicly available literature on mycotoxins for similar food products. To obtain a current, thorough, and comprehensive understanding of publicly available studies that support the safe use of mycoprotein, literature searches were conducted using both PubMed and GoogleScholar to identify relevant mycotoxins. The search terms "mycoprotein AND mycotoxins", "*Fusarium* species AND mycotoxins" and "*Fusarium compactum* AND mycotoxins" were used to identify published studies conducted with mycotoxin to February 8, 2025.

Mycotoxins, a type of secondary metabolite produced by fungi, have a wide range of toxicity. Once these toxins enter the food chain in large quantities, they are known to cause serious harm to animals, humans, and the entire biological environment. Therefore, the detection of mycotoxins is essential to ensure food safety and human health.

More than 400 varieties of mycotoxins have been identified to date. However, according to a previous work that shows the distribution of the varieties of mycotoxin production in *Fusarium* species, different *Fusarium* species have various abilities to produce mycotoxins (Munkvold, 2021; Wu, 2013). The three predominant classes of mycotoxins synthesized by *Fusarium* species are trichothecenes

(specifically, diacetoxyscirpenol), beauvericin, and enniatins, together with fumonisins and cyclonerodiol, which were reported from *Fusarium compactum*. Moreover, a type of zearalenone analog, deoxynivalenol, is consistently regarded as a significant concern in the realm of food safety (Ekwomadu, 2021).

Fusarium spp. reported by O'Donnell et al. (2018), among others, varies in the particular subsets of known mycotoxins that those closely-related strains produce. The potential and actual variation in the coding for and production of toxic secondary metabolites in *Fusarium* can readily be evaluated based on generally recognized bioinformatics and analytical methods, as has been conducted on *F. compactum*. As such, these sources do not contradict our determination that *F. compactum*, *F. strain flavolapis* and *F. venenatum* have the same toxicological potential or lack thereof. Rather, the references cited are corroborative of the analysis as discussed in Section 6.4, and instead show that:

- Each biosynthetic gene complex (BGC) produces a different mycotoxin;
- New mycotoxins have historically evolved by duplication and divergence of known BGCs;
- Genetic variation within a BGC can cause differences in mycotoxin expression levels, loss of mycotoxin production, or loss of certain decorations to the core chemical backbone (Proctor et al., 2006; Suga et al., 2018; Sultana et al. 2019); and
- Genetic variation within a BGC does not demonstrate the production of novel mycotoxins outside of those already discovered.

MoreMeat has conducted extensive analysis to determine if any mycotoxins are produced by *F. compactum* MM-135 using both genetic sequencing and analytical testing of the finished product (Section 2.3.3.1). Literature search, genome sequencing, and analysis indicates the potential for the formation of the mycotoxins diacetoxyscirpenol, beauvericin, and enniatins. Analytical testing was then used to determine whether *F. compactum* MM-135 produces these mycotoxins under the specific manufacturing conditions used for MoreMeat's mycoprotein. This testing has demonstrated that these mycotoxins are not present at levels that would represent a safety concern.

As discussed in Section 3.5, the only mycotoxins for which there is the potential for dietary exposure from consuming MoreMeat's mycoprotein are enniatins and beauvericin. A summary of the potential dietary intakes is provided in Table 27. Dietary exposure to mycotoxins from consuming MoreMeat's mycoprotein is well below the safe levels of enniatins and beauvericin as discussed in further details below.

6.7.1 Diacetoxyscirpenol (DAS)

Although the literature search indicated the potential for formation of diacetoxyscirpenol, this substance was not detected in analytical testing of three representative lots of MoreMeat's mycoprotein finished product. Diacetoxyscirpenol is one of the most toxic of the trichothecene mycotoxins

(Whitacre, 2014). It is produced by certain species of *Fusarium* (e.g., *F. compactum*, *F. poae*, *F. semitectum*, *F. moniliforme*, *F. sporotrichioides*, *F. acuminatum*, *F. culmorum*, *F. crookwellense*, *F. venenatum*, *F. sambucinum*, *F. graminearum*, *F. equiseti*, *F. solani*, *F. roseum*, *F. tricinctum*, *F. avenaceum*, *F. langsethiae*, and *F. clamydosporum*) (Omurtag et al., 2007, Schollenberger et al., 2007). MoreMeat conducted a bioinformatics analysis on genomic sequences of the production strain. They found that the strain lacks some critical genes encoding for the enzymes involved in the synthesis of the mycotoxin DAS. Therefore, it is believed that MoreMeat's production strain is not able to synthesize DAS.

In section 3.5.1, MoreMeat calculated the possible maximum mean and 90th percentile EDIs of 0.013 µg/kg bw/day and 0.030 µg/kg bw/day, respectively (assuming 70 kg bw/person), derived from the detection limit of DAS. This DAS level is well below the highest acute and chronic dietary exposures evaluated by the CONTAM Panel, which were estimated to be 0.8 and 0.49 µg DAS/kg bw/ day, respectively.

In summary, the probability of DAS formation in mycoprotein derived from *F. compactum* is minimum according to genome sequencing and analytical test results. Hence, DAS is not included in the product specification, but MoreMeat monitors DAS level in its mycoprotein product biannually. Additionally, a stringent quality assurance program is in place to ensure product quality in the event of a failed DAS test. (See Section 2.3.3.1 for details)

6.7.2 Enniatins

Enniatins are emerging mycotoxins synthesized by toxigenic microfungi mainly belonging to the *Fusarium spp.*, but also other genera such as *Alternaria*, *Halosarpheia*, and *Verticillium* (Pallares, 2020). These filamentous fungi are widespread pathogens of crops and represent a critical concern for food and feed safety worldwide. To date, 29 ENN analogs have been isolated and characterized, and, among them, ENNs A, A1, B, and B1 are most frequently found in cereals with cyclic hexadepsipeptides (Meca, 2012).

Studies on enniatins by Meca et al. (2012) used duodenal fluid from the simulated gastrointestinal digestion of 3 g of wheat crispy bread spiked with enniatin A, A1, B, and B1 at 1.5 and 3.0 µMol/g to evaluate the intestinal absorption (transport profile) and bioavailability of these mycotoxins using a two-compartment Caco-2 cell system. However, generally, the calculated enniatins bioavailability percentages were below or around 50%, thus, half of the amount of the enniatins digested was not transported across the intestinal barrier (Gautier, 2020).

The enniatins have ionophore properties and their toxicity mechanism is related to their ability to transport cations across the cell membrane. Enniatins are cytotoxic, as they decrease cell viability, promote cell cycle arrest, and increase apoptosis and the generation of reactive oxygen species in several cell lines. They also cause changes at the transcriptomic level and have immunomodulatory effects *in vitro* and *in vivo* (Hasuda, 2024). Toxicokinetic findings are limited, and despite the

established toxicity in laboratory settings, no regulations or risk evaluations have been undertaken owing to the absence of *in vivo* information.

In 2018, EFSA conducted studies for *in vivo* toxicity and genotoxicity of beauvericin and enniatins. For enniatins, the results support a genotoxic effect in bone marrow and liver cells after acute treatment, but not after repeated exposure. Immunotoxic enniatins effects were observed in both genders, suggestive of a suppressive/inhibiting activity particularly evident in males. Based on the repeated-dose enniatins study, the no observed adverse effect level (NOAEL) for females is 0.18 mg/kg bw/day (histomorphometrical effects on thymus, uterus and spleen). In male mice, the NOAEL is 1.8 mg/kg bw/day (enterocyte vacuolization in duodenum, increased Reactive Oxygen Species, and reduced glutathione brain levels). The maternal NOAEL is 1.8 mg/kg bw/day (decreased white pulp area and increased red/white pulp area ratio in spleen) and the developmental NOAEL is 18 mg/kg bw/day (Maranghi, 2018). The average enniatins levels determined in three representative lots of MoreMeat's mycoprotein result in a calculated mean and 90th percentile EDI of 0.001 µg/kg bw/day and 0.003 µg/kg bw/day, respectively, which are well below NOAEL.

Additionally, the calculated dietary exposure to enniatins from consumption of MoreMeat's mycoprotein is much lower than the CONTAM Panel estimated total chronic and acute dietary exposures to enniatins, with mean exposure ranged from 0.42 to 1.82 µg/kg bw/day and the 95th percentile exposure ranged from 0.91 to 3.28 µg/kg bw/day. The highest acute exposure estimates of enniatins were 4.67 µg/kg bw/day (mean) and 10.1 µg/kg bw/day (95th percentile) (EFSA, 2014). Therefore, the safety margins currently in place are not significantly altered by MoreMeat's mycoprotein when compared to existing consumption levels.

In summary, enniatins were identified in MoreMeat's mycoprotein ingredient and were previously found to be toxic in *in vitro* assays. However, published data indicate that half of the amount of digested enniatins are not transported across the intestinal barrier in the duodenal fluid from the simulated gastrointestinal digestion. Importantly, the resulting dietary exposure amount of enniatins from MoreMeat's mycoprotein is at a low level compared to the CONTAM Panel exposure estimates. Consequently, there is reasonable certainty of no safety risks from the consumption of mycoprotein derived from *F. compactum* MM-135 in relation to any anticipated enniatins residues. Furthermore, MoreMeat has established maximum specifications for enniatins to minimize the risk of increased consumer exposure.

6.7.3 Beauvericin

Beauvericin is a common mycotoxin produced by various *Fusarium spp.* and occurs naturally on corn and corn-based foods and feeds. Currently, the EFSA has issued relevant risk assessment reports on the contamination of beauvericin in cereals and their products. However, no regulatory or expert authority has established a specific threshold of safety concern for beauvericin.

An evaluation of beauvericin on oral toxicity was implemented by EFSA in 2018, and is composed of:

1) *in vitro* and acute *in vivo* genotoxicity tests; and 2) a repeated-dose oral toxicity study focused on genotoxic, immune, endocrine, nervous endpoints and the reproductive/developmental toxicity screening with mice. All the genotoxicity endpoints yielded negative results excluding a Comet assay in duodenum and kidney after repeated doses with 0.1, 1, and 10 mg/kg bw/day beauvericin. Beauvericin immunotoxicity was observed in female mice, concentrated in the number and functional activity of effector T cells in the spleen (Maranghi, 2018). Based on the repeated-dose beauvericin study, the NOAEL for female mice is 1 mg/kg bw/day (increased thyroid pycnotic nuclei and endometrial hyperplasia). In males, the NOAEL is 0.1 mg/kg bw/day (reduced colloid and altered T4 serum levels). Maternal NOAEL is 0.1 mg/kg bw/day (increased thymus weight) and developmental NOAEL is 10 mg/kg bw/day (Maranghi, 2018).

Based upon the LD₅₀ values of beauvericin, adverse health effects from acute exposure to beauvericin in farm and companion animals are unlikely, and chronic exposure data from poultry indicate that adverse health effects from beauvericin and enniatins are unlikely. Thus, the CONTAM Panel concluded that acute exposure to beauvericin does not indicate concern for human health (Benfor, 2014).

As described in Table 27, very low levels of beauvericin are detected in MoreMeat's mycoprotein, resulting in calculated mean and 90th percentile EDIs of 0.007 µg/kg bw/day and 0.015 µg/kg bw/day, respectively. Considering the minimal increase to existing beauvericin dietary exposure due to the consumption of MoreMeat's mycoprotein, we determine that there is a reasonable certainty of no safety risks from the consumption of mycoprotein derived from *F. compactum* MM-135 in relation to any anticipated beauvericin residues. Furthermore, MoreMeat has established maximum specifications for beauvericin to minimize the risk of increased consumer exposure.

6.7.4 Other Commonly Occurring Mycotoxins

FDA has established action or advisory levels for various commonly occurring mycotoxins, such as aflatoxins. Except for diacetoxyscirpenol, none of these mycotoxins were predicted to be present in MoreMeat's mycoprotein based on 3-batch analytical testing (Table 6) and the genetic analysis indicating that these mycotoxins are not expected to be produced by *F. compactum*. Therefore, there is reasonable certainty of no harm from these mycotoxins from the consumption of MoreMeat's mycoprotein under the proposed conditions of use.

6.8 Nitrates

Ammonia is used as fermentation ingredient, that means nitrates maybe generated during the fermentation process and are the only possible source of input identified in mycoprotein production. Nitrogen is an essential building block for the natural synthesis of amino acids and proteins.

Concerns about the safety of nitrate exposure have resulted in reviews by several authoritative bodies, including JECFA and FDA. The established safety levels for nitrate exposure compared with

estimated nitrates exposures from MoreMeat's mycoprotein under the proposed conditions of use are not anticipated to present a safety risk.

JECFA evaluated the safety of nitrates in conjunction with the safety of nitrites and nitrosamines on the basis that, although nitrate can generally be considered to be of relatively low toxicity, nitrate can be metabolized in the human body to form nitrite and N-nitroso compounds can also be formed from nitrite and N-nitrosatable compounds under certain conditions (JECFA, 1995). In its most recent evaluation in 2002, JECFA retained the acceptable daily intake (ADI) for nitrate of 0-3.7 mg nitrate ion/kg bw/day, which was first established in 1995 (JECFA, 2002). This ADI specifically addressed concerns about the potential conversion of nitrates to nitrites. A reevaluation conducted by the EFSA in 2017 maintained the ADI for nitrate at 3.7 mg/kg bw/day¹⁰.

As calculated in Section 3.6, the exposure to nitrates from consumption of MoreMeat's mycoprotein is estimated to be 0.592 mg/person/day (mean) and 1.304 mg/person/day (90th percentile). This exposure is more than 400 - fold below the upper - bound ADI established by JECFA, giving a reasonable certainty of no harm from nitrates from the consumption of mycoprotein derived from *F. compactum*.

In addition to the JECFA review, FDA and the U.S. Environmental Protection Agency (EPA) have established safety thresholds for the presence of nitrates in drinking water. The drinking water standard is 10 mg/L, measured as nitrogen (21 CFR 165.110). The National Academies of Sciences, Engineering, and Medicine has determined that an adequate daily fluid intake is about 15.5 cups (3.7 L) of fluids for men and about 11.5 cups (2.7 L) of fluids per day for women, of which about 20% is absorbed from food¹¹.

Assuming a fluid intake entirely from the tap or bottled plain water, the daily consumption of nitrate-nitrogen for men would be 29.6 mg/person/day (equivalent to 130 mg nitrate ion/person/day, as 1 mg nitrate-nitrogen = 4.4 mg nitrate). The consumption of nitrates from MoreMeat's mycoprotein under the proposed uses (i.e., 90th percentile EDI of 1.304 mg/person/day nitrate) is well below this level. EPA also has established an oral reference dose (RfD) for chronic exposures to nitrate-nitrogen of 1.6 mg/kg bw/day, based on critical effects seen in infants (equivalent to about 7.0 mg nitrate ion/kg bw/day) (Jonas, 2001). As noted in Section 3.6, the 90th percentile EDI for nitrates is estimated to be 0.019 mg/kg bw/day from consumption of MoreMeat mycoprotein under the proposed conditions of use. On these bases, there is reasonable certainty of no harm from the calculated exposure to nitrates from consumption of MoreMeat's mycoprotein.

¹⁰ Data available at:

https://www.efsa.europa.eu/sites/default/files/corporate_publications/files/nitrates-nitrites-170614.pdf. Accessed: 2024/09/17

¹¹ Data available at: <http://www.nationalacademies.org/hmd/Reports/2004/Dietary-Reference-Intakes-Water-Potassium-Sodium-Chloride-and-Sulfate.aspx> Accessed: 2025/02/08)

6.9 Ribonucleic Acid (RNA)

Another important consideration in the production of any mycoprotein intended for human consumption is the nucleic acid content of the final product. Upon ingestion, purinic nucleic acids are ultimately metabolized to the relatively insoluble uric acid. If uric acid levels exceed the excretion capacity of the kidneys, the excess acid crystallizes in joints and soft tissues, leading to gout and kidney stones (Sinskey, 1975).

The production organism *F. compactum* was isolated from the soil as a wild strain, thus genetic modification concerns need not be addressed. During its production, the biomass of *F. compactum* is subjected to heat treatment. By rapidly heating the fungal biomass in the broth to temperatures of 70–90°C, and maintaining it for 20–30 min, the RNA content of the mycoprotein is reduced to less than 2%. The thermal treatment acts by degrading the RNA into monomers that diffuse out of the cells (Raats, 2007) and are regularly consumed as a significant component of the human diet.

Importantly, the analysis of the 3 representative batches demonstrates an average RNA level of 0.02% (Table 5), which is much lower than the 2% specification limit, and at a level similar to the RNA present in mycoprotein product described in GRN 000904 (page 48).

As a core component in the structure of DNA and RNA, nucleic acids are abundant in all living organisms. Yeast (baking) has a relatively high nucleic acid content, of approximately 10% of the dry weight (Kihlberg R, 1972). Commonly consumed pig and beef have nucleic acid content of around 1–4% and 2–6% of the dry weight, respectively (GRN 000904, page 48). In addition, MoreMeat’s mycoprotein has the same maximum limit of RNA compared with the mycoprotein ingredients described in GRN 000904 and GRN 000945 (which received “no question” letters from FDA). The level of RNA in MoreMeat’s mycoprotein product is similar to or lower than levels present in common foods and reported in previous GRAS notifications; therefore, there is reasonable certainty of no harm from dietary exposure to RNA from consumption of MoreMeat’s mycoprotein.

6.10 Pesticides

As MoreMeat’s mycoprotein is produced using highly purified plant-based raw materials and is manufactured using a fermentation method, there is no reasonable expectation that any pesticide residues would be present.

6.11 Adverse Effects

To obtain a current, thorough, and comprehensive understanding of publicly available studies that support the safe use of mycoprotein, literature searches were conducted using both PubMed and GoogleScholar to identify adverse effects that have been published. The search terms “mycoprotein

AND adverse effects” and “fungal protein AND adverse effects” were used to identify published adverse effect of mycoprotein as of February 8, 2025. Special attention was paid to papers published between May 2020 and February 2025 due to the fact the GRN 0000945 was submitted in 2020; however, no recent publications related to subject of adverse effects from the consumption of mycoprotein were identified.

Relevant publications that discussed or mentioned adverse effects of mycoprotein, and which were also identified and discussed on page 27 of GRN 945, include:

- five papers that discussed nutrition and satiety (Derbyshire, 2018; Bottin, 2012; Bottin, 2014; Bottin, 2016; Williamson, 2006);
- six papers focused on adverse reactions, hypersensitivity, and intolerances after consuming mycoprotein (Katona, 2002; Hoff, 2003c; Van Durme P, 2003; Dzeladini, 2017; Jacobson, 2018; Finnigan, 2019); and
- seven papers (including a dose response study, a randomized controlled trial, and a pilot study) which focused on glycemia/insulinemia, blood lipids, and dietary fiber (Bottin, 2014; Williamson, 2006; Bottin, 2011; Dunlop, 2017; Denny, 2008; Ruxton, 2012; Finnigan, 2019).

Gastrointestinal Symptoms (Nausea, Vomiting, Diarrhea, Abdominal Cramps).

Mycoprotein’s nutritional composition and structure were the basis of research projects investigating the effect of mycoprotein on human health. With billions of mycoprotein-based products consumed over a 30+ year period in numerous countries around the world, there are very few published cases of intolerance. Four clinical cases have been reported, which all suggest that patients who are allergic to mold might react adversely to mycoprotein due to cross-reactivity (Katona, 2002; Hoff, 2003c; Van Durme P, 2003; Dzeladini, 2017).

As noted on page 28 of GRN 000945 and a publication written by Finnigan et al. (2019), the mycoprotein manufacturer’s database indicates that 92% of reported illnesses are associated with GI symptoms. In 2011, an Expert Panel was convened by Marlow Foods to review adverse reactions related to mycoprotein and potential causes. The Panel concluded that gastrointestinal (GI) symptoms were likely to be related to the high fiber content of mycoprotein. The panel hypothesized that in certain individuals or under certain conditions, consuming mycoprotein could speed up the normal transit of foods from the small to the large intestine. This could, in turn, cause the fiber in mycoprotein to be fermented very rapidly in the large intestine, leading to symptoms of gastrointestinal distress of the type reported by some consumers. Those at risk from this type of gut response may have an imbalance in their normal gut bacteria, an unusual dietary intake of fiber (too low or too high), or may suffer from irritable bowel syndrome (York Health Economics Consortium, 2015).

In summary, while adverse effects of GI symptoms, allergic reactions, and hyperinsulinemia from consumption of mycoprotein were identified, only a small portion of consumers suffered from these

effects. Therefore, the risks of adverse effects of mycoprotein are unlikely to the population at large. Given the substantial similarity of MoreMeat's mycoprotein to those described in GRNs 000904 and 000945, the risk of adverse effects from the consumption of MoreMeat's mycoprotein is no greater than the previous GRAS notified mycoproteins that have received "no questions" letters from FDA. As a precautionary step, MoreMeat shall ensure proper product labeling that clearly indicates that the product contains proteins derived from *Fusarium* fungi.

6.12 Expert Panel Findings of Safety of MoreMeat's Mycoprotein

An independent evaluation of the safety and GRAS status of the intended use of MoreMeat's mycoprotein ingredient was conducted by an Expert Panel convened by ACC. The Panel consisted of Margitta Dziwenka, DVM, DABT, Katrina Emmel, Ph.D., and Jianjun (Gene) Yang, Ph.D. The Expert Panel reviewed MoreMeat's dossier as well as other publicly available information available to them. The individuals who served as Expert Panelists are qualified to evaluate the safety of foods and food ingredients by merit of scientific training and experience. The GRAS Expert Panel report is provided in Appendix 5.

6.13 Conclusion of GRAS Determination

Pursuant to 21 CFR § 170.30 (a)(b) of the Code of Federal Regulations (CFR) (U.S. FDA, 2018b), MoreMeat has concluded that, under the proposed conditions of use, mycoprotein derived from *Fusarium compactum* MM-135, is GRAS. This conclusion is supported by the rigorous scientific assessment of the extensive and thorough information and data as described above and as summarized as follows:

1. The subject of this GRAS determination, mycoprotein, is derived from *Fusarium compactum* MM-135.
 - a. *Fusarium compactum* MM-135 is naturally found in soil. The production organism is not genetically modified.
 - b. DNA sequence comparison shows that *F. compactum* MM-135 shares the greatest sequence homology with *F. compactum* LLC 2178, with a similarity of 99.82%.
 - c. The phylogenetic tree shows that *F. compactum* is located between *F. venenatum* and *F. strain flavolapis*, a position closer to *F. venenatum*, which implies that *F. compactum* is much closer to *F. venenatum* than *F. strain flavolapis*, in terms of genetic similarity.
2. The subject of this GRAS determination is manufactured in compliance with cGMP using processing aids and food contact substances that conform to the conditions of use specified in Title 21 of the U.S. Code of Federal Regulations and/or Food Chemicals Codex (FCC) specifications.
3. Process procedures and product specifications are in place to control pivotal quality attributes that ensure a consistent, safe, food-grade finished ingredient.

4. Mycotoxins that were identified by both genetic sequencing and analytical testing are beauvericin and enniatins. These mycotoxins are monitored in every batch of the final product and maximum specifications have been established.
5. Other commonly occurring food mycotoxins are non-detectable and genetic analysis indicates that these mycotoxins are not expected to be produced by *F. compactum*, and hence, there is reasonable certainty of no harm from these mycotoxins.
6. *F. compactum*, *F. strain flavolapis*, and *F. venenatum* have the same toxicological potential or lack thereof, when grown under controlled conditions, and the read across to pivotal data cited in GRN 000091 for *F. compactum* offers sound scientific support for MoreMeat's conclusion that the estimated consumption of MoreMeat's mycoprotein is safe to a reasonable certainty.
7. MoreMeat conducted toxicology studies and demonstrated that mycoprotein derived from *F. compactum* is not genotoxic based on results of an unpublished Ames test, an *in vivo* mammalian erythrocyte micronucleus test, and an *in vitro* mammalian chromosome aberration test. Additionally, MoreMeat's mycoprotein does not result in any adverse effects based on results of an unpublished 14-day study.
8. The published toxicology studies conducted by the notifiers of GRN 000091 and GRN 000904 demonstrate that the mycoprotein derived from *Fusarium spp.* is not genotoxic and does not result in adverse effects at high intake levels.
9. Numerous toxicology studies of mycoprotein derived from *Fusarium spp.* were conducted, including: acute and chronic toxicity; genotoxicity studies; and developmental and reproductive toxicity studies. In these reports, no evidence of toxicity was noted; therefore, mycoprotein is concluded to be non-genotoxic and non-toxicogenic.
10. Mycoproteins derived from *Fusarium spp.* are GRAS for use in conventional foods. The intended uses of MoreMeat's mycoprotein derived from *F. compactum* is substantially equivalent to mycoprotein derived from *F. strain flavolapis* and *F. venenatum* as described in GRN 000904 (page 20) and GRN 000945 (page 22-25).
11. Among the total population (ages 2 years and older) and on a body weight basis, the mean and 90th percentile intakes of MoreMeat mycoprotein were determined to be 1.5 g/kg bw/day and 3.4 g/kg bw/day, respectively, based on the proposed conditions of use.
12. The quality of protein present in MoreMeat's mycoprotein was assessed, and the PDCAAS results indicate that MoreMeat's mycoprotein is a source of quality protein.
13. MoreMeat's mycoprotein is unlikely to present an elevated risk for food allergy based on history of use, a literature review, and the lack of high identity matches to abundant and potent allergens. MoreMeat also confirms that it has concluded that food containing mycoprotein produced from *Fusarium compactum* is unlikely to represent any additional risk of food allergy beyond what is currently available in the human food supply.

The extensive and thorough information and data described above support the conclusion that MoreMeat's mycoprotein derived from *Fusarium compactum* is generally recognized as safe for use

as a food ingredient and a source of macronutrients intended for use as meat, dairy, and flour alternatives. No evidence in the available information on mycoprotein derived from *Fusarium compactum* demonstrates or suggests reasonable grounds to suspect a hazard to the public when it is used at levels that might reasonably be expected from the proposed uses in selected conventional foods.

Part 7 LIST OF SUPPORTING DATA AND INFORMATION IN THE GRAS NOTICE

7.1 List of Acronyms and References

7.1.1 List of Acronyms

ALT	alanine aminotransferase
ASP	aspartate aminotransferase
ATP	adenosine triphosphate
BAM	Bacteriological Analytical Manual
BUN	blood urea nitrogen
bw	body weight
°C	degrees Celsius
CAS	Chemical Abstracts Service
CDP	cytidine diphosphate
CFR	Code of Federal Regulations
CFU	colony forming unit
cGMP	current good manufacturing practices
DNA	deoxyribonucleic acid
ECG	electrocardiogram
EDI	estimated daily intake
EEG	electroencephalogram
EFSA	European Food Safety Authority
FAERS	FDA Adverse Events Reporting System
FCC	Food Chemicals Codex
FDA	U.S. Food and Drug Administration
FFDCA	Federal Food, Drug, and Cosmetic Act
FR	Federal Register
FSANZ	The Food Standards Agency of Australia and New Zealand
FTC	U.S. Federal Trade Commission
F.W.	formula weight
g	gram
GRAS	generally recognized as safe
GRN	GRAS notice
HDT	highest dose tested
HED	human equivalent dose
JECFA	Joint WHO/FAO Expert Committee on Food Additives
kcal	kilocalorie
kg	kilogram
mg	milligram
mL	milliliter
NDI	new dietary ingredient

NHP	Natural Health Products
NLEA	Nutrition Labeling and Education Act
NOAEL	no observed adverse effect level
OECD	Organization for Economic Cooperation and Development
ppm	parts per million
RCT	randomized clinical trial
SBP	systolic blood pressure
USDA	U.S. Department of Agriculture
WBC	white blood cell
WHO/FAO	World Health Organization/Food and Agriculture Organization

7.1.2 List of References

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Anchor Center for Certification (ACC)

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7.1.3 List of Appendix

Appendix 1 Food Safety System Certification FSSC 22000 Certificate

Appendix 2 Representative Food Codes for Proposed Food Uses of MoreMeat Mycoprotein in the U.S.

Appendix 3 Dr. Goodman's Evaluation on MoreMeat's Mycoprotein

Appendix 4 PDCAAS Report for MoreMeat's Mycoprotein

Appendix 5 Expert Panel Statement for MoreMeat's Mycoprotein

DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved:OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER	DATE OF RECEIPT
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3. Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): _____	
4. For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person luqin yin		Position or Title Regulatory Affairs Director	
	Organization (<i>if applicable</i>) MoreMeat (Guangzhou) Biotech Co.,Ltd			
	Mailing Address (<i>number and street</i>) Room 101, Building C1, No. 11, Kaiyuan Avenue, Huangpu District, Guangzhou City, Guangdong Province, China			
City Guangzhou		State or Province Guangdong	Zip Code/Postal Code 510000	Country China
Telephone Number 020-82000063		Fax Number	E-Mail Address info-reg@moremeat.com	
1b. Agent or Attorney (if applicable)	Name of Contact Person Stella Si		Position or Title Managing Director	
	Organization (<i>if applicable</i>) Anchor Center for Certification			
	Mailing Address (<i>number and street</i>) No. 1295 Chuan Qiao Road, Building 2, Suite 302			
City Shanghai		State or Province Shanghai	Zip Code/Postal Code 201206	Country China
Telephone Number 02150320118		Fax Number	E-Mail Address stella.si@acclab.cn	

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Mycoprotein

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway

☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)*

☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))*

☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8))

☐ Yes *(Proceed to Item 8)*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

MoreMeat's mycoprotein is intended for use in processed food products (excluding infant formulas and meat and poultry products) to increase the level of protein in a product, serving as a substitute for other animal or vegetable protein.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☒ Yes ☐ No

Did you include this other information in the list of attachments?

☒ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that MoreMeat (Guangzhou) Biotech Co., Ltd

(name of notifier)

has concluded that the intended use(s) of Mycoprotein

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. MoreMeat (Guangzhou) Biotech Co., Ltd agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

Room 101, Building C1, No. 11, Kaiyuan Avenue, Huangpu District, Guangzhou City, Guangdong Province, China

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Luqin Yin

数字签名者: Luqin Yin
日期: 2025.02.24 16:56:04 +08'00'

Printed Name and Title

Luqin Yin, Regulatory Affairs Director

Date (mm/dd/yyyy)

02/23/2025

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	MoreMeatGRASDossier-V29FDAGRAS-signed.pdf	Submission
	Appendix1FSSC22000Certificate.pdf	Submission
	Appendix2RepresentativeFoodCodesforProposedFoodUsesofMoreMeatMycoproteininthelU.S..pdf	Submission
	Appendix3Dr.GoodmansEvaluationonMoreMeatsMycoprotein.pdf	Submission
	Appendix4PDCAASReportforMoreMeatsMycoprotein.pdf	Submission
	Appendix5ExpertPanelStatementforMoreMeatsMycoprotein.pdf	Submission

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.

Certificate CN24/00006363

The Food Safety Management System of

MoreMeat (Guangzhou) Biotech Co.,Ltd

COID: CHN-1-0587-046482

Room 101, Building C1, No. 11, Kaiyuan Avenue, Huangpu District, Guangzhou City, Guangdong Province, China

has been assessed and determined to comply with the requirements of

Food Safety System Certification FSSC 22000

Certification scheme for food safety management systems consisting of the following elements: ISO 22000:2018, ISO/TS22002-1:2009 and Additional FSSC 22000 requirements (version 6).

This certificate is applicable for the scope of

Production of Mycoprotein for food processing for export, the main process including fermentation and sterilization

EXCLUSIONS: N/A

Food Category: CIV - Processing of ambient stable products

This certificate is valid from 30 September 2024 until 29 September 2027 and remains valid subject to satisfactory surveillance audits.

Date of Certification decision 30 September 2024

Issue Date: 11 October 2024

Issue 1. Certified since 30 September 2024

Date of the last unannounced audit: N/A

At least one (1) surveillance audit is required to be undertaken unannounced after the initial certification audit and within each three (3) year period thereafter.

Authorised by

Liz Moran

Business Manager

SGS United Kingdom Ltd

Rossmore Business Park, Ellesmere Port, Cheshire, CH65 3EN, UK

t +44 (0)151 350-6666 - www.sgs.com



The authenticity of this certificate can be verified in the FSSC 22000 database of certified organizations available on www.fssc.com.



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Appendix 3

Representative Food Codes for Proposed Food Uses of MoreMeat Mycoprotein in the U.S.

Baked Goods and Baking Mixes

[MoreMeat Mycoprotein] = 50%

51000100	Bread, NS as to major flour
51000110	Bread, NS as to major flour, toasted
51000180	Bread, made from home recipe or purchased at a bakery, NS as to major flour
51000190	Bread, made from home recipe or purchased at a bakery, toasted, NS as to major flour
51000200	Roll, NS as to major flour
51000300	Roll, hard, NS as to major flour
51000400	Roll, bran, NS as to type of bran
51101000	Bread, white
51101010	Bread, white, toasted
51101050	Bread, white, made from home recipe or purchased at a bakery
51101060	Bread, white, made from home recipe or purchased at a bakery, toasted
51102010	Bread, white with whole wheat swirl
51102020	Bread, white with whole wheat swirl, toasted
51105010	Bread, Cuban
51105040	Bread, Cuban, toasted
51106010	Bread, native, water, Puerto Rican style
51106020	Bread, native, water, toasted, Puerto Rican style
51106200	Bread, lard, Puerto Rican style
51106210	Bread, lard, toasted, Puerto Rican style
51107010	Bread, French or Vienna
51107040	Bread, French or Vienna, toasted
51108010	Focaccia, Italian, plain
51108100	Bread, naan
51109010	Bread, Italian, Grecian, Armenian
51109040	Bread, Italian, Grecian, Armenian, toasted
51109100	Bread, pita
51109150	Bread, pita with fruit
51111010	Bread, cheese
51111040	Bread, cheese, toasted
51113010	Bread, cinnamon
51113100	Bread, cinnamon, toasted
51115010	Bread, cornmeal and molasses
51115020	Bread, cornmeal and molasses, toasted

51119010	Bread, egg, Challah
51119040	Bread, egg, Challah, toasted
51121015	Garlic bread, NFS
51121025	Garlic bread, from fast food / restaurant
51121035	Garlic bread, from frozen
51121045	Garlic bread, with parmesan cheese, from fast food / restaurant
51121055	Garlic bread, with parmesan cheese, from frozen
51121065	Garlic bread, with melted cheese, from fast food / restaurant
51121075	Garlic bread, with melted cheese, from frozen
51121110	Bread, onion
51121120	Bread, onion, toasted
51122000	Bread, reduced calorie and/or high fiber, white or NFS
51122010	Bread, reduced calorie and/or high fiber, white or NFS, toasted
51122100	Bread, reduced calorie and/or high fiber, white or NFS, with fruit and/or nuts
51122110	Bread, reduced calorie and/or high fiber, white or NFS, with fruit and/or nuts, toasted
51122300	Bread, white, special formula, added fiber
51122310	Bread, white, special formula, added fiber, toasted
51123020	Bread, high protein, toasted
51127010	Bread, potato
51127020	Bread, potato, toasted
51129010	Bread, raisin
51129020	Bread, raisin, toasted
51133010	Bread, sour dough
51133020	Bread, sour dough, toasted
51134000	Bread, sweet potato
51134010	Bread, sweet potato, toasted
51135000	Bread, vegetable
51135010	Bread, vegetable, toasted
51136000	Bruschetta
51140100	Bread, dough, fried
51150000	Roll, white, soft
51153000	Roll, white, hard
51154010	Roll, white, hot dog bun
51154100	Roll, white, hamburger bun
51154510	Roll, diet
51154550	Roll, egg bread
51154600	Roll, cheese
51155000	Roll, French or Vienna

51156500	Roll, garlic
51157000	Roll, white, hoagie, submarine
51158100	Roll, Mexican, bolillo
51159000	Roll, sour dough
51160000	Roll, sweet, no frosting
51160100	Roll, sweet, cinnamon bun, no frosting
51160110	Roll, sweet, cinnamon bun, frosted
51160200	Pan dulce, NFS
51161000	Pan Dulce, with fruit, no frosting
51161020	Roll, sweet, with fruit, frosted
51161030	Roll, sweet, with fruit, frosted, diet
51161050	Roll, sweet, frosted
51161250	Pan Dulce, no topping
51161270	Pan Dulce, with sugar topping
51161280	Pan Dulce, with raisins and icing
51165000	Coffee cake, yeast type
51166000	Croissant
51166100	Croissant, cheese
51166200	Croissant, chocolate
51166500	Croissant, fruit
51167000	Brioche
51168000	Bread, Spanish coffee
51180010	Bagel
51180030	Bagel, with raisins
51180080	Bagel, with fruit other than raisins
51182010	Bread stuffing
51182020	Bread stuffing made with egg
51183990	Breadsticks, NFS
51184000	Breadsticks, hard, NFS
51184100	Breadsticks, hard, reduced sodium
51184200	Breadsticks, soft, NFS
51184210	Breadsticks, soft, fast food / restaurant
51184220	Breadsticks, soft, frozen
51184230	Breadsticks, soft, with parmesan cheese, fast food / restaurant
51184260	Breadsticks, soft, stuffed or topped with melted cheese
51185000	Croutons
51186010	Muffin, English
51186100	Muffin, English, with raisins

51186130	Muffin, English, cheese
51186160	Muffin, English, with fruit other than raisins
51187000	Melba toast
51187020	Anisette toast
51188100	Pannetone
51188500	Zwieback toast
51300050	Bread, whole grain white
51300060	Bread, whole grain white, toasted
51300100	Bagel, whole grain white
51300110	Bread, whole wheat
51300120	Bread, whole wheat, toasted
51300140	Bread, whole wheat, made from home recipe or purchased at bakery
51300150	Bread, whole wheat, made from home recipe or purchased at bakery, toasted
51300175	Bread, chappatti or roti
51300180	Bread, puri
51300185	Bread, paratha
51300210	Bread, whole wheat, with raisins
51300220	Bread, whole wheat, with raisins, toasted
51300300	Bread, sprouted wheat
51300310	Bread, sprouted wheat, toasted
51301010	Bread, wheat or cracked wheat
51301020	Bread, wheat or cracked wheat, toasted
51301040	Bread, wheat or cracked wheat, made from home recipe or purchased at bakery
51301050	Bread, wheat or cracked wheat, made from home recipe or purchased at bakery, toasted
51301120	Bread, wheat or cracked wheat, with raisins
51301130	Bread, wheat or cracked wheat, with raisins, toasted
51301510	Bread, wheat or cracked wheat, reduced calorie and/or high fiber
51301520	Bread, wheat or cracked wheat, reduced calorie and/or high fiber, toasted
51301540	Bread, French or Vienna, whole wheat
51301550	Bread, French or Vienna, whole wheat, toasted
51301600	Bread, pita, whole wheat
51301620	Bread, pita, wheat or cracked wheat
51301700	Bagel, wheat
51301750	Bagel, whole wheat
51301800	Bagel, wheat, with raisins
51301805	Bagel, whole wheat, with raisins
51301820	Bagel, wheat, with fruit and nuts
51301900	Bagel, wheat bran

51302500	Muffin, English, wheat bran
51302520	Muffin, English, wheat bran, with raisins
51303010	Muffin, English, wheat or cracked wheat
51303030	Muffin, English, whole wheat
51303050	Muffin, English, wheat or cracked wheat, with raisins
51303070	Muffin, English, whole wheat, with raisins
51303100	Muffin, English, whole grain white
51306000	Breadsticks, hard, whole wheat
51320010	Roll, wheat or cracked wheat
51320060	Roll, wheat or cracked wheat, hot dog bun
51320070	Roll, wheat or cracked wheat, hamburger bun
51320500	Roll, whole wheat
51320550	Roll, whole wheat, hot dog bun
51320560	Roll, whole wheat, hamburger bun
51320700	Roll, whole grain white
51320710	Roll, whole grain white, hot dog bun
51320720	Roll, whole grain white, hamburger bun
51401010	Bread, rye
51401020	Bread, rye, toasted
51401030	Bread, marble rye and pumpernickel
51401040	Bread, marble rye and pumpernickel, toasted
51401200	Muffin, English, rye
51404010	Bread, pumpernickel
51404020	Bread, pumpernickel, toasted
51404500	Bagel, pumpernickel
51404550	Muffin, English, pumpernickel
51407010	Bread, black
51407020	Bread, black, toasted
51420000	Roll, rye
51421000	Roll, pumpernickel
51501010	Bread, oatmeal
51501020	Bread, oatmeal, toasted
51501040	Bread, oat bran
51501050	Bread, oat bran, toasted
51501080	Bagel, oat bran
51502010	Roll, oatmeal
51503000	Muffin, English, oat bran
51503040	Muffin, English, oat bran, with raisins

51601010	Bread, multigrain, toasted
51601020	Bread, multigrain
51601210	Bread, multigrain, with raisins
51601220	Bread, multigrain, with raisins, toasted
51602010	Bread, multigrain, reduced calorie and/or high fiber
51602020	Bread, multigrain, reduced calorie and/or high fiber, toasted
51620000	Roll, multigrain
51620020	Roll, multigrain, hot dog bun
51620030	Roll, multigrain, hamburger bun
51630000	Bagel, multigrain
51630100	Bagel, multigrain, with raisins
51630200	Muffin, English, multigrain
51801010	Bread, barley
51801020	Bread, barley, toasted
51804010	Bread, soy
51804020	Bread, soy, toasted
51805010	Bread, sunflower meal
51805020	Bread, sunflower meal, toasted
51806010	Bread, rice
51806020	Bread, rice, toasted
51807000	Injera, Ethiopian bread
51808000	Bread, gluten free
51808010	Bread, gluten free, toasted
51808050	Breadsticks, hard, gluten free
51808100	Roll, gluten free
52101000	Biscuit, NFS
52102040	Biscuit, from refrigerated dough
52103000	Biscuit, from fast food / restaurant
52104010	Biscuit, home recipe
52104040	Biscuit, wheat
52104100	Biscuit, cheese
52104200	Biscuit with fruit
52105100	Scone
52105200	Scone, with fruit
52201000	Cornbread, prepared from mix
52202060	Cornbread, made from home recipe
52204000	Cornbread stuffing
52206010	Cornbread muffin, stick, round

52206060	Cornbread muffin, stick, round, made from home recipe
52207010	Corn flour patty or tart, fried
52208010	Corn pone, baked
52208020	Corn pone, fried
52208760	Gordita/sope shell, plain, no filling
52209010	Hush puppy
52211010	Johnnycake
52213010	Spoonbread
52215000	Tortilla, NFS
52215100	Tortilla, corn
52215200	Tortilla, flour
52215260	Tortilla, whole wheat
52215300	Taco shell, corn
52215350	Taco shell, flour
52220110	Arepa Dominicana
52301000	Muffin, NFS
52302010	Muffin, fruit
52302020	Muffin, fruit, low fat
52302500	Muffin, chocolate chip
52302600	Muffin, chocolate
52303010	Muffin, whole wheat
52303500	Muffin, wheat
52304000	Muffin, whole grain
52304010	Muffin, wheat bran
52304040	Muffin, bran with fruit, lowfat
52304100	Muffin, oatmeal
52304150	Muffin, oat bran
52306010	Muffin, plain
52306300	Muffin, cheese
52306500	Muffin, pumpkin
52306550	Muffin, zucchini
52306700	Muffin, carrot
52311010	Popover
52401000	Bread, Boston Brown
52403000	Bread, nut
52404060	Bread, pumpkin
52405010	Bread, fruit
52407000	Bread, zucchini

52408000	Bread, Irish soda
53100100	Cake or cupcake, NFS
53101100	Cake, angel food
53102200	Cake or cupcake, apple
53102700	Cake or cupcake, banana
53102800	Cake or cupcake, Black Forest
53103000	Cake, Boston cream pie
53104260	Cake or cupcake, carrot
53104400	Cake or cupcake, coconut
53104500	Cheesecake, plain
53104550	Cheesecake, fruit
53104600	Cheesecake, chocolate
53105262	Cake or cupcake, chocolate with white icing, bakery
53105264	Cake or cupcake, chocolate with white icing, from mix
53105270	Cake or cupcake, chocolate with chocolate icing, bakery
53105272	Cake or cupcake, chocolate with chocolate icing, from mix
53105275	Cake or cupcake, chocolate, no icing
53105300	Cake or cupcake, German chocolate
53105310	Cake or cupcake, gluten free
53105396	Cake, chocolate, flourless
53106500	Cake, cream
53108200	Snack cake, chocolate
53109200	Snack cake, white
53110000	Cake, fruit cake
53111000	Cake or cupcake, gingerbread
53113000	Cake, jelly roll
53114100	Cake or cupcake, lemon
53115200	Cake or cupcake, marble
53115450	Cake or cupcake, peanut butter
53116000	Cake, pound
53116510	Cake or cupcake, pumpkin
53116520	Cake or cupcake, red velvet
53117200	Cake or cupcake, spice
53118100	Cake, sponge
53118110	Cake or cupcake, strawberry
53118500	Cake, torte
53118550	Cake, tres leche
53119000	Cake, pineapple, upside down

53120270	Cake or cupcake, white with white icing, bakery
53120272	Cake or cupcake, white with white icing, from mix
53121270	Cake or cupcake, white with chocolate icing, bakery
53121272	Cake or cupcake, white with chocolate icing, from mix
53121275	Cake or cupcake, white, no icing
53123070	Cake, strawberry shortcake
53124110	Cake or cupcake, zucchini
53200100	Cookie, batter or dough, raw
53201000	Cookie, NFS
53202000	Cookie, almond
53203000	Cookie, applesauce
53203500	Cookie, biscotti
53204000	Cookie, brownie, NS as to icing
53204010	Cookie, brownie, without icing
53204100	Cookie, brownie, with icing or filling
53204840	Cookie, brownie, reduced fat, NS as to icing
53205250	Cookie, butterscotch, brownie
53205260	Cookie, bar, with chocolate
53206000	Cookie, chocolate chip
53206020	Cookie, chocolate chip, made from home recipe or purchased at a bakery
53206030	Cookie, chocolate chip, reduced fat
53206100	Cookie, chocolate chip sandwich
53206500	Cookie, chocolate, made with rice cereal
53206550	Cookie, chocolate, made with oatmeal and coconut, no bake
53207000	Cookie, chocolate or fudge
53208000	Cookie, marshmallow, chocolate-covered
53208200	Cookie, marshmallow pie, chocolate covered
53209005	Cookie, chocolate, with icing or coating
53209010	Cookie, sugar wafer, chocolate-covered
53209015	Cookie, chocolate sandwich
53209020	Cookie, chocolate sandwich, reduced fat
53209100	Cookie, chocolate, sandwich, with extra filling
53209500	Cookie, chocolate and vanilla sandwich
53210000	Cookie, chocolate wafer
53210900	Cookie, graham cracker with chocolate and marshmallow
53211000	Cookie bar, with chocolate, nuts, and graham crackers
53215500	Cookie, coconut
53220000	Cookie, fruit-filled bar

53220030	Cookie, fig bar
53222010	Cookie, fortune
53222020	Cookie, cone shell, ice cream type, wafer or cake
53223000	Cookie, gingersnaps
53223100	Cookie, granola
53224000	Cookie, ladyfinger
53224250	Cookie, lemon bar
53225000	Cookie, macaroon
53226000	Cookie, marshmallow, with coconut
53226500	Cookie, marshmallow, with rice cereal, no bake
53226550	Cookie, marshmallow, with rice cereal and chocolate chips
53226600	Cookie, marshmallow and peanut butter, with oat cereal, no bake
53228000	Cookie, meringue
53230000	Cookie, molasses
53231000	Cookie, Lebkuchen
53231400	Cookie, multigrain, high fiber
53233000	Cookie, oatmeal
53233010	Cookie, oatmeal, with raisins
53233040	Cookie, oatmeal, reduced fat, NS as to raisins
53233050	Cookie, oatmeal sandwich, with creme filling
53233060	Cookie, oatmeal, with chocolate chips
53233080	Cookie, oatmeal sandwich, with peanut butter and jelly filling
53233100	Cookie, oatmeal, with chocolate and peanut butter, no bake
53234000	Cookie, peanut butter
53234100	Cookie, peanut butter, with chocolate
53234250	Cookie, peanut butter with rice cereal, no bake
53235000	Cookie, peanut butter sandwich
53235500	Cookie, with peanut butter filling, chocolate-coated
53235600	Cookie, Pfeffernusse
53236000	Cookie, Pizzelle
53236100	Cookie, pumpkin
53237000	Cookie, raisin
53237010	Cookie, raisin sandwich, cream-filled
53237500	Cookie, rum ball, no bake
53238000	Cookie, sandwich-type, not chocolate or vanilla
53239000	Cookie, shortbread
53239010	Cookie, shortbread, reduced fat
53239050	Cookie, shortbread, with icing or filling

53239100	Pocky
53240000	Cookie, animal
53240010	Cookie, animal, with frosting or icing
53241500	Cookie, butter or sugar
53241510	Marie biscuit
53241600	Cookie, butter or sugar, with fruit and/or nuts
53242000	Cookie, sugar wafer
53242500	Cookie, toffee bar
53243000	Cookie, vanilla sandwich
53243010	Cookie, vanilla sandwich, extra filling
53243050	Cookie, vanilla sandwich, reduced fat
53244010	Cookie, butter or sugar, with chocolate icing or filling
53244020	Cookie, butter or sugar, with icing or filling other than chocolate
53246000	Cookie, tea, Japanese
53247000	Cookie, vanilla wafer
53247050	Cookie, vanilla wafer, reduced fat
53247500	Cookie, vanilla with caramel, coconut, and chocolate coating
53251100	Cookie, rugelach
53260030	Cookie, chocolate chip, reduced sugar
53260200	Cookie, oatmeal, reduced sugar
53260300	Cookie, sandwich, reduced sugar
53260400	Cookie, sugar or plain, sugar free
53260500	Cookie, sugar wafer, sugar free
53260600	Cookie, peanut butter, sugar free
53261000	Cookie, gluten free
53270100	Cookies, Puerto Rican style
53300100	Pie, NFS
53300200	Tart, all types
53301000	Pie, apple
53301070	Pie, apple, fast food
53303500	Pie, berry
53304000	Pie, blueberry
53305000	Pie, cherry
53305700	Pie, lemon
53305800	Pie, key lime
53307000	Pie, peach
53312000	Pie, strawberry
53341000	Pie, banana cream

53342000	Pie, chocolate cream
53343000	Pie, coconut cream
53344000	Pie, custard
53344300	Dessert pizza
53346000	Pie, peanut butter cream
53346500	Pie, pineapple cream
53347000	Pie, pumpkin
53360000	Pie, sweet potato
53365000	Pie, vanilla cream
53381000	Pie, lemon meringue
53385000	Pie, pecan
53385500	Pie, oatmeal
53391000	Pie shell
53391100	Pie shell, graham cracker
53410100	Cobbler, apple
53410300	Cobbler, berry
53410500	Cobbler, cherry
53410800	Cobbler, peach
53415100	Crisp, apple
53415110	Fritter, plain
53415120	Fritter, fruit
53415300	Crisp, berry
53415400	Crisp, cherry
53415500	Crisp, peach
53420000	Cream puff, eclair, custard or cream filled, NS as to icing
53420100	Cream puff, eclair, custard or cream filled, not iced
53420200	Cream puff, eclair, custard or cream filled, iced
53420210	Cream puff, eclair, custard or cream filled, iced, reduced fat
53420250	Cream puff, no filling or icing
53420400	Sopaipilla, without syrup or honey
53420410	Sopaipilla with syrup or honey
53430100	Crepe, chocolate filled
53430200	Crepe, fruit filled
53430700	Tamale, sweet
53440000	Strudel, apple
53440300	Strudel, berry
53440500	Strudel, cherry
53440600	Strudel, cheese

53440700	Strudel, peach
53441110	Baklava
53441210	Basbousa
53450000	Turnover, fruit
53452100	Pastry, fruit-filled
53452120	Pastry, made with bean or lotus seed paste filling, baked
53452130	Pastry, made with bean paste and salted egg yolk filling, baked
53452150	Pastry, Chinese, made with rice flour
53452170	Pastry, cookie type, fried
53452200	Pastry, Italian, with cheese
53452400	Pastry, puff
53452420	Pastry, puff, custard or cream filled, iced or not iced
53452450	Cheese pastry puffs
53452500	Pastry, mainly flour and water, fried
53453150	Empanada, fruit
53500100	Breakfast pastry, NFS
53510000	Danish pastry, plain or spice
53510100	Danish pastry, with fruit
53511000	Danish pastry, with cheese
53520000	Doughnut, NFS
53520100	Doughnut, cake type, plain
53520120	Doughnut, chocolate
53520130	Doughnut, cake type, powdered sugar
53520135	Doughnut, cake type, with icing
53520140	Doughnut, cake type, chocolate icing
53520160	Doughnut, chocolate, with chocolate icing
53520170	Doughnut holes
53520200	Churros
53520510	Beignet
53521110	Doughnut, yeast type
53521130	Doughnut, yeast type, with chocolate icing
53521140	Doughnut, jelly
53521210	Doughnut, custard-filled
53521230	Doughnut, custard-filled, with icing
53530000	Breakfast tart
53530010	Breakfast tart, lowfat
53610100	Coffee cake, crumb or quick-bread type
53610170	Coffee cake, crumb or quick-bread type, with fruit

53610200	Coffee cake, crumb or quick-bread type, cheese-filled
53710400	Cereal or granola bar (General Mills Fiber One Chewy Bar)
53710500	Cereal or granola bar (Kellogg's Nutri-Grain Cereal Bar)
53710502	Cereal or granola bar (Kellogg's Nutri-Grain Yogurt Bar)
53710504	Cereal or granola bar (Kellogg's Nutri-Grain Fruit and Nut Bar)
53710600	Milk 'n Cereal bar
53710700	Cereal or granola bar (Kellogg's Special K bar)
53710800	Cereal or granola bar (Kashi Chewy)
53710802	Cereal or granola bar (Kashi Crunchy)
53710810	Cereal or granola bar (KIND Fruit and Nut Bar)
53710900	Cereal or granola bar (General Mills Nature Valley Chewy Trail Mix)
53710902	Cereal or granola bar, with yogurt coating (General Mills Nature Valley Chewy Granola Bar)
53710904	Cereal or granola bar (General Mills Nature Valley Sweet and Salty Granola Bar)
53710906	Cereal or granola bar (General Mills Nature Valley Crunchy Granola Bar)
53711000	Cereal or granola bar (Quaker Chewy Granola Bar)
53711002	Cereal or granola bar (Quaker Chewy 90 Calorie Granola Bar)
53711004	Cereal or granola bar (Quaker Chewy 25% Less Sugar Granola Bar)
53711006	Cereal or granola bar (Quaker Chewy Dipp's Granola Bar)
53711100	Cereal or granola bar (Quaker Granola Bites)
53712000	Snack bar, oatmeal
53712100	Cereal or Granola bar, NFS
53712200	Cereal or granola bar, lowfat, NFS
53712210	Cereal or granola bar, nonfat
53713000	Cereal or granola bar, reduced sugar, NFS
53713010	Cereal or granola bar, fruit and nut
53713100	Cereal or granola bar, peanuts , oats, sugar, wheat germ
53714200	Cereal or granola bar, chocolate coated, NFS
53714210	Cereal or granola bar, with coconut, chocolate coated
53714220	Cereal or granola bar with nuts, chocolate coated
53714230	Cereal or granola bar, oats, nuts, coated with non-chocolate coating
53714250	Cereal or granola bar, coated with non-chocolate coating
53714300	Cereal or granola bar, high fiber, coated with non-chocolate yogurt coating
53714400	Cereal or granola bar, with rice cereal
53714500	Breakfast bar, NFS
53714510	Breakfast bar, date, with yogurt coating
53714520	Breakfast bar, cereal crust with fruit filling, lowfat
53720100	Nutrition bar (Balance Original Bar)
53720200	Nutrition bar (Clif Bar)

53720210	Nutrition bar (Clif Kids Organic Zbar)
53720300	Nutrition bar (PowerBar)
53720400	Nutrition bar (Slim Fast Original Meal Bar)
53720500	Nutrition bar (Snickers Marathon Protein Bar)
53720600	Nutrition bar (South Beach Living Meal Bar)
53720610	Nutrition bar (South Beach Living High Protein Bar)
53720700	Nutrition bar (Tiger's Milk)
53720800	Nutrition bar (Zone Perfect Classic Crunch)
53729000	Nutrition bar or meal replacement bar, NFS
54001000	Crackers, NFS
54102010	Graham crackers
54102015	Graham crackers (Teddy Grahams)
54102020	Graham crackers, chocolate covered
54102050	Crackers, oatmeal
54102060	Crackers, Cuban
54102100	Graham crackers, reduced fat
54102200	Graham crackers, sandwich, with filling
54103000	Crackers, breakfast biscuit
54200100	Crackers, butter, reduced sodium
54201010	Crackers, matzo, reduced sodium
54202020	Crackers, saltine, reduced sodium
54204020	Crackers, wheat, reduced sodium
54204030	Crackers, woven wheat, reduced sodium
54301010	Crackers, butter, plain
54301020	Crackers, butter, flavored
54301030	Crackers, butter (Ritz)
54301100	Crackers, butter, reduced fat
54304000	Crackers, cheese
54304005	Crackers, cheese (Cheez-It)
54304020	Crackers, cheese (Goldfish)
54304100	Crackers, cheese, reduced fat
54304110	Crackers, cheese, reduced sodium
54304150	Crackers, cheese, whole grain
54305010	Crackers, crispbread
54305020	Crackers, flatbread
54307000	Crackers, matzo
54308000	Crackers, milk
54313000	Crackers, oyster

54318000	Chips, rice
54318500	Rice cake
54319000	Crackers, rice
54319005	Crackers, rice and nuts
54319020	Popcorn cake
54319500	Rice paper
54325000	Crackers, saltine
54325010	Crackers, saltine, reduced fat
54325060	Crackers, saltine, multigrain
54326000	Crackers, multigrain
54328000	Crackers, sandwich
54328100	Crackers, sandwich, peanut butter filled
54328105	Crackers, sandwich, peanut butter filled (Ritz)
54328110	Crackers, sandwich, reduced fat, peanut butter filled
54328120	Crackers, whole grain, sandwich, peanut butter filled
54328200	Crackers, sandwich, cheese filled
54328210	Crackers, sandwich, cheese filled (Ritz)
54336000	Crackers, water
54336100	Crackers, wonton
54337010	Crackers, woven wheat
54337020	Crackers, woven wheat, plain (Triscuit)
54337030	Crackers, woven wheat, flavored (Triscuit)
54337060	Crackers, woven wheat, reduced fat
54338000	Crackers, wheat
54338010	Crackers, wheat, plain (Wheat Thins)
54338020	Crackers, wheat, flavored (Wheat Thins)
54338100	Crackers, wheat, reduced fat
54340100	Crackers, gluten free, plain
54340110	Crackers, gluten free, flavored
54402700	Pita chips
54440010	Bagel chips
54440020	Cracker chips
55100005	Pancakes, NFS
55100010	Pancakes, plain, frozen
55100020	Pancakes, fruit, frozen
55100025	Pancakes, chocolate, frozen
55100030	Pancakes, whole grain, frozen
55100050	Pancakes, plain, fast food / restaurant

55100055	Pancakes, fruit, fast food / restaurant
55100060	Pancakes, chocolate, fast food / restaurant
55100065	Pancakes, whole grain, fast food / restaurant
55100080	Pancakes, school
55101000	Pancakes, plain
55101015	Pancakes, plain, reduced fat
55103000	Pancakes, fruit
55103020	Pancakes, pumpkin
55103100	Pancakes, chocolate
55105200	Pancakes, whole grain
55105205	Pancakes, whole grain, reduced fat
55106000	Pancakes, gluten free
55200010	Waffle, NFS
55200020	Waffle, plain, frozen
55200030	Waffle, plain, reduced fat
55200040	Waffle, fruit, frozen
55200050	Waffle, chocolate, frozen
55200060	Waffle, whole grain, frozen
55200070	Waffle, whole grain, reduced fat
55200080	Waffle, whole grain, fruit, frozen
55200100	Waffle, plain, fast food / restaurant
55200110	Waffle, chocolate, fast food / restaurant
55200120	Waffle, fruit, fast food / restaurant
55200130	Waffle, whole grain, fast food / restaurant
55200200	Waffle, school
55201000	Waffle, plain
55203000	Waffle, fruit
55203600	Waffle, chocolate
55203700	Waffle, cinnamon
55205000	Waffle, whole grain
55208000	Waffle, gluten free
55300010	French toast, NFS
55300020	French toast, frozen
55300050	French toast, fast food / restaurant
55300060	French toast, school
55301000	French toast, plain
55301015	French toast, whole grain
55301025	French toast, gluten free

55301031	French toast sticks
55301040	French toast sticks, fast food / restaurant
55301048	French toast sticks, school
55400010	Crepe, NFS
55401000	Crepe, plain
55501000	Chinese pancake
55610300	Dumpling, no meat
55702100	Dosa, plain
55801000	Funnel cake with sugar
55801010	Funnel cake with sugar and fruit
58124210	Pastry, cheese-filled

Dairy product analogs

Non-dairy milk, creamer, cheese, and milk-based meal replacements
[MoreMeat Mycoprotein] = 50%

11300100	Non-dairy milk, NFS
11320000	Soy milk, sweetened
11320100	Soy milk, unsweetened
11321000	Soy milk, chocolate
11350000	Almond milk, sweetened
11350010	Almond milk, chocolate
11350020	Almond milk, unsweetened
11350040	Almond milk, NFS
11360000	Rice milk
11360200	Oat milk
11370000	Coconut milk
11513385	Chocolate milk, Nesquik, made with non-dairy milk
11514150	Hot chocolate / cocoa, dry mix , made with non-dairy milk
11514325	Hot chocolate / cocoa, reduced sugar, made with non-dairy milk
11514360	Hot chocolate / cocoa, dry mix, reduced sugar, made with non-dairy milk
12200100	Coffee creamer, NFS
12210200	Coffee creamer, liquid
12210210	Coffee creamer, liquid, flavored
12210260	Coffee creamer, liquid, fat free
12210270	Coffee creamer, liquid, fat free, flavored
12210280	Coffee creamer, liquid, fat free, sugar free, flavored
12210310	Coffee creamer, liquid, sugar free, flavored

12210400	Coffee creamer, powder
12210420	Coffee creamer, powder, flavored
12210430	Coffee creamer, powder, fat free
12210505	Coffee creamer, powder, sugar free, flavored
12210520	Coffee creamer, soy, liquid
12320100	Sour cream, imitation
14502000	Imitation cheese
42402010	Coconut cream, canned, sweetened
95101000	Nutritional drink or shake, ready-to-drink (Boost)
95101010	Nutritional drink or shake, ready-to-drink (Boost Plus)
95102000	Nutritional drink or shake, ready-to-drink (Carnation Instant Breakfast)
95103000	Nutritional drink or shake, ready-to-drink (Ensure)
95103010	Nutritional drink or shake, ready-to-drink (Ensure Plus)
95104000	Nutritional drink or shake, ready-to-drink, sugar free (Glucerna)
95106000	Nutritional drink or shake, ready-to-drink (Muscle Milk)
95106010	Nutritional drink or shake, ready-to-drink, light (Muscle Milk)
95110000	Nutritional drink or shake, ready-to-drink (Slim Fast)
95110010	Nutritional drink or shake, ready-to-drink, sugar free (Slim Fast)
95110020	Nutritional drink or shake, high protein, ready-to-drink (Slim Fast)
95120000	Nutritional drink or shake, ready-to-drink, NFS
95120010	Nutritional drink or shake, high protein, ready-to-drink, NFS
95120020	Nutritional drink or shake, high protein, light, ready-to-drink, NFS
95120050	Nutritional drink or shake, liquid, soy-based

Non-dairy yogurt

[MoreMeat Mycoprotein] = 70%

42401100	Yogurt, coconut milk
42401200	Yogurt, almond milk

Fats and Oils, Condiments and Sauces

[MoreMeat Mycoprotein] = 30%

14650100	Cheese sauce
27150210	Fish sauce
28500000	Gravy, poultry
28500040	Gravy, beef
28501010	Gravy, beef, fat free
28501110	Gravy, poultry, fat free

28520000	Gravy, made with soy sauce
28520010	Gravy, NFS
28520100	Oyster sauce
28522000	Mole sauce
41420100	Miso sauce
41420110	Miso
41420200	Natto
41420250	Hoisin sauce
41420300	Soy sauce
41420350	Soy sauce, reduced sodium
41420380	Yogurt, soy
41420400	Teriyaki sauce
41420410	Teriyaki sauce, reduced sodium
41420450	Worcestershire sauce
42204050	Peanut sauce
42204100	Gravy, vegetarian
74401010	Ketchup
74401110	Ketchup, reduced sodium
74402010	Tomato chili sauce
74402210	Taco sauce
74402250	Enchilada sauce, red
74402260	Enchilada sauce, green
74402350	Salsa verde or salsa, green
74403000	Hot Thai sauce
74404010	Spaghetti sauce
74404020	Spaghetti sauce with added vegetables
74404050	Spaghetti sauce, reduced sodium
74404060	Spaghetti sauce, fat free
74404090	Vodka sauce with tomatoes and cream
74406010	Barbecue sauce
74406060	Buffalo sauce
74406100	Steak sauce
74406500	Cocktail sauce
75506010	Mustard
75506100	Honey mustard dip
75511010	Hot pepper sauce
81102000	Margarine, NFS
81102010	Margarine, stick

81102020	Margarine, tub
81103090	Butter replacement, liquid
81104010	Margarine, light
81106010	Butter replacement, powder
81200100	Oil or table fat, NFS
81203000	Shortening, NS as to vegetable or animal
81301000	Garlic sauce
81301020	Lemon-butter sauce
81302010	Hollandaise sauce
81302040	Sandwich spread
81302050	Tartar sauce
81302060	Horseradish sauce
81302070	Pesto sauce
81308100	Fry sauce
81308200	Sauce, NFS
81312100	Curry sauce
83100100	Salad dressing, NFS, for salads
83100200	Salad dressing, NFS, for sandwiches
83101000	Blue or roquefort cheese dressing
83101600	Bacon and tomato dressing
83102000	Caesar dressing
83103000	Coleslaw dressing
83104000	French or Catalina dressing
83105500	Honey mustard dressing
83106000	Italian dressing, made with vinegar and oil
83107000	Mayonnaise, regular
83108000	Vegan mayonnaise
83109000	Russian dressing
83110000	Mayonnaise-type salad dressing
83112000	Avocado dressing
83112400	Creamy Italian dressing
83112950	Poppy seed dressing
83112990	Sesame dressing
83113500	Ranch dressing
83114000	Thousand Island dressing
83115000	Fruit dressing
83200100	Salad dressing, light, NFS
83201000	Blue or roquefort cheese dressing, light

83201500	Creamy Italian dressing, light
83202020	French or Catalina dressing, light
83203000	Caesar dressing, light
83204000	Mayonnaise, light
83204030	Mayonnaise, reduced fat, with olive oil
83204050	Mayonnaise-type salad dressing, light
83204500	Honey mustard dressing, light
83205450	Italian dressing, light
83205560	Ranch dressing, light
83206500	Sesame dressing, light
83207000	Thousand Island dressing, light
83208500	Korean dressing or marinade
83300100	Blue or roquefort cheese dressing, fat free
83300200	Caesar dressing, fat free
83300250	Creamy Italian dressing, fat free
83300400	French or Catalina dressing, fat free
83300500	Honey mustard dressing, fat free
83300600	Italian dressing, fat free
83300700	Mayonnaise, fat free
83300750	Ranch dressing, fat free
83300900	Salad dressing, fat free, NFS
83301000	Thousand Island dressing, fat free
89901010	Cream sauce, for use with vegetables
89901020	Cheese sauce, for use with vegetables
89901030	Gravy, for use with vegetables
89901040	Soy based sauce, for use with vegetables
89901050	Tomato sauce, for use with vegetables
91361010	Sweet and sour sauce
91361050	Duck sauce

Grain products and pastas

[MoreMeat Mycoprotein] = 50%

56104000	Pasta, vegetable, cooked
56112000	Noodles, cooked
56113000	Noodles, whole grain, cooked
56116000	Noodles, chow mein
56116990	Long rice noodles, made from mung beans, cooked

56117090	Rice noodles, cooked
56130000	Pasta, cooked
56132990	Pasta, whole grain, cooked
56140100	Pasta, gluten free

Plant protein products

[MoreMeat Mycoprotein] = 90%

27250520	Imitation crab meat
27450130	Crab salad made with imitation crab
27564420	Hot dog sandwich, vegetarian, on bun
27564430	Hot dog sandwich, vegetarian, on bread
33001010	Egg substitute, omelet, scrambled, or fried, fat added
33001050	Egg substitute, omelet, scrambled, or fried, no added fat
33401000	Egg substitute, omelet, scrambled, or fried, with cheese
33401100	Egg substitute, omelet, scrambled, or fried, with meat
33401200	Egg substitute, omelet, scrambled, or fried, with vegetables
33401300	Egg substitute, omelet, scrambled, or fried, with cheese and meat
33401400	Egg substitute, omelet, scrambled, or fried, with cheese and vegetables
33401500	Egg substitute, omelet, scrambled, or fried, with meat and vegetables
33401600	Egg substitute, omelet, scrambled, or fried, with cheese, meat, and vegetables
41420010	Soybean curd
41420050	Soybean curd cheese
41421010	Soybean curd, deep fried
41421020	Soybean curd, breaded, fried
41425010	Vermicelli, made from soybeans
41440000	Textured vegetable protein, dry
41480020	Frozen dessert, non-dairy
41810200	Bacon strip, meatless
41810400	Breakfast link, pattie, or slice, meatless
41810600	Chicken, meatless, NFS
41810610	Chicken, meatless, breaded, fried
41811400	Hot dog, vegetarian
41811600	Luncheon slice, meatless-beef, chicken, salami or turkey
41811800	Meatball, meatless
41811890	Veggie burger patty, no bun
41812600	Vegetarian, fillet
41901010	Veggie burger, on bun

41901020	Veggie burger, on bun, with cheese
41901030	Falafel sandwich
42203200	Soy nut butter
59003000	Meat substitute, cereal- and vegetable protein-based, fried
95201000	Nutritional powder mix (Carnation Instant Breakfast)
95201010	Nutritional powder mix, sugar free (Carnation Instant Breakfast)
95201200	Nutritional powder mix (EAS Whey Protein Powder)
95201300	Nutritional powder mix (EAS Soy Protein Powder)
95201500	Nutritional powder mix, high protein (Herbalife)
95201600	Nutritional powder mix (Isopure)
95202000	Nutritional powder mix (Muscle Milk)
95202010	Nutritional powder mix, light (Muscle Milk)
95210000	Nutritional powder mix (Slim Fast)
95210010	Nutritional powder mix, sugar free (Slim Fast)
95210020	Nutritional powder mix, high protein (Slim Fast)
95220000	Nutritional powder mix, NFS
95220010	Nutritional powder mix, high protein, NFS
95230000	Nutritional powder mix, whey based, NFS
95230010	Nutritional powder mix, protein, soy based, NFS
95230020	Nutritional powder mix, protein, light, NFS
95230030	Nutritional powder mix, protein, NFS

Processed fruit and fruit juices

[MoreMeat Mycoprotein] = 60%

11553120	Fruit smoothie, with whole fruit and dairy, added protein
64134020	Fruit smoothie, with whole fruit, no dairy, added protein
64134025	Fruit smoothie, with whole fruit, non-dairy
64134030	Fruit smoothie juice drink, no dairy
64134100	Fruit smoothie, light
64134200	Fruit smoothie, bottled
78101110	Fruit and vegetable smoothie, added protein
78101115	Fruit and vegetable smoothie, non-dairy
78101118	Fruit and vegetable smoothie, non-dairy, added protein
78101120	Fruit and vegetable smoothie, bottled
78101125	Fruit and vegetable smoothie, no dairy
92510610	Fruit juice drink
92530510	Cranberry juice drink, with high vitamin C

92530610	Fruit juice drink, with high vitamin C
92530950	Vegetable and fruit juice drink, with high vitamin C
92531030	Fruit juice drink (Sunny D)
92550030	Fruit juice drink, with high vitamin C, light
92550035	Fruit juice drink, light
92550040	Fruit juice drink, diet
92550110	Cranberry juice drink, with high vitamin C, light
92550200	Grape juice drink, light
92550350	Orange juice beverage, 40-50% juice, light
92550360	Apple juice beverage, 40-50% juice, light
92550370	Lemonade, fruit juice drink, light
92550380	Pomegranate juice beverage, 40-50% juice, light
92550400	Vegetable and fruit juice drink, with high vitamin C, diet
92550405	Vegetable and fruit juice drink, with high vitamin C, light
92552030	Fruit juice drink (Capri Sun)
92582110	Fruit juice drink, added calcium (Sunny D)

Processed vegetable and vegetable juices

[MoreMeat Mycoprotein] = 60%

74302000	Tomato juice cocktail
75132000	Mixed vegetable juice
78101130	Vegetable smoothie

Soups and soup mixes

[MoreMeat Mycoprotein] = 60%

28315050	Soup, beef, canned
28340600	Soup, chicken, canned
41601030	Soup, bean, canned
41601160	Soup, bean, canned, reduced sodium
41603000	Soup, lentil, canned
41603005	Soup, lentil, canned, reduced sodium
58403010	Soup, chicken noodle, canned
58403060	Soup, chicken, canned, reduced sodium
74602010	Soup, tomato, canned
74602200	Soup, tomato, canned / carton, reduced sodium
75649010	Soup, vegetable, canned

75649040	Soup, vegetable, canned, reduced sodium
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Other

Frozen or shelf stable prepared meals; Quick-frozen prepared cuisine products
[MoreMeat Mycoprotein] = 60%

28101000	Frozen dinner, NFS
28110000	Beef dinner, NFS, frozen meal
28110150	Beef with vegetable, diet frozen meal
28110300	Salisbury steak dinner, NFS, frozen meal
28110310	Salisbury steak with gravy, potatoes, vegetable, frozen meal
28110330	Salisbury steak with gravy, whipped potatoes, vegetable, dessert, frozen meal
28110380	Salisbury steak with gravy, macaroni and cheese, frozen meal
28110510	Beef, sliced, with gravy, potatoes, vegetable, frozen meal
28110660	Meatballs, Swedish, in gravy, with noodles, diet frozen meal
28113140	Beef with spaetzle or rice, vegetable, frozen meal
28140100	Chicken dinner, NFS, frozen meal
28140710	Chicken, fried, with potatoes, vegetable, frozen meal
28140720	Chicken patty, or nuggets, boneless, breaded, potatoes, vegetable, frozen meal
28141010	Chicken, fried, with potatoes, vegetable, dessert, frozen meal, large meat portion
28141201	Teriyaki chicken with rice and vegetable, diet frozen meal
28141250	Chicken with rice and vegetable, diet frozen meal
28141650	Chicken and vegetables au gratin with rice, diet frozen entree
28143020	Chicken and vegetable entree with rice, diet frozen meal
28143080	Chicken with noodles and cheese sauce, diet frozen meal
28143150	Chicken and vegetable entree with noodles, diet frozen meal
28143200	Chicken in soy-based sauce, rice and vegetables, frozen meal
28144100	Chicken and vegetable entree with noodles and cream sauce, frozen meal
28145110	Turkey with vegetable, stuffing, diet frozen meal
28145210	Turkey with gravy, dressing, potatoes, vegetable, frozen meal
28150510	Fish in lemon-butter sauce with starch item, vegetable, frozen meal
28154010	Shrimp and vegetables in sauce with noodles, diet frozen meal
28160300	Meat loaf dinner, NFS, frozen meal
28160310	Meat loaf with potatoes, vegetable, frozen meal
42303100	Peanut butter and jelly sandwich, frozen commercial product without crusts
58106200	Pizza, cheese, from frozen, thin crust
58106205	Pizza, cheese, from frozen, thick crust
58106300	Pizza, cheese, with vegetables, from frozen, thin crust
58106305	Pizza, cheese with vegetables, from frozen, thick crust

58106512	Pizza with pepperoni, from frozen, thin crust
58106514	Pizza with pepperoni, from frozen, medium crust
58106516	Pizza with pepperoni, from frozen, thick crust
58106602	Pizza with meat other than pepperoni, from frozen, thin crust
58106604	Pizza with meat other than pepperoni, from frozen, medium crust
58106606	Pizza with meat other than pepperoni, from frozen, thick crust
58106700	Pizza with meat and vegetables, from frozen, thin crust
58106702	Pizza with meat and vegetables, from frozen, medium crust
58106705	Pizza with meat and vegetables, from frozen, thick crust
58126100	Turnover or hot pocket, NFS
58126105	Turnover or hot pocket, meatless
58126160	Turnover or hot pocket, pizza pocket
58126290	Turnover or lean pocket
58126400	Turnover or breakfast pocket, egg
58130016	Lasagna with meat, frozen
58145112	Macaroni or noodles with cheese, made from packaged mix
58145113	Macaroni or noodles with cheese, canned
58145117	Macaroni or noodles with cheese, Easy Mac type
58145119	Macaroni or noodles with cheese, made from reduced fat packaged mix
58146223	Pasta with tomato-based sauce, ready-to-heat
58146303	Pasta with tomato-based sauce, and added vegetables, ready-to-heat
58146323	Pasta with tomato-based sauce and meat, ready-to-heat
58146333	Pasta with tomato-based sauce, meat, and added vegetables, ready-to-heat
58146343	Pasta with tomato-based sauce and poultry, ready-to-heat
58146353	Pasta with tomato-based sauce, poultry, and added vegetables, ready-to-heat
58146363	Pasta with tomato-based sauce and seafood, ready-to-heat
58146373	Pasta with tomato-based sauce, seafood, and added vegetables, ready-to-heat
58146383	Pasta with cream sauce, ready-to-heat
58146393	Pasta with cream sauce and added vegetables, ready-to-heat
58146403	Pasta with cream sauce and meat, ready-to-heat
58146413	Pasta with cream sauce, meat, and added vegetables, ready-to-heat
58146423	Pasta with cream sauce and poultry, ready-to-heat
58146433	Pasta with cream sauce, poultry, and added vegetables, ready-to-heat
58146443	Pasta with cream sauce and seafood, ready-to-heat
58146453	Pasta with cream sauce, seafood, and added vegetables, ready-to-heat
58146603	Pasta, whole grain, with tomato-based sauce, ready-to-heat
58146613	Pasta, whole grain, with tomato-based sauce and added vegetables, ready-to-heat
58146623	Pasta, whole grain, with tomato-based sauce and meat, ready-to-heat

58146633	Pasta, whole grain, with tomato-based sauce, meat, and added vegetables, ready-to-heat
58146643	Pasta, whole grain, with tomato-based sauce and poultry, ready-to-heat
58146653	Pasta, whole grain, with tomato-based sauce, poultry, and added vegetables, ready-to-heat
58146663	Pasta, whole grain, with tomato-based sauce and seafood, ready-to-heat
58146673	Pasta, whole grain, with tomato-based sauce, seafood, and added vegetables, ready-to-heat
58146683	Pasta, whole grain, with cream sauce, ready-to-heat
58146693	Pasta, whole grain, with cream sauce, and added vegetables, ready-to-heat
58146703	Pasta, whole grain, with cream sauce and meat, ready-to-heat
58146713	Pasta, whole grain, with cream sauce, meat, and added vegetables, ready-to-heat
58146723	Pasta, whole grain, with cream sauce and poultry, ready-to-heat
58146733	Pasta, whole grain, with cream sauce, poultry, and added vegetables, ready-to-heat
58146743	Pasta, whole grain, with cream sauce and seafood, ready-to-heat
58146753	Pasta, whole grain, with cream sauce, seafood, and added vegetables, ready-to-heat
58302080	Noodles with vegetables in tomato-based sauce, diet frozen meal
58304010	Spaghetti and meatballs dinner, NFS, frozen meal
58304060	Spaghetti with meat sauce, diet frozen meal
58310310	Pancakes and sausage, frozen meal

**Evaluation of the Allergenicity Assessment of MoreMeat 's Mycoprotein
from *Fusarium compactum* MM-135**

Completed On

12 November 2024

Performed by

RE Goodman Consulting, LLC

8110 Dougan Circle

Lincoln, NE 68516

Client

MoreMeat (Guangzhou) Biotech Co.,Ltd

Room 101, Building C1, No. 11, Kaiyuan Avenue,

Huangpu District, Guangzhou City,

Guangdong Province, China

I have carefully reviewed the documents on MoreMeat's Mycoprotein product that were provided to me by the Anchor Center for Certification (ACC) of Shanghai, China. That includes the overall description in the GRAS dossier, as well as the In Vitro digestibility report, the literature research associated with *Fusarium sp.* and *Neospora crassa* including reports and papers on Quorn.

My professional experiences and knowledge about allergy began in 1980-1985 when I worked at Hollister-Stier Laboratories in Spokane WA, USA describing and characterizing allergen extracts for diagnosis. Then a PhD and postdoctoral studies in immunology. I worked at Monsanto from 1997-2004 on the allergenicity evaluation of genetically modified crops. I represented biotechnology companies as a non-governmental organization representative during the CODEX Alimentarius Committee meeting in 2001 in Vancouver, Canada. In 2004 I accepted a faculty position at the University of Nebraska-Lincoln in the Food Allergy Research and Resource Program in the Dept. of Food Science and Technology and managed the www.AllergenOnline.org database from 2004-2023. In this database we review peer reviewed publications demonstrating IgE binding or more importantly allergic reactivity to allergenic sources and specifically to proteins in the sources. The database includes FASTA amino acid sequence comparisons for potential risks of cross-reactivity. We have included a celiac protein database of glutenins and gliadins for risks assessment. I was the Chairman of the WHO/IUIS Allergen nomenclature subcommittee from 2014 to 2022 where short names are given to proteins from food, airway, dermal and injectable proteins that submitters claim IgE antibody binding from subjects with suspected allergies. It is not a risk assessment database. I am a Fellow in the American Academy of Allergy, Asthma and Immunology since 2002 and member of the European Academy of Allergy and Clinical Immunology.

My laboratory has performed bioinformatics analysis and literature searches on many Novel Foods and GMOs. We have evaluated serum IgE antibody binding on proteins suspected of allergic cross-reactivity and performed basophil histamine release to evaluate whether there are at least two IgE epitopes on some proteins. We have also tested the stability of proteins in Simulated Gastric Fluid and Simulated Intestinal Fluid models.

As discussed by MoreMeat, the genus *Fusarium* has many species and types, a few have been associated with some allergies, mostly from humans with lung or airway infections of some species of the genus. As shown by gene and protein comparisons, many proteins, usually including house-keeping proteins that are enzymes or functional membrane bound proteins are highly conserved in sequences of *Fusarium sp.*, *Aspergillus sp.*, *Penicillium sp.* and other fungi. Literature searches did not identify clinical allergy from the related *Fusarium* grouped species. The long history of the population in the United Kingdom consuming Quorn, for 30 years and in the United States for more than 10 years with only rare reports of symptoms that might be food allergy have not been proven. Some proteins from these fungi have been reported to be targets of IgE binding in some studies, but usually without inhibition assays or controls of whole material extracts leaves an incomplete picture of allergy. There is some proof that these fungi can decorate some of their proteins with complex carbohydrates that are listed as CCD, cross-reactive carbohydrates. Those do not cause allergic reactions as binding to IgE on basophils and mast cells is really low affinity and rare.

There are four *Fusarium* sp. proteins listed in the WHO/IUIS Allergen Nomenclature database, but those are common housekeeping proteins. AllergenOnline lists the four from WHO/IUIS plus the P2 acidic ribosomal protein as an apparent allergen from Quorn. This was determined by IgE binding protein from serum of one subject with asthma reaction after consumption of Quorn. Hoff et al. (2003) describes this protein as a common minor allergen of a number of mold species including *Fusarium culmorum* protein, though Quorn is from *Fusarium venenatum*. Symptoms from ingestion in the UK were reported as 1 in 370,000 and in Switzerland as 1 in 80,000 subjects with symptoms of gastric cramping beginning one hour to four hours after ingestion. Based on skin prick tests of weak reactions and low level RAST IgE binding, they found apparent reactivity across a wide range of fungi. The one subject had a history of 30 years of perennial asthma. They performed double blind food challenge with Quorn and also clear testing with cloned Fus c 1 and with Quorn extract. The IgE binding was cross-reactive but not equal and there is no connection with understanding clinical allergic risks. The low reactivity of Quorn which is a high protein content mycoprotein food ingredient indicates little risk for the population.

The genome of the species is referenced but not in a way that allows it to be seen. The description of FASTA results states that no matches were identified to clinically severe allergens such as the 2S albumins of peanuts or tree nuts, tropomyosin of crustacean shellfish, parvalbumin of fish or beta-lactoglobulin of cow's milk. In addition, the only reference to a described allergen in *Fusarium* is the acidic ribosomal protein P2. And the relevance of that protein causing clinical allergic responses has not been verified. 12,548 proteins predicted from the genome were compared to AllergenOnline.org version 22. Protein sequences of this species were obtained computer predictions of DNA sequences. The predicted protein amino acid sequences were also searched against the whole protein database using BLASTP, to evaluate similarity of the proteins to housekeeping proteins that are shared by most fungi. Proteins that have high identity matches to Quorn proteins should be safe as few people are allergic to Quorn and the causative allergen(s) have not been identified. Only 33 proteins have less than 60% identity to Quorn proteins and those have high identities to other proteins that are from housekeeping genes. BLASTP was also used to compare proteins to human proteins (*Homo sapiens*). Some of the *Fusarium* proteins including Fus p 4 had 61% identity to a human protein, and other proteins had modest identities to human proteins. Those are not like to be significant allergens.

Tests of stability of the mycoproteins in SGF showed that most of the proteins were digested in pepsin in relatively short times. While that does not guarantee safety from allergenicity, it is a good indication that most of the proteins are relatively instable in conditions broadly mimicking the stomach and are less likely to be available for absorption and allergy in the small intestine. A similar assessment was tested using Simulated Intestinal Fluid, which also showed good digestion. These tests indicate the proteins are relatively unstable under conditions that mimic human food protein digestion.

My overall conclusion is that this mycoprotein does not represent a likely risk for food allergy based on history from literature, and lack of high identity matches to abundant and potent allergens.

Reference: Hoff M, Trueb RM, Ballmer-Weber BK, Vieths S, Wuethrich B. Immediate-type hypersensitivity reaction to ingestion of mycoprotein (Quorn) in a patient allergic to molds caused by acidic ribosomal protein P2. J Allergy Clin Immunol 2003. 111(5):1106-1110.

Signed,



Date: 12/5/2024

Report No.:A2240749328101001

Test Report

Sample Name: Mycoprotein (Fermented mycelium protein from
Fusarium compactum MM-135)

Customer: MoreMeat(Guangzhou)Biotech Co.,Ltd

Test Type: Commission Test



Centre Testing International (Guangzhou)Co., Ltd.

www.cti-cert.com



Verification
Code: UYDO

Test Report

Report No. :A2240749328101001

Page 2 of 9

Sample Information	Sample Name	Mycoprotein (Fermented mycelium protein from Fusarium compactum MM-135)		CTI Sample Code	HJQ04188001
	Sample Brand	/		Specification	500g/bag
	Production Date/Exp. Date	2024.07.10/2025.07.09		Lot No.	MM13524002
	Sample Quantity	5 bags		Sample State	powder
	Manufacturer	MoreMeat(Guangzhou)Biotech Co.,Ltd			
	Address of Manufacturer	Room 101, Building C1, No. 11, Kaiyuan Avenue, Huangpu District, Guangzhou City, Guangdong Province			
Customer Information	Customer	MoreMeat(Guangzhou)Biotech Co.,Ltd			
	Address	Room 101, Building C1, No. 11, Kaiyuan Avenue, Huangpu District, Guangzhou City, Guangdong Province			
Test Information	Sample Received Date	2024/12/02	Sample Tested Date	2024/12/02 ~ 2025/02/06	
	Test Item(s)	True Protein Digestibility			
	Test Standards	The test program provided by the customer was based on AOAC991.29			
Test Conclusion	According to the requirements of the customer, the true protein digestibility (TD) of the sample was tested, and the protein digestibility corrected amino acid score (PDCAAS) of the sample was calculated based on the amino acid content in the protein provided by the customer. Test conclusion: The true protein digestibility of the sample was 90.79%. Base on the amino acid requirements of children aged 1-2 years in the 2013 FAO food and nutrition paper NO.92 report, PDCAAS was 0.93; base on the amino acid requirements of children aged 3-10 years, PDCAAS was 1.00; base on the amino acid requirements of adults aged 18 years and above, PDCAAS was 1.00. The Special Seal for Inspection Approve date:2025/02/12				
Note(s)	-----				

Compiled by



Reviewed by



Approved by



Approved Signatory:Pan Qingling

Centre Testing International (Guangzhou)Co., Ltd.

No.203, Bohua 1st Road, Huangpu District, Guangzhou,Guangdong,China



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1.Material

1.1 Sample:

Sample name: Mycoprotein (Fermented mycelium protein from Fusarium compactum MM-135)

1.2 Main materials and instruments:

①Rat metabolic cage

Brand: Suzhou Fengshi Experimental Animal Equipment Co., Ltd.

Model: DXL-D

1.3 Laboratory animals and feeding environment

1.3.1 Laboratory animals

Species: Rat

Strains: Sprague Dawley (SD)

Level: SPF grade

Gender: Male

Age: 21-30 days

Quantity: 18

Purchase weight: 45-55g

Starting weight of the experiment:65.2-69.8g

Provided by: Guangdong Medical Laboratory Animal Center

Laboratory animal production License No.: SCXK (Guangdong) 2023-0035

Quality Certificate No.: 44829700046211

1.3.2 Feeding environment

Temperature: 18°C-26°C

Relative humidity:40%-70%

Light: 12h alternating light and dark

Laboratory animal Environmental Use License No.: SYXK (Guangdong) 2024-0317

Feed source: Guangdong Medical Laboratory Animal Center

Feed production License No.: Guangdong Feed License (2024) 05073

Feed Certificate No.: 202412130077

2.Method

18 weaned SD rats aged 21 to 30 days were housed in rat cages at room temperature of 18 °C -26 °C, relative humidity of 40% -70%, with alternating lighting for 12 hours and free access to sterile pure water. After passing the 5-day quarantine, the animals were randomly divided into 3 groups based on their weight: standard control casein group, negative control protein free group, and sample group, with 6 animals in each group.

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During the test period, rats were kept in metabolic cages, and feed was weighed and feces were collected. Each group was given corresponding customized feed, free drinking water, and weighed daily. The customized feed for the standard control group uses casein as the protein source, while the customized feed for the negative control group has no protein (e.g., protein was replaced with cornstarch). The customized feed for the sample group used protein from the sample as the protein source, with the same nutritional components. The test diets and the positive control (casein) diet were formulated to contain 10% protein (N x6.25) and levels of vitamins, minerals and calories to satisfy the requirements of the rats. The negative control (protein free) diet was formulated to contain the same levels of all nutrients except protein (e.g., protein was replaced with cornstarch). Diets were custom-made by the Guangdong Provincial Medical Laboratory Animal Center. Each group was fed with 15 grams per day of the corresponding custom diets till the end of the study and had free access to water, with daily weighing. Collect feces every 24 hours during Day 5-Day 9. At the end of the collection period, feces from each animal were mixed, dried and analyzed for nitrogen content. Kjeldahl method was used to determine the nitrogen content in feces. True protein digestibility (TD) was calculated as follows:

$$TD(\%) = \frac{Ni - (Fn - Mn)}{Ni} \times 100$$

Ni represents the nitrogen intake (g) of animals in the sample or standard control group, Fn represents the fecal nitrogen excretion (g) of animals in the sample or standard control group, and Mn represents the fecal nitrogen excretion (g) of animals in the negative control group (endogenous nitrogen content).

-Amino acid score (AAS) = [amino acid content in sample protein (mg/g)]/[corresponding essential amino acid content in FAO/WHO scoring standard model (mg/g)]

-PDCAAS=AAS×TD (%)

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3.Result

Table 1. Diet Consumption Statistics (N=6)

Group	Average Diet Consumption (g) ^a	Measured Protein Concentration in Prepared Diet (%) ^b	Average Protein Consumption (g) ^c	Average Intake of Dietary Nitrogen (g) ^d
standard control group	74.0	10.88	8.05	1.24
negative control group	46.7	0.63	0.29	0.05
sample group	71.8	11.38	8.17	1.26

Note: a. The average diet consumption of each animal during the collection period.

b. Protein concentration in prepared diet = Nitrogen content of feed as determined by Kjeldahl method × 6.25

c. Average protein consumption (g)= Average diet consumption × Protein concentration

d. Average intake of dietary nitrogen (g)=average protein consumption (g)/6.25

Table 2. Fecal nitrogen content (N=6)

Group	Average Total Fecal Excretion (g) ^a	Nitrogen concentration in Feces (%) ^b	Average Total Nitrogen Excretion (g) ^c
standard control group	4.097	2.52	0.103
negative control group	2.334	1.20	0.028
sample group	4.011	3.58	0.144

Note: a. The average total fecal excretion of each animal during the collection period.

b. Mean nitrogen concentration in feces of each animal.

c. Average fecal nitrogen excretion=average total fecal excretion × nitrogen concentration in feces.

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Table 3. Amino Acid Scores (AAS) of the Standard Control Group

Amino Acid	TEST (mg/g protein)	REF (mg/g protein)	Amino Acid Score
Histidine	26	18 ^a	1.44
		16 ^b	1.63
		15 ^c	1.73
Isoleucine	48	31 ^a	1.55
		30 ^b	1.60
		30 ^c	1.60
Leucine	88	63 ^a	1.40
		61 ^b	1.44
		59 ^c	1.49
Lysine	74	52 ^a	1.42
		48 ^b	1.54
		45 ^c	1.64
Methionine + Cystine ^d	30	25 ^a	<u>1.20</u>
		23 ^b	<u>1.30</u>
		22 ^c	<u>1.36</u>
Phenylalanine + Tyrosine	103	46 ^a	2.24
		41 ^b	2.51
		38 ^c	2.71
Threonine	38	27 ^a	1.41
		25 ^b	1.52
		23 ^c	1.65
Valine	57	41 ^a	1.39
		40 ^b	1.43
		39 ^c	1.46
Tryptophan	12	7.0 ^a	1.71
		6.6 ^b	1.82
		6.0 ^c	2.00

Note: a. Amino acid requirements for children aged 1-2 in the FAO food and nutrition paper NO.92.

b. Amino acid requirements for children aged 3-10 in the FAO food and nutrition paper NO.92.

c. Amino acid requirements for adults aged 18 and above in the FAO food and nutrition paper NO.92.

d. First limiting amino acid.

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Table 4. Amino acid fractions (AAS) of sample groups

Amino Acid	TEST (g/100g sample)	Actual content (mg/g protein) ^a	REF (mg/g protein)	Amino Acid Score
Histidine	1.10	21.32	18 ^b	1.08
			16 ^c	1.21
			15 ^d	1.29
Isoleucine	2.30	44.57	31 ^b	1.44
			30 ^c	1.49
			30 ^d	1.49
Leucine	4.05	78.49	63 ^b	1.25
			61 ^c	1.29
			59 ^d	1.33
Lysine	3.71	71.90	52 ^b	1.38
			48 ^c	1.50
			45 ^d	1.60
Methionine + Cystine ^e	1.31	25.39	25 ^b	<u>1.02</u>
			23 ^c	<u>1.10</u>
			22 ^d	<u>1.15</u>
Phenylalanine + Tyrosine	3.81	73.84	46 ^b	1.61
			41 ^c	1.80
			38 ^d	1.94
Threonine	2.36	45.74	27 ^b	1.69
			25 ^c	1.83
			23 ^d	1.99
Valine	2.70	52.33	41 ^b	1.28
			40 ^c	1.31
			39 ^d	1.34
Tryptophan	0.65	12.60	7.0 ^b	1.80
			6.6 ^c	1.91
			6.0 ^d	2.10

Note: a Actual content (mg/g protein) = TEST (g/100g sample) x 10 / Protein content of sample (51.6%).

b. Amino acid requirements for children aged 1-2 in the FAO food and nutrition paper NO.92.

c. Amino acid requirements for children aged 3-10 in the FAO food and nutrition paper NO.92.

d. Amino acid requirements for adults aged 18 and above in the FAO food and nutrition paper NO.92.

e. First limiting amino acid.

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Table 5. Protein Digestibility Corrected Amino Acid Score (PDCAAS)

Group	True Digestibility (TD, %) ^a	Amino Acid Score	Actual PDCAAS ^b	PDCAAS ^c
standard control group	93.95	1.20 ^d	1.13	1.00
		1.30 ^e	1.23	1.00
		1.36 ^f	1.29	1.00
sample group	90.79	1.02 ^d	0.93	0.93
		1.10 ^e	1.00	1.00
		1.15 ^f	1.04	1.00

Note: a. $TD(\%) = \frac{Ni - (Fn - Mn)}{Ni} \times 100$

- b. Calculate actual PDCAAS = AAS x TD
- c. Correct the PDCAAS to 1 for actual values greater than 1
- d. Amino acid score calculated based on amino acid requirements for children aged 1-2 in the FAO food and nutrition paper NO.92
- e. Amino acid score calculated based on amino acid requirements for children aged 3-10 in the FAO food and nutrition paper NO.92
- f. Amino acid score calculated based on amino acid requirements for adults aged 18 and above in the FAO food and nutrition paper NO.92

4. Conclusion

The true protein digestibility of the sample was 90.79%. Base on the amino acid requirements of children aged 1-2 years in the 2013 FAO food and nutrition paper NO.92 report, PDCAAS was 0.93; base on the amino acid requirements of children aged 3-10 years, PDCAAS was 1.00; base on the amino acid requirements of adults aged 18 years and above, PDCAAS was 1.00.

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Generally Recognized As Safe (GRAS) Panel Consensus on the Safety and GRAS Status of MoreMeat's Mycoprotein for Use as an Ingredient in Foods

EXPERT PANEL STATEMENT

An independent panel of experts (GRAS Panel), qualified by scientific training and experience to assess the safety of food and food ingredients, was convened by the Anchor Center for Certification (ACC) on behalf of MoreMeat (Guangzhou) Biotech Co., Ltd (MoreMeat). The panel conducted a critical and comprehensive evaluation of the available data and relevant information to determine whether MoreMeat mycoprotein, when used in selected conventional human foods, is safe and Generally Recognized as Safe (GRAS) under the conditions of its intended use. This determination was made in accordance with scientific procedures outlined in the Federal Food, Drug, and Cosmetic Act (FFDCA), as specified in Title 21 of the Code of Federal Regulations (21 CFR) §170.30(b).

The GRAS Expert Panel convened on December 12, 2024, and unanimously concluded that mycoprotein, as specified in the document "GRAS Determination for the Use of MoreMeat Mycoprotein in Selected Conventional Foods," is Generally Recognized as Safe (GRAS) based on scientific procedures. This conclusion applies when mycoprotein is used as an ingredient in conventional foods as a protein source and is manufactured in compliance with cGMP. A summary of the basis for the GRAS Expert Panel's conclusion is provided herein.

Summary and Basis for a GRAS Conclusion

MoreMeat's mycoprotein is produced by fermentation of *Fusarium compactum* MM-135. The finished product contains $\geq 43\%$ protein (on a dried basis). MoreMeat's mycoprotein is intended for use in processed food products (excluding infant formulas and meat or poultry products) as a protein source, serving as a substitute for other animal- or plant-based proteins.

The GRAS Panel independently, collectively, and critically examined MoreMeat's GRAS dossier for mycoprotein as well as publicly available scientific information and data compiled from the literature and other published sources and searches of published scientific literature conducted through December 2024. The data evaluated by the GRAS Panel included information pertaining to the methods of manufacturing and product specifications, analytical data, intended use levels, consumption estimates, and comprehensive literature on the safety of mycoprotein.

Composition

MoreMeat's mycoprotein derived from *F. compactum* is substantially similar to the mycoprotein derived from *Fusarium venenatum* and *Fusarium flavolapis*, as described in GRN 91 and GRN 904. MoreMeat's mycoprotein is a dietary source of protein, and its product specifications were developed based on the monograph for mycoprotein derived from *F. venenatum* included in the 14th Edition of the Food and Chemical Codex (FCC 14th).

Analytical data provided in MoreMeat's GRAS dossier for mycoprotein confirms three non-consecutive batches meet or exceed the finished product specifications. In addition, MoreMeat has established sufficient and appropriate maximum specification for potential contaminants, including RNA, mycotoxins, heavy metals, and microbes.

The Expert Panel discussed the potential of the presence of diacetoxyscirpenol (DAS) in the finished product. While DAS was not detected in representative batches, and while genome sequencing does not indicate an increased likelihood of DAS formation, the Expert Panel notes that MoreMeat plans to monitor DAS levels in their mycoprotein at least biannually and has a response plan in place should detectable levels of DAS be found.

MoreMeat has also presented data on amino acid composition, nutrient composition, nitrate and nitrite content, and protein quality in the GRAS dossier. The Expert Panel has reviewed the data and concurs that MoreMeat's mycoprotein is a dietary source of amino acids and nutrients, and—based on the reported Protein Digestibility Adjusted Amino Acid Score (PDCAAS)—a high-quality protein.

Additionally, the Expert Panel notes that while a 3-month accelerated stability study has concluded, long-term stability testing for MoreMeat's mycoprotein is ongoing. The available stability data presented in MoreMeat's GRAS dossier indicate that the finished product is stable under the recommended storage conditions (i.e., frozen). While MoreMeat has concluded that these results support a 12-month shelf-life when stored in a vacuum-sealed aluminum bag at -18°C, the Expert Panel notes that the shelf-life should be re-evaluated by MoreMeat as additional long-term stability data becomes available.

Production Strain

MoreMeat's mycoprotein is manufactured from the fermentation of non-genetically modified *F. compactum* MM-135, which is naturally found in soil. MoreMeat states that *F. compactum* is non-pathogenic and non-toxicogenic.

MoreMeat collected TEF1, RPB1, RPB2, and ITS sequences from all strains of multiple subspecies in the FUSARIOID-ID database (www.fusarium.org) and analyzed the evolutionary position of the *F. compactum* MM-135 by constructing a multigene phylogenetic tree. Results of DNA sequence comparison shows that *F. compactum* MM-135 shares the greatest sequence homology with *F. compactum* LLC 2178 with a similarity of 99.82%.

MoreMeat also performed a DNA sequence comparison between *F. compactum* MM-135, *F. venenatum* (production organism described in GRN 945 and GRN 91), and *Fusarium strain flavolapis* (production organism described in GRN 904), demonstrating that TEF1, RPB1, and ITS genes of *F. compactum* MM-135 have a similarity greater than 80% to their counterparts in *F. venenatum* and *Fusarium strain flavolapis*, respectively.

Manufacturing Process

MoreMeat's mycoprotein is manufactured under current Good Manufacturing Practices (cGMP) with food grade raw materials and processing aids. MoreMeat states that none of the raw materials used in the production process are derived from major allergens. Furthermore, MoreMeat states that the mycoprotein is produced with aseptic technology under sterile conditions.

As detailed in Table 3 of MoreMeat's GRAS dossier, the production process consistently yields a reproducible product. In addition, the mycoprotein has been extensively analyzed to ensure compliance with MoreMeat's established specifications, including protein content, moisture level, total fiber, ash, heavy metals, RNA, and mycotoxins. Microbiological analyses include total aerobic count, yeasts and molds, *Escherichia coli*, *Salmonella*, *Listeria monocytogenes*, *Staphylococcus aureus*, and viable fusarium.

Product Specifications

MoreMeat has established specifications for mycoprotein that are tailored to their finished product while also analogous to the current industry standards for mycoprotein derived from *F. venenatum* and *Fusarium strain flavolapis* (as outlined in FCC 14 in the monograph for mycoprotein derived from *F. venenatum*, as well as GRNs 91, 904, and 945). The GRAS Panel reviewed batch analysis results for 3 non-consecutive batches of MoreMeat's mycoprotein as presented in Table 5 of the GRAS dossier, and notes that the ingredient is well-characterized, and that the analytical data obtained using internationally accepted methods and internally validated methods (as outlined in Table 4) includes protein, moisture, total fiber, ash, mycotoxins, microbiological, RNA, and heavy metal content.

Proposed Uses and Estimated Intake

MoreMeat's mycoprotein is intended for use as a food ingredient and macronutrient across a variety of food categories. The range of intended incorporation rates for MoreMeat's mycoprotein in these foods is detailed in Table 13 of the GRAS dossier. Since MoreMeat's mycoprotein primarily consists of water (~75%), protein (~15%), and fiber (~10%), the intended use levels are expressed on a wet weight basis, with the anticipated minimum and maximum levels provided.

Among the total population (ages 2 years and older), the mean and 90th percentile consumer-only intakes of MoreMeat's mycoprotein were determined to be 94 g/day and 207 g/day, respectively. Male teenagers (ages 12 to 19 years) were determined to have the greatest consumer-only mean and 90th percentile intakes of MoreMeat's mycoprotein, at 105 g/day and 252 g/day, respectively. The lowest consumer-only mean and 90th percentile intakes were for female adults (ages 20 years and older), at 88 g/day and 194 g/day, respectively.

Among the total population (ages 2 years and older) and on a body weight basis, the mean and 90th percentile intakes of MoreMeat's mycoprotein were determined to be 1.5 g/kg bw/day and 3.4 g/kg bw/day, respectively. Children (ages 2-5 years) were determined to have the greatest

mean and 90th percentile consumer-only intakes at 5.3 g/kg bw/day and 11.9 g/kg bw/day, respectively. The lowest mean and 90th percentile consumer-only intakes were for male adults (ages 20 years and older), at 1.1 g/kg bw/day and 2.7 g/kg bw/day, and female adults (ages 20 years and older) at 1.2 g/kg bw/day and 2.6 g/kg bw/day, respectively. The Expert Panel concurs with MoreMeat's conclusion that these estimates are conservative, as it is unlikely that MoreMeat's mycoprotein will be incorporated into every product in the marketplace at the maximum proposed uses levels.

MoreMeat also calculated the estimated daily intake of protein from its mycoprotein from the conservative 90th percentile consumer-only estimate of the daily consumption of MoreMeat's mycoprotein from the proposed uses is 207 g/day for the total population (2 years and older). Assuming a maximum protein content of 15% for MoreMeat's mycoprotein, the conservative 90th percentile consumer-only estimate of the daily consumption of protein from the proposed uses of MoreMeat's mycoprotein is 31 g/day for the total population (ages 2 years and older). This estimated 90th percentile consumer-only intake of protein from all proposed uses for the total population is lower than or within the range of the recommended and/or background intake levels of protein in the diet.

To further evaluate the quality of MoreMeat's mycoprotein, a protein digestibility-corrected amino acid score (PDCAAS) test report was included in Appendix 5 of the GRAS dossier. The Expert Panel notes that the PDCAAS for MoreMeat's mycoprotein is 1.0 for children (ages 3 to 10 years old) and adults (ages 18 years and older). The highest possible PDCAAS score is 1.0, which indicates that MoreMeat's mycoprotein is a source of quality protein.

The estimated daily intake for fiber was also calculated for MoreMeat's mycoprotein. MoreMeat's mycoprotein contains ~10% dietary fiber; therefore, the conservative consumer-only mean and 90th percentile estimate of the daily consumption of dietary fiber from MoreMeat's mycoprotein is ~9 g/day and 21 g/day, respectively, for the total population (2 years and older). This estimated 90th percentile consumer-only intake of dietary fiber from all proposed uses for the total population is lower than or within the range of the recommended and/or background intake levels of dietary fiber in the diet.

To further evaluate the safety of dietary exposure of MoreMeat's mycoprotein, the estimated dietary exposure to mycotoxins and nitrates are presented in the GRAS dossier. MoreMeat has identified DAS, enniatin, and beauvericin as potential mycotoxins in mycoprotein. Enniatins and beauvericin were identified and quantified in analyses of three non-consecutive batches of mycoprotein. MoreMeat has established maximum specifications for enniatin and beauvericin, thereby ensuring that consumer exposure remains consistently within safe levels. The Expert Panel notes that DAS was not detected in the three non-consecutive representative batches of mycoprotein. As previously noted herein, the Expert Panel discussed the potential of the presence of DAS in MoreMeat's mycoprotein. While DAS was not detected in the representative batches, and while genome sequencing does not indicate an increased likelihood of DAS formation, the Expert Panel notes that MoreMeat intends to monitor DAS levels in their mycoprotein at least biannually and have a response plan in place should detectable levels of DAS be found.

The EDIs for nitrates was also calculated for MoreMeat's mycoprotein using a mean nitrates residue of 6.3 mg/kg (wet weight) and the mean and 90th percentile intakes for mycoprotein of 94 and 207 g/day, respectively. Thus, the mean and 90th percentile intakes for nitrates were determined to be 0.324 mg/person/day and 0.714 mg/person/day, respectively, or 0.005 mg/kg bw/day and 0.010 mg/kg bw/day, respectfully, for a 70-kg individual. The EDIs for nitrates are lower than or within the range of the recommended and/or background intake levels of nitrates in the diet.

Pivotal Safety Data

Limited toxicological and safety studies are available for mycoprotein derived from *F. compactum*. However, as demonstrated in the data presented in the GRAS dossier, MoreMeat's mycoprotein derived from *F. compactum* is substantially equivalent to the mycoprotein derived from *F. venenatum* (as described in GRN 945 and GRN 91) and *F. strain flavolapis* (as described in GRN 904).

MoreMeat summarized and compared the DNA sequencing, manufacturing process, and product specifications of the three mycoproteins, and concluded that its mycoprotein is quantitatively and qualitatively equivalent to the subject ingredients of GRN 904 and GRN 945, respectively. Furthermore, based on the precedence of read-across approach to support GRAS conclusions for novel proteins and other complex biological substances, MoreMeat applied the same approach of referencing pivotal studies and data from GRN 91 and GRN 904 to support a GRAS conclusion for *F. compactum*. Based upon published and peer-reviewed data on *Fusarium* toxicity and *F. compactum* toxicity, as well as the published data on *F. venenatum* and *F. strain flavolapis* toxicity, MoreMeat has determined that *F. compactum*, *F. venenatum*, and *F. strain flavolapis* have the same toxicological potential or lack thereof, when grown under controlled conditions, and that the read-across to pivotal data cited in GRN 91 for *F. compactum* and GRN 904 for *F. strain flavolapis* offers sound scientific support for MoreMeat's conclusion that the consumption of MoreMeat's mycoprotein is safe under the proposed uses.

The published toxicology studies performed with mycoprotein derived from *F. venenatum* and *F. strain flavolapis* include: a bacterial reverse mutation test; an *in vivo* mammalian erythrocyte test; an *in vitro* mammalian chromosome aberration test; a 14-day oral range-finding study in rats; an estrogenic potential study on mouse and non-RNA reduced mycoprotein study on Landrace pig; a 22-week oral feeding study in rats; a 13-week oral feeding study in rats; a 13-week oral feeding study in baboons; two 90-day oral feeding studies in rats; a 1-year oral feeding study in dogs; a 2-year carcinogenicity study in rats; a 2-generation reproductive and developmental toxicity study in rats; and teratogenicity studies in rats and rabbits. These published toxicology studies were previously reviewed in GRN 91 and GRN 904, and demonstrate that mycoprotein derived from *Fusarium spp.* is not genotoxic and does not result in adverse effects at high intake levels.

In addition, unpublished toxicology studies commissioned by MoreMeat provide corroborative evidence for establishing the safety of MoreMeat's mycoprotein. MoreMeat has assessed their mycoprotein for potential genotoxic effects using an AMES test, an *in vivo* Mammalian Erythrocyte Micronucleus study, and an *in vitro* Mammalian Chromosome Aberration test.

Based upon the results presented in the GRAS dossier, the Expert Panel concurs with MoreMeat's conclusion that their mycoprotein is non-genotoxic and non-mutagenic under the tested conditions.

In addition, MoreMeat conducted a 14-days range finding toxicity study on their mycoprotein preparation. While the study is unpublished, the Expert Panel notes that it was completed in a GLP compliant lab and follows OECD guideline No. 407. Furthermore, the results presented in the GRAS dossier indicate that there are no toxicological concerns related to mycoprotein administration under the study conditions.

The Expert Panel has reviewed the totality of toxicity data, both published and corroborative, and concur with MoreMeat's conclusion that consumption of mycoprotein is reasonably expected to be safe under the proposed conditions outlined in the GRAS dossier.

MoreMeat conducted an allergenicity evaluation of its mycoprotein and also contracted immunologist Dr. Rick Goodman to conduct an independent allergenicity evaluation. Following a thorough evaluation of the product information, the *in vitro* digestibility report, the published literature regarding *Fusarium sp.* – including reports and papers on mycoprotein derived from *F. venenatum*— both MoreMeat and Dr. Goodman concluded that MoreMeat's mycoprotein does not represent a likely risk for food allergy. The Expert Panel reviewed the information provided in MoreMeat's GRAS dossier in concert with Dr. Goodman's evaluation report (available as Appendix 3 in the GRAS dossier), and concur with the conclusion that there is no expectation of increased risk for food allergy from the consumption of MoreMeat's mycoprotein.

The Expert Panel reviewed the safety aspects of nutrition and digestibility, mycotoxins, nitrates, ribonucleic acid (RNA), and reported adverse effects. The digestibility test conducted by MoreMeat indicated that over 94% of the proteins are digested by pepsin in relatively short times. Thus, MoreMeat's mycoprotein is expected to have good digestion following consumption and it is very unlikely that usual intakes of mycoprotein from the proposed uses would result in either adverse gastrointestinal or physiological effects. MoreMeat states that the only mycotoxins for which there is the potential for dietary exposure from consuming MoreMeat's mycoprotein are enniatins and beauvericin; however, the Expert Panel notes that DAS must also be considered and should be monitored on at least a biannual bases.

We, the GRAS Panel, unanimously conclude that the proposed use of MoreMeat's mycoprotein as an ingredient in conventional foods, at levels specified herein, up to consumer-only mean and 90th percentile intakes of 94 g/day and 207 g/day, respectively, for the total population (ages 2 years and older), is safe and GRAS based on scientific procedures and supported by a history of safe use.

We, the undersigned independent GRAS Panel, have critically evaluated the safety of MoreMeat's mycoprotein, a non-animal and non-plant mycoprotein, and unanimously conclude that the scientific data, information, and methods described herein establish that the ingredient, when produced in accordance with current Good Manufacturing Practice and meeting the specifications outlined in the supporting dossier, meets the standard of reasonable certainty of no harm and is Generally Recognized As Safe under 21 CFR.

In accordance with Title 21 of the Code of Federal Regulations, Part 170, Subpart E on Generally Recognized as Safe (GRAS) Notices, we, the Expert Panel, have evaluated the scientific evidence and agree with MoreMeat (Guangzhou) Biotech Co., Ltd's conclusion that MoreMeat's mycoprotein is GRAS for use as an ingredient in foods at the levels specified herein. In our opinion, other experts qualified to assess the safety of food ingredients would concur with this conclusion. Therefore, MoreMeat's mycoprotein is excluded from the definition of a food additive and may be used in the U.S. without requiring the promulgation of a food additive regulation by the FDA under 21 CFR.

Margitta Dziwenka, DVM, DABT

President

Toxalta Consulting LLC,



Signature _____ Date: December 12, 2024

Katrina Emmel, Ph.D.

President

KemmelCal Inc.



Signature: _____ Date: December 12, 2024

Jianjun (Gene) Yang, Ph.D.

President

Brandywine BioSciences, LLC



Signature: _____ Date: December 12, 2024

Biographical Sketch of the Panelists

Margitta Dziwenka, DVM, DABT

Dr. Dziwenka holds a Doctor of Veterinary Medicine degree from the University of Saskatchewan and is a Diplomat with the American Board of Toxicology. She has over 27 years of experience as a practicing veterinarian and laboratory animal research experience. Dr. Dziwenka has worked in research, preclinical regulatory toxicology, safety evaluation of human food and animal feed additives, GRAS dossier, and other regulatory filing dossier preparation for those 27 years. Dr. Dziwenka also has extensive experience as an Expert Panelist. She has experience in the management of preclinical contract research organizations and in the role of Study Director and Study Monitor for GLP-compliant preclinical studies and has assisted in the design of numerous preclinical studies. Dr. Dziwenka is experienced in the writing of position papers, safety opinions for a variety of products, manuscripts for publication and critical review of study results used to support regulatory submissions and safety conclusions.

Katrina Emmel, Ph.D.

Dr. Katrina Emmel is the founder and president of KemmelCal Inc. With an MS in Biochemistry (WPI) and a Ph.D. in Chemistry (Northern Illinois University), Dr. Emmel started her career as an analytical chemist, performing routine LC-MS analysis in a clinical laboratory, and then developing analytical methods for a dietary supplement and food manufacturer. She was involved in preparing the first Generally Recognized as Safe (GRAS) Notification for Enzyme Modified Steviol Glycosides on behalf of NOW Foods. As an independent consultant since 2013, Dr. Emmel has been involved in a number of GRAS evaluations as a chemist, technical writer, project manager, Expert Panelist, and as Chair of the Expert Panel. She has also served as an Expert Witness to support patent litigation activities for food ingredients.

Jianjun (Gene) Yang, Ph.D.

Dr. Yang has led a distinguished career in food ingredients innovation. He obtained his Ph.D. from University of Kentucky, and then completed his NIH fellowship at Cornell University. He had worked for almost 20 years at the DuPont Central Research and Development as Principal Research Scientist, leading the DuPont Biotech Lab in many innovation projects for food ingredients and industrial materials. He had also worked as the leader and advisor in other SynBio and pharmaceutical companies. Dr. Yang has many U.S. and international patents, patent applications, and peer-reviewed publications. His expertise includes genetic engineering, protein engineering, and metabolic engineering.